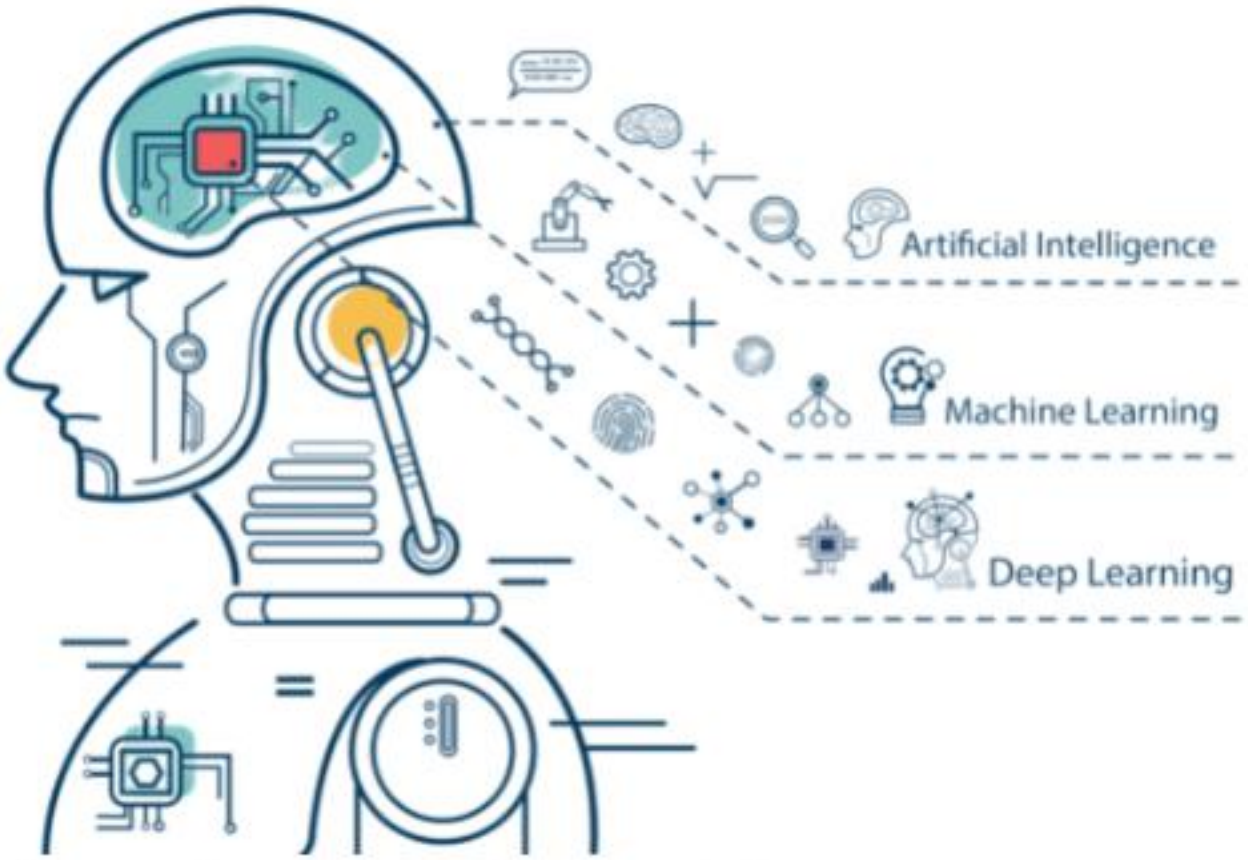
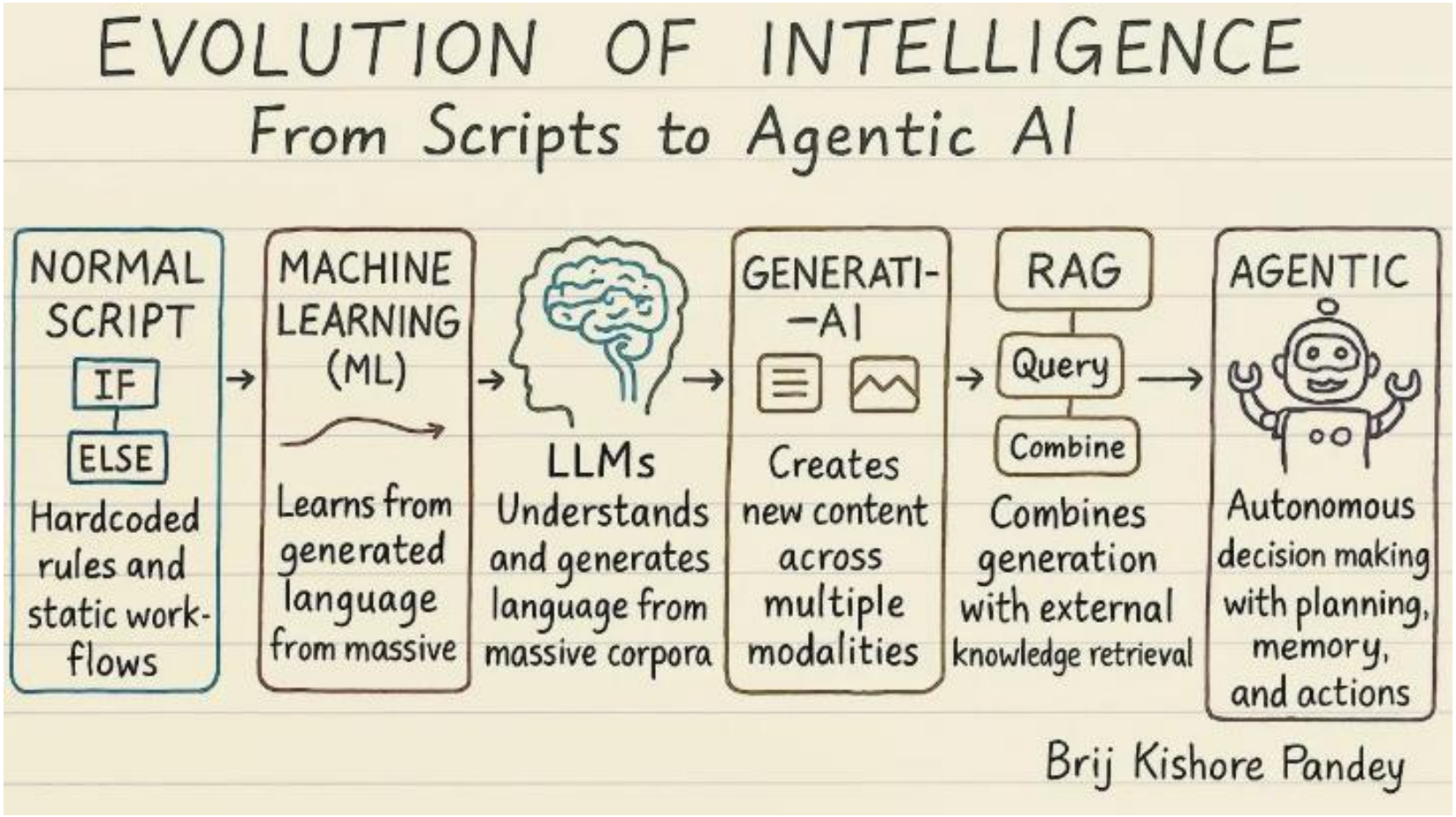
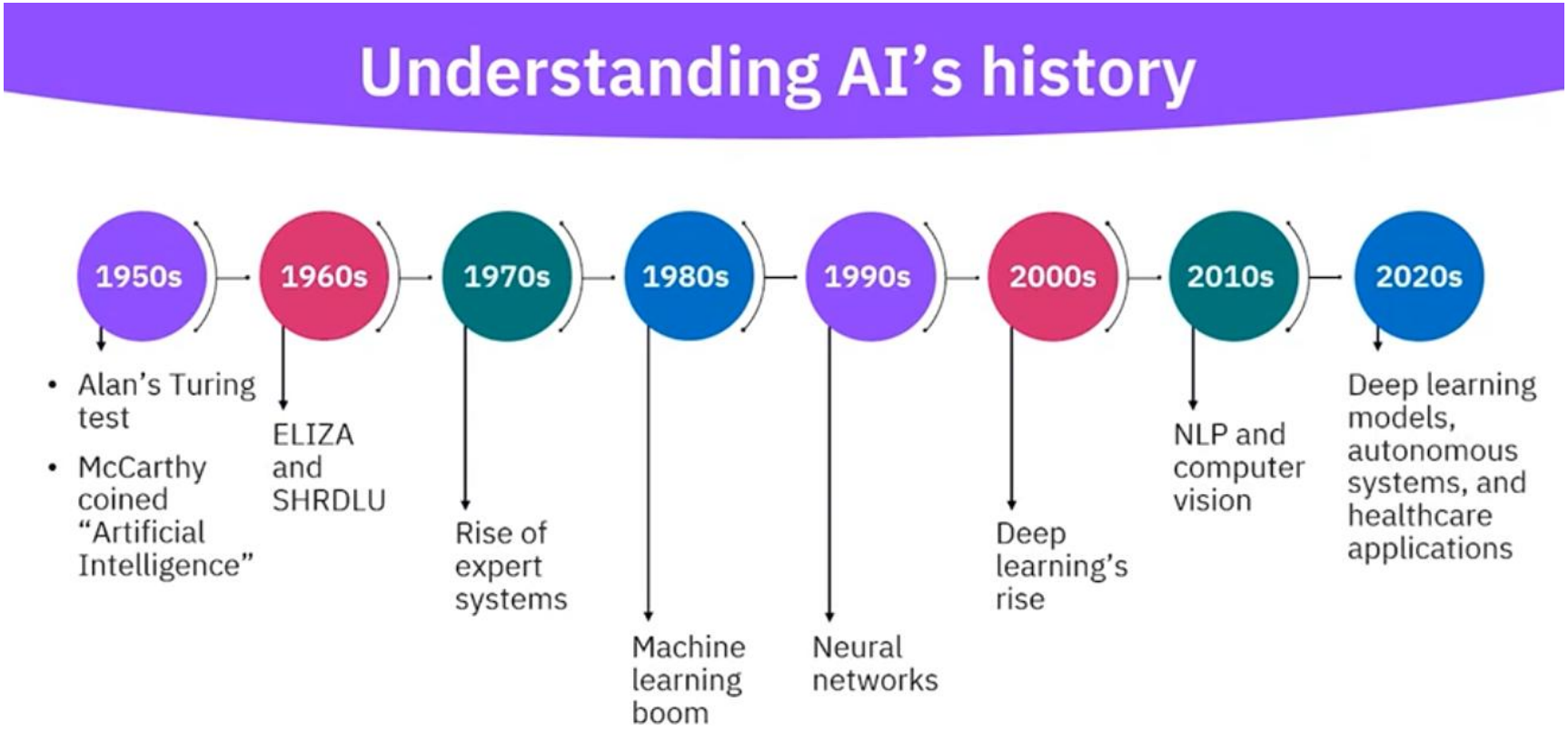
Evolution of Intelligence: From Data to Generative AI



Evolution of Intelligence: From Data to Generative AI



Evolution of Intelligence: From Data to Generative AI

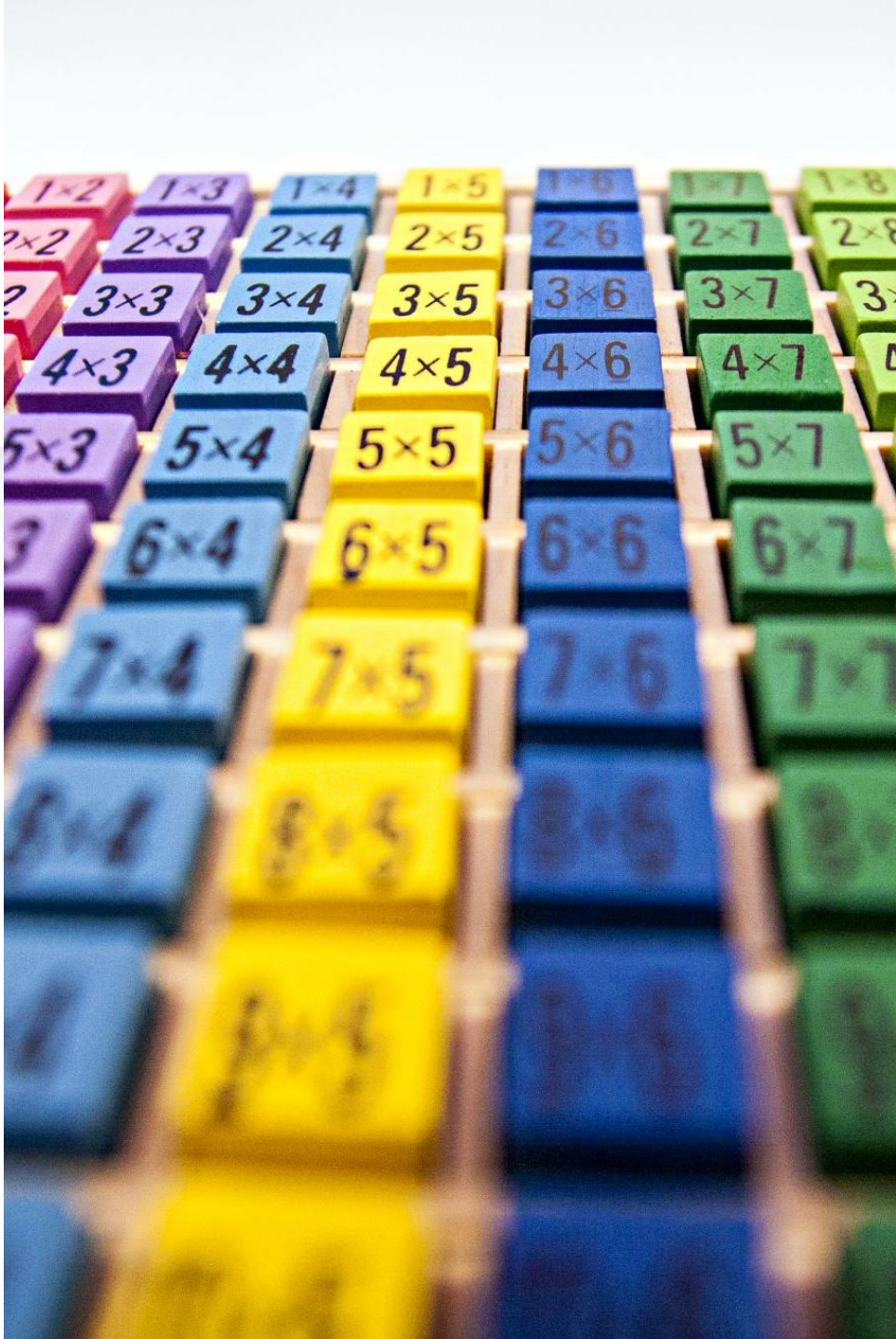


Contents

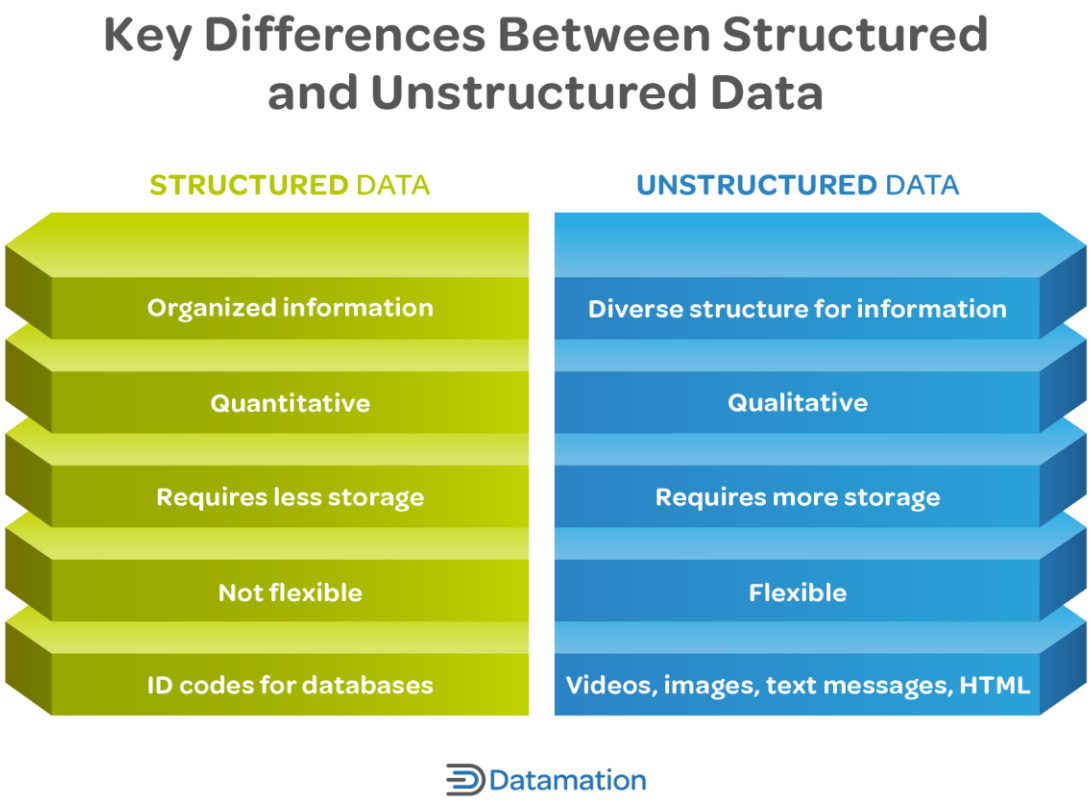
- **Data Foundation**
  The journey begins with raw data—structured and unstructured—serving as the backbone for all analytical and intelligent systems.
- **Descriptive & Predictive Analytics**
  Leveraging statistical methods and business intelligence tools to understand historical patterns and forecast future trends.
- **Machine Learning (ML)**
  Algorithms learn from data to make predictions or decisions without being explicitly programmed. This marks the shift from rule-based systems to data-driven intelligence.
- **Deep Learning (DL)**
  A subset of ML that uses neural networks with multiple layers to model complex patterns, especially in image, speech, and text data.
- **Artificial Intelligence (AI)**
  Encompasses ML and DL, aiming to simulate human-like intelligence in tasks such as reasoning, perception, and decision-making.
- **Generative AI (GenAI)**
  The latest frontier—models that not only analyze but **create** content. GenAI systems like large language models and diffusion models generate text, images, code, and more, revolutionizing productivity and creativity.

Introduction to Data & Variables

- Data: Raw facts and figures collected for analysis.
- Variables: Attributes that can take different values.
- Types of Variables: Quantitative (numerical), Qualitative (categorical).



Structured Vs Unstructured Data



WHAT IS DATA MINING?

A set of techniques to extract valuable information from large data volume

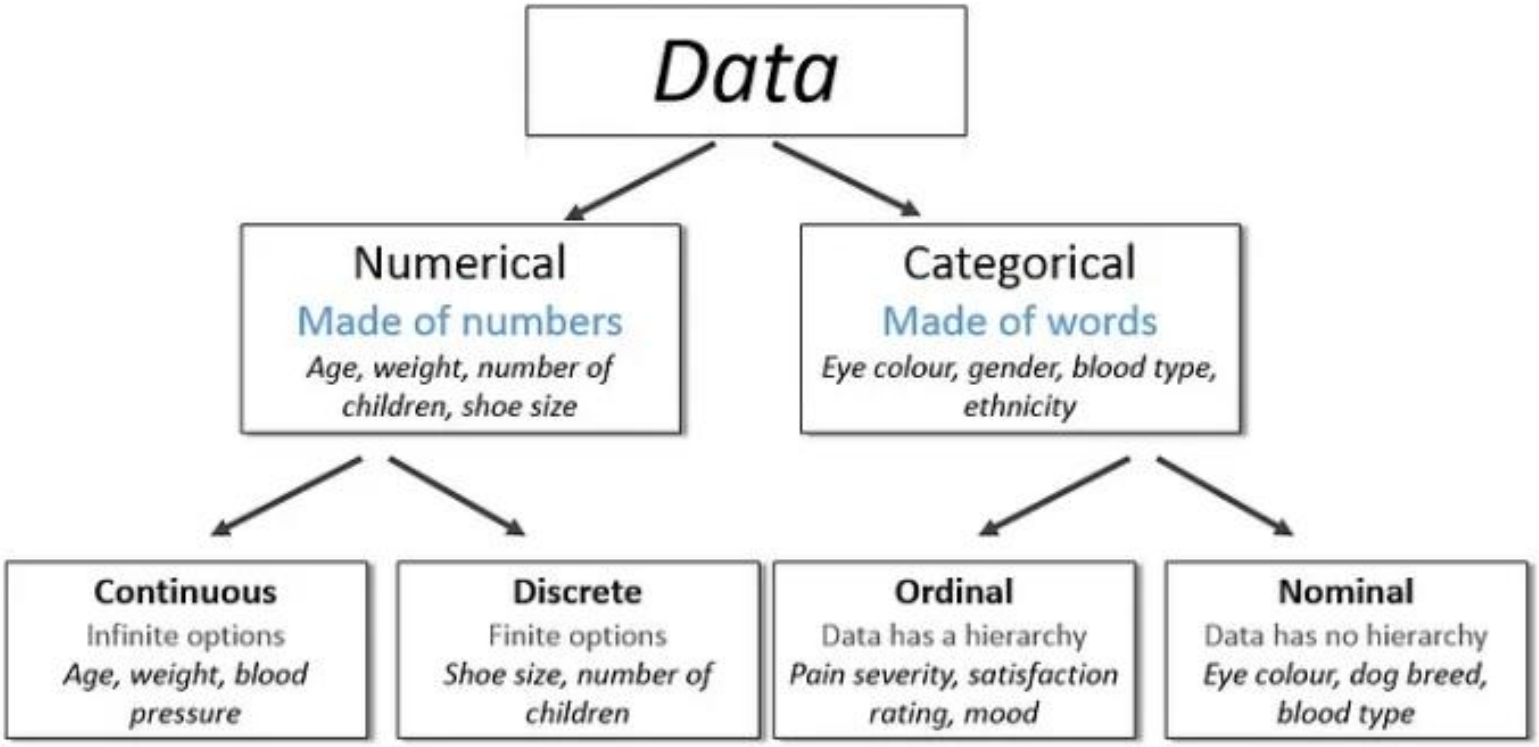
Apply statistics or artificial Inteligence to extract valuable information from large database



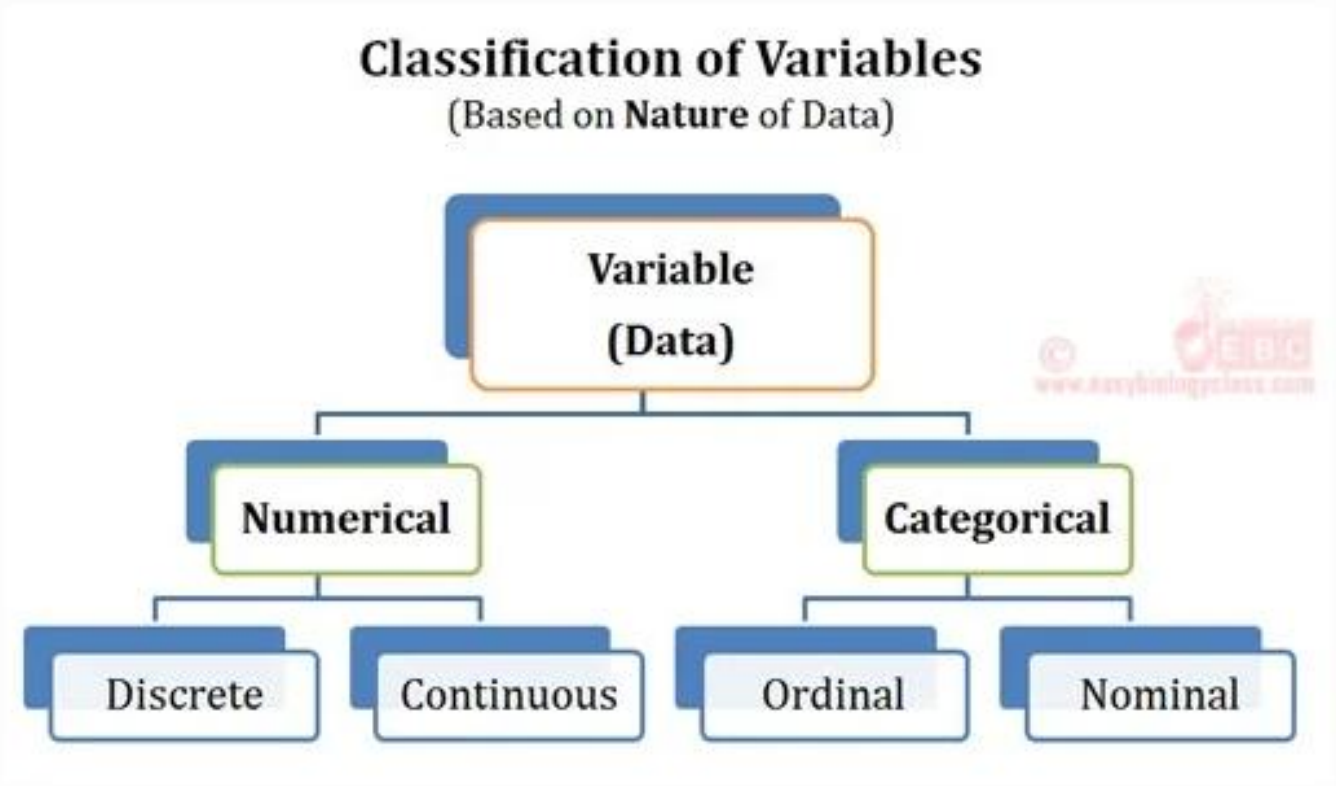
The 4 categories information gathered from Data mining



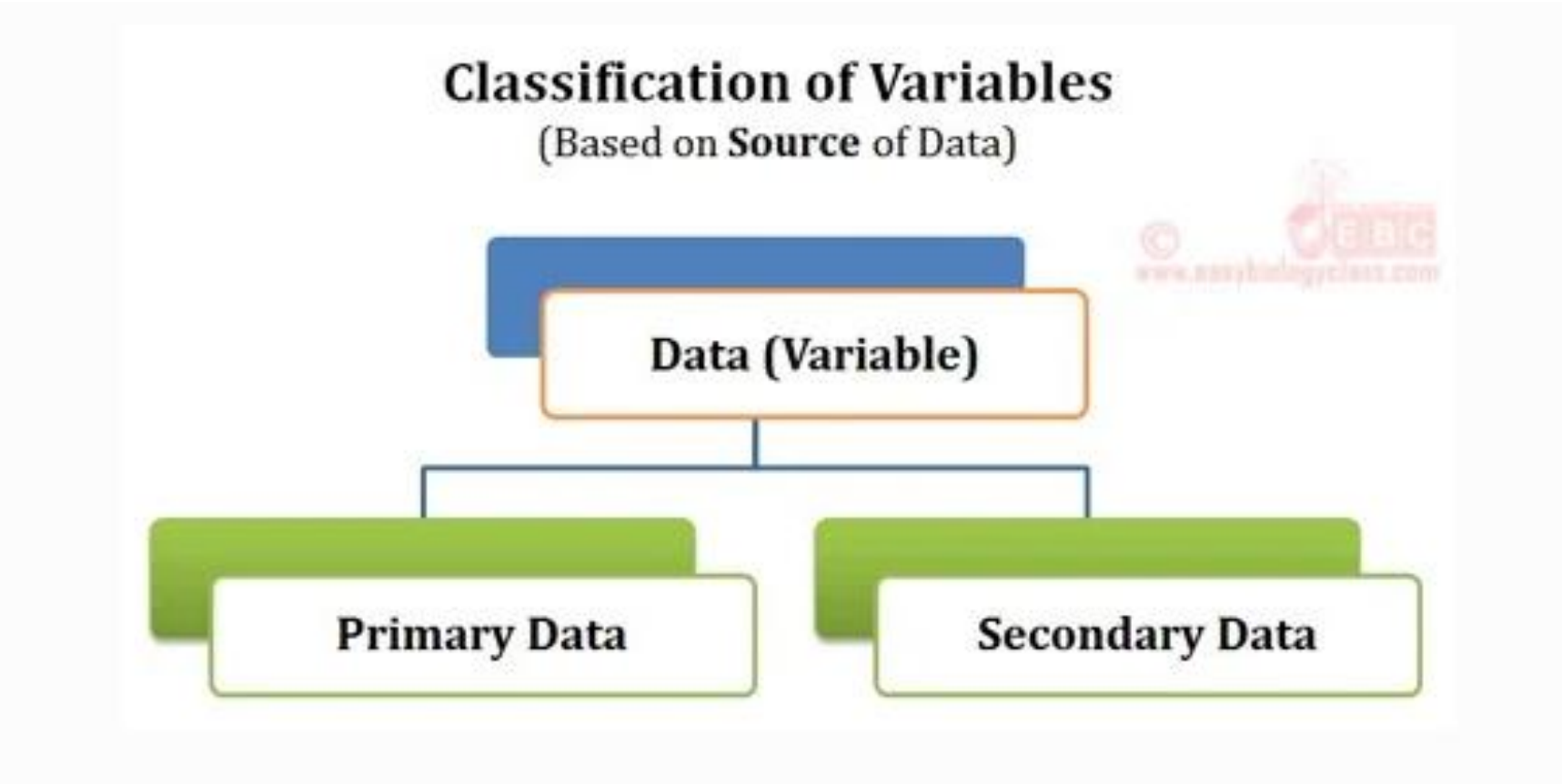
Types of Data



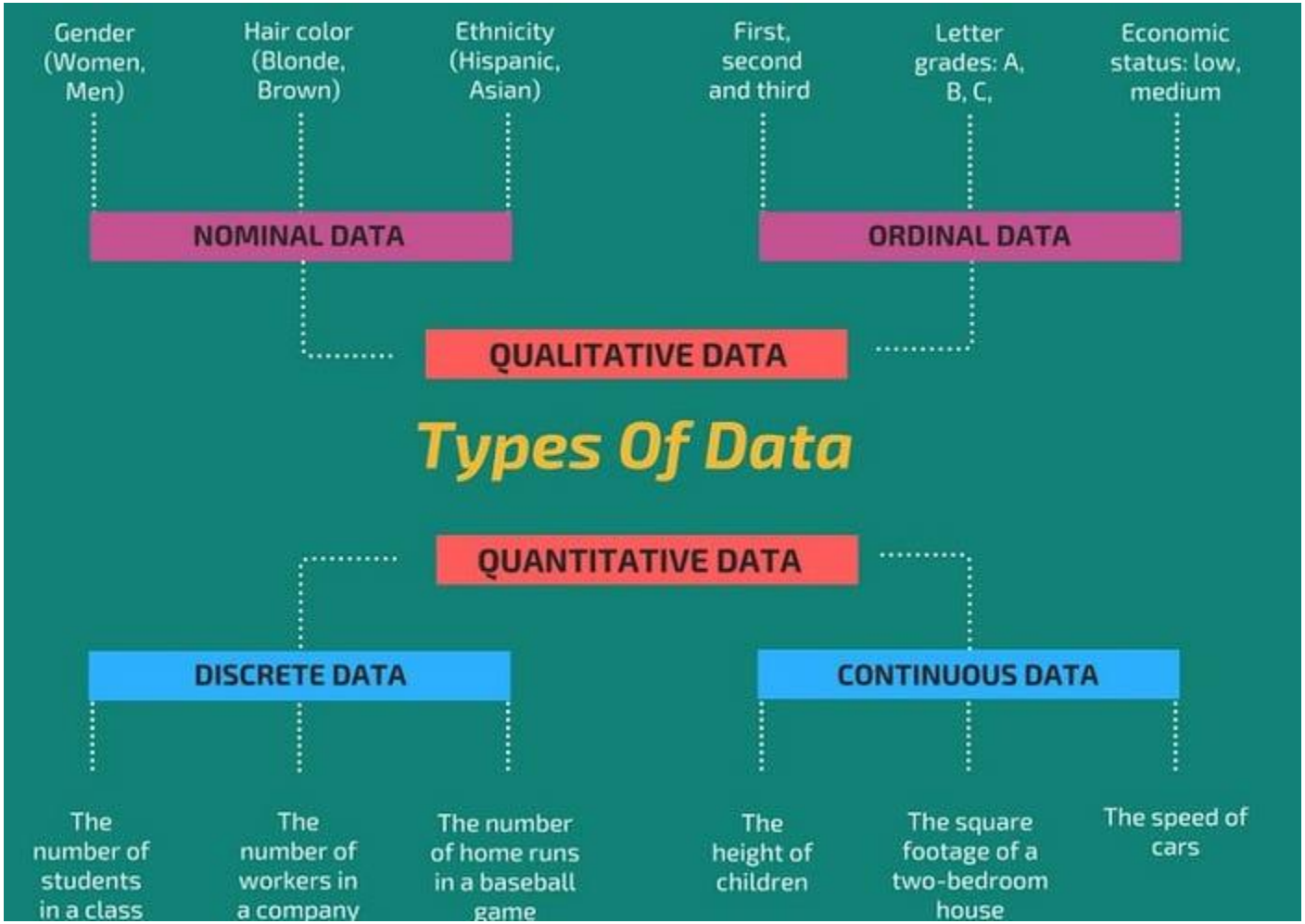
Classification of variables



Classification of variables



Types of Data



Dependent vs. Independent Variables



Independent Variable: Input or cause used to predict outcomes.

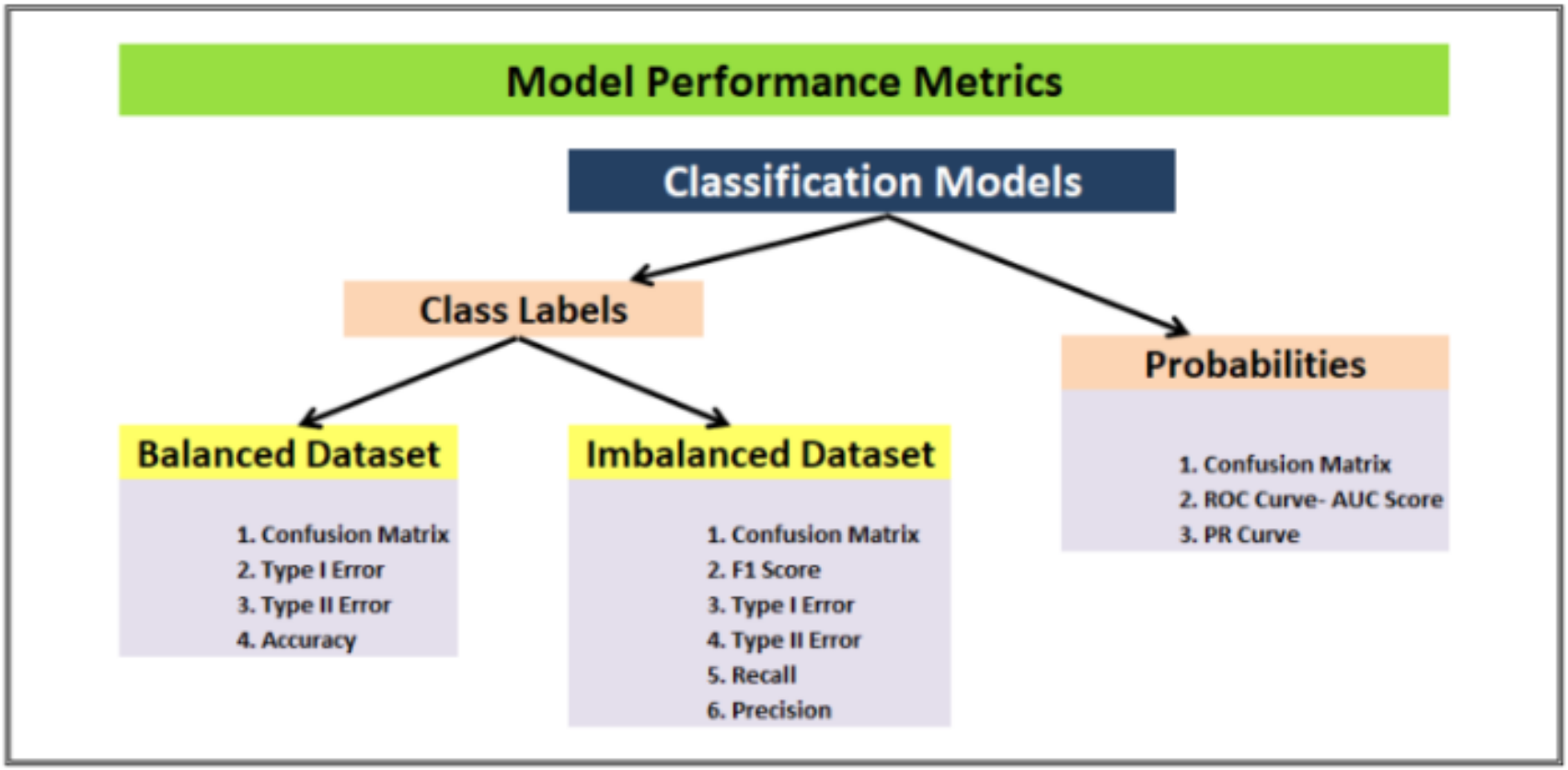


Dependent Variable: Output or effect being predicted.



Example: Advertising Spend (Independent) → Sales (Dependent).

Model Performance Metrics



Details on Metrics

Accuracy
Measures how many emails were correctly classified overall.
✅ Good when spam and non-spam emails are balanced.
❗ Misleading if 95% of emails are not spam—your model could be 95% accurate by always predicting "Not Spam."

Precision
Of all emails predicted as spam, how many were actually spam?
✅ High precision = fewer false alarms (important if marking a legit email as spam is costly).

Recall
Of all actual spam emails, how many did the model catch?
High recall = fewer missed spam emails.
❗ Important if missing a spam email could lead to phishing or malware exposure.

F1 Score
Balances precision and recall.
✅ Useful when you need a trade-off between catching spam and avoiding false positives.

Logarithmic Loss (Log Loss)
Evaluates the confidence of the model's predictions.
✅ Penalizes incorrect predictions more if the model was confident.
❗ Helps in tuning models that output probabilities (e.g., logistic regression).

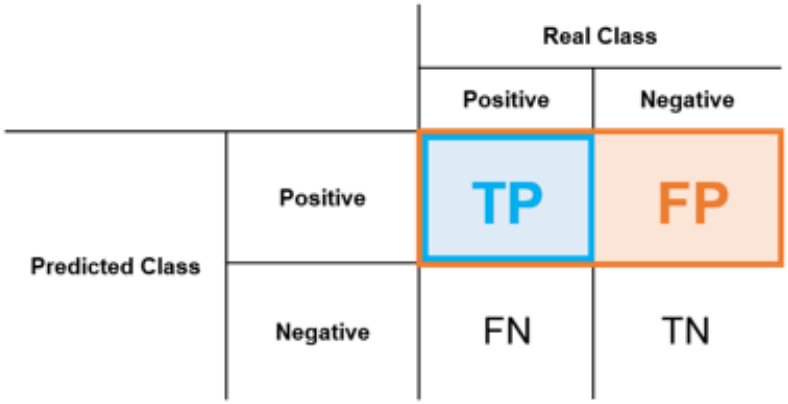
AUC & ROC Curve
Measures how well the model distinguishes between spam and not spam across all thresholds.
✅ AUC close to 1 = excellent model.
❗ Useful for comparing models regardless of classification threshold.

Precision and Recall: Balancing the Trade-offs

Precision

Of all positive predictions, how many are really positive?

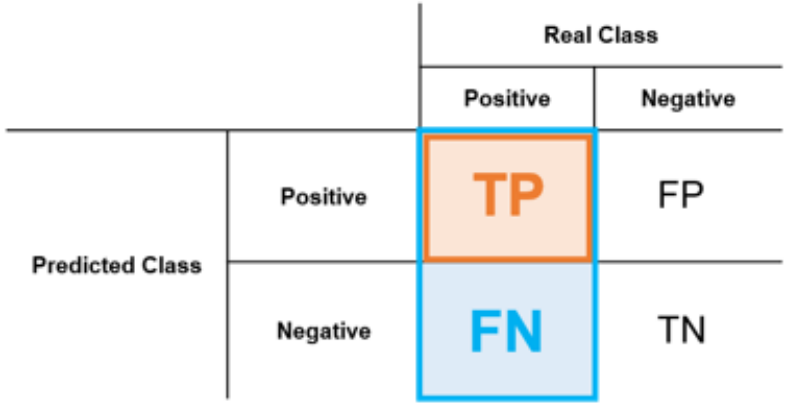
$$\frac{TP}{TP + FP}$$



Recall

Of all real positive cases, how many are predicted positive?

$$\frac{TP}{TP + FN}$$



F1 Score: Harmonizing Precision and Recall



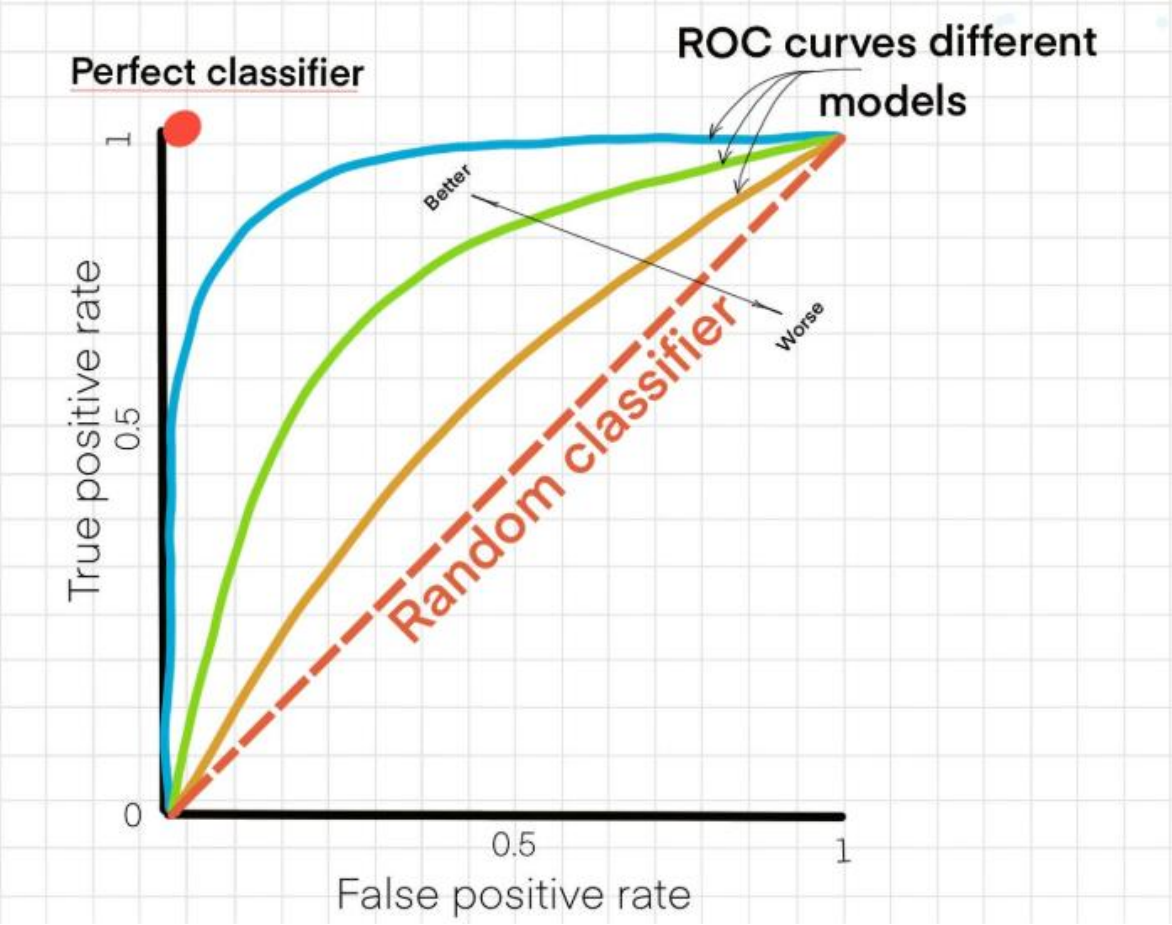
$$F1 = \frac{2 * precision * recall}{precision + recall}$$

$$F1 = \frac{2 \times 0.3 \times 0.1}{0.3 + 0.1} \quad \therefore F1=0.15$$

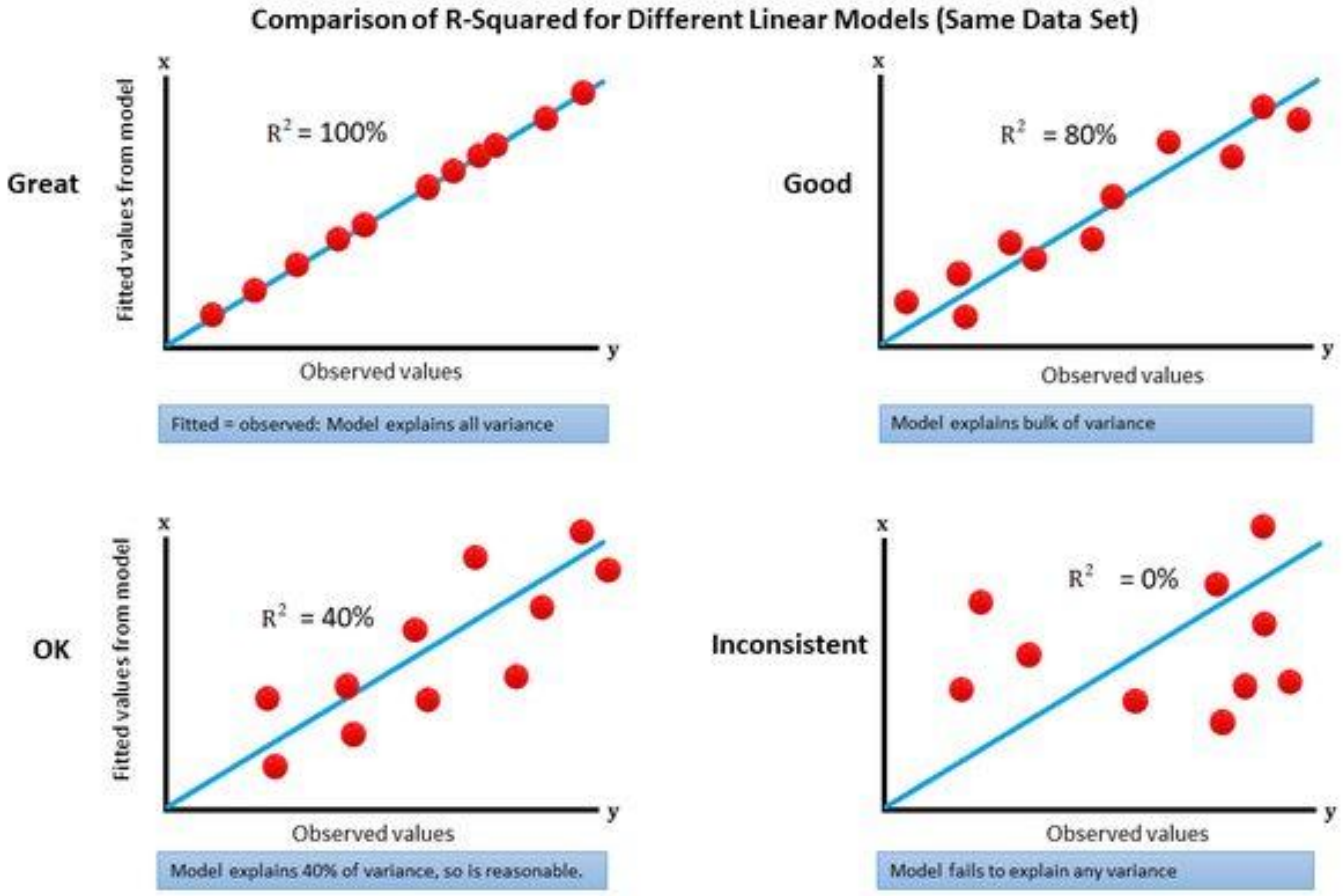
Harmonic mean is conservative mean compared to Arithmetic meand and geometric mean.

It means that Harmonic mean is nearest to the smallest of the input numbers.

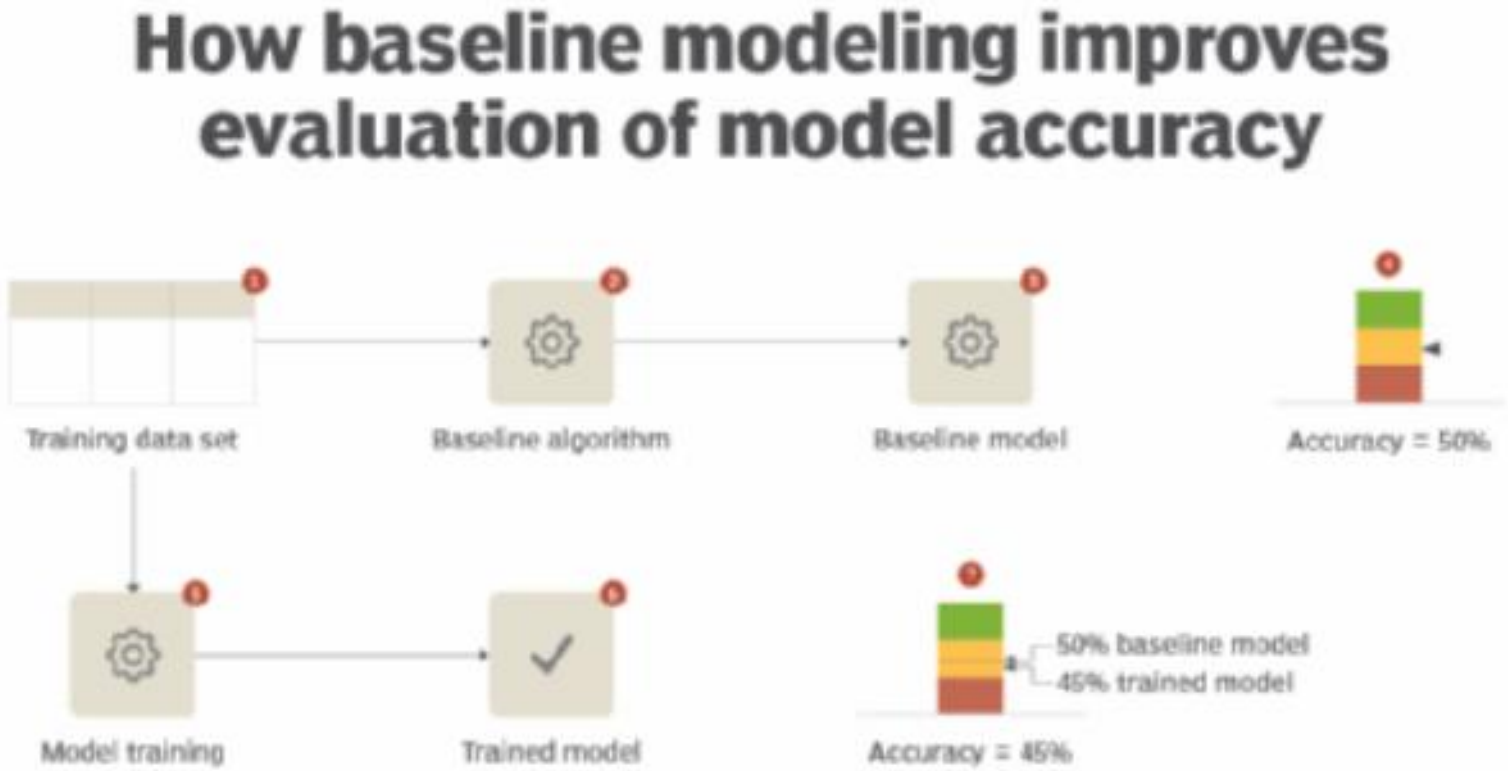
ROC Curve and AUC: Evaluating Classifier Performance

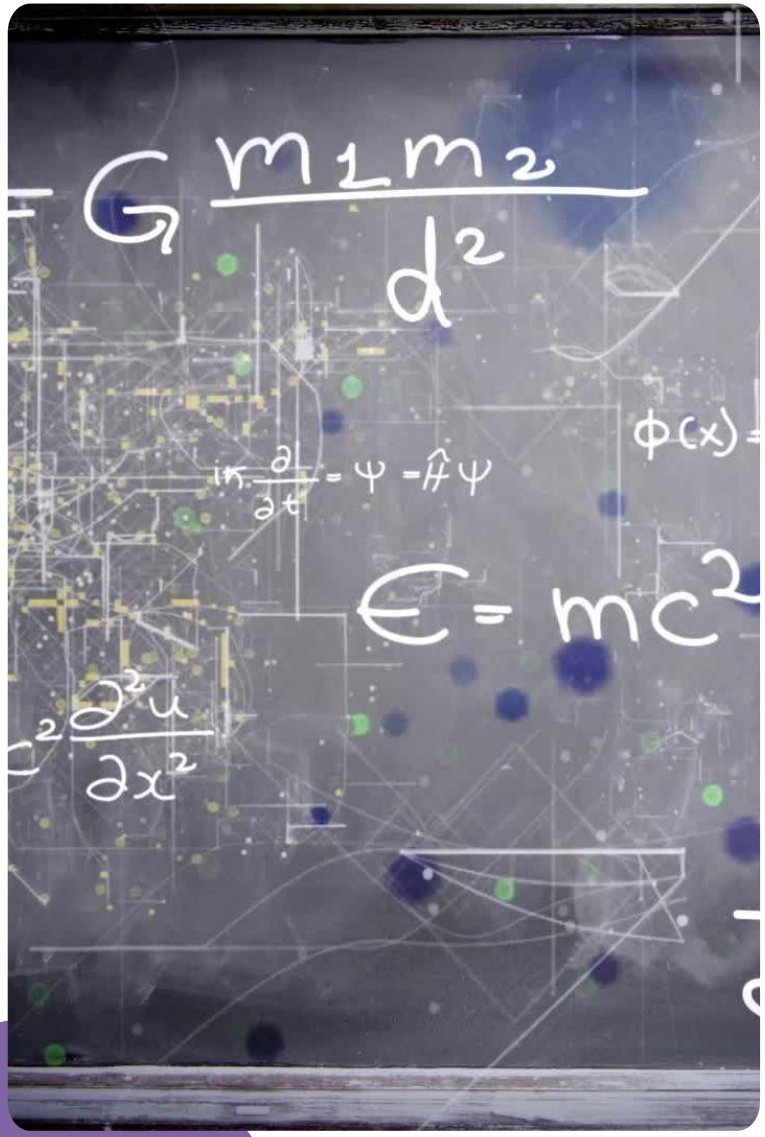


Mean Squared Error and R^2: Key in Regression Analysis



Model Performance Metrics





Measures of Central Tendency

- Mean: Average value. Formula: Mean = $\Sigma x_i / n$
- Median: Middle value when data is sorted.
- Mode: Most frequently occurring value.

Measures of Dispersion

Variance: Average squared deviation from the mean.

Formula: Variance = $\Sigma(x_i - \mu)^2 / n$

Standard Deviation: Square root of variance.

Indicates how spread out the data is.

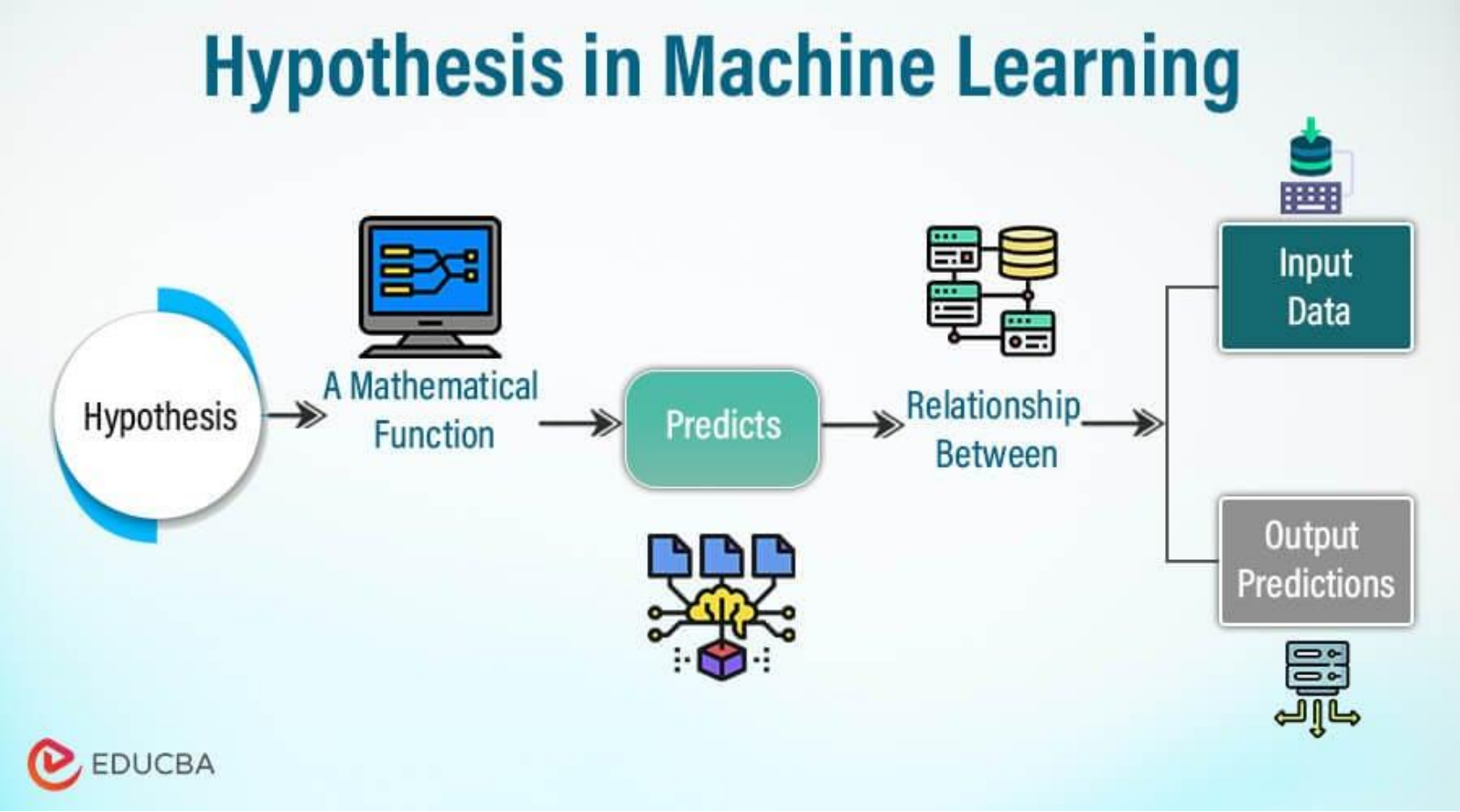
Population vs. Sample

Population: Entire group of interest.

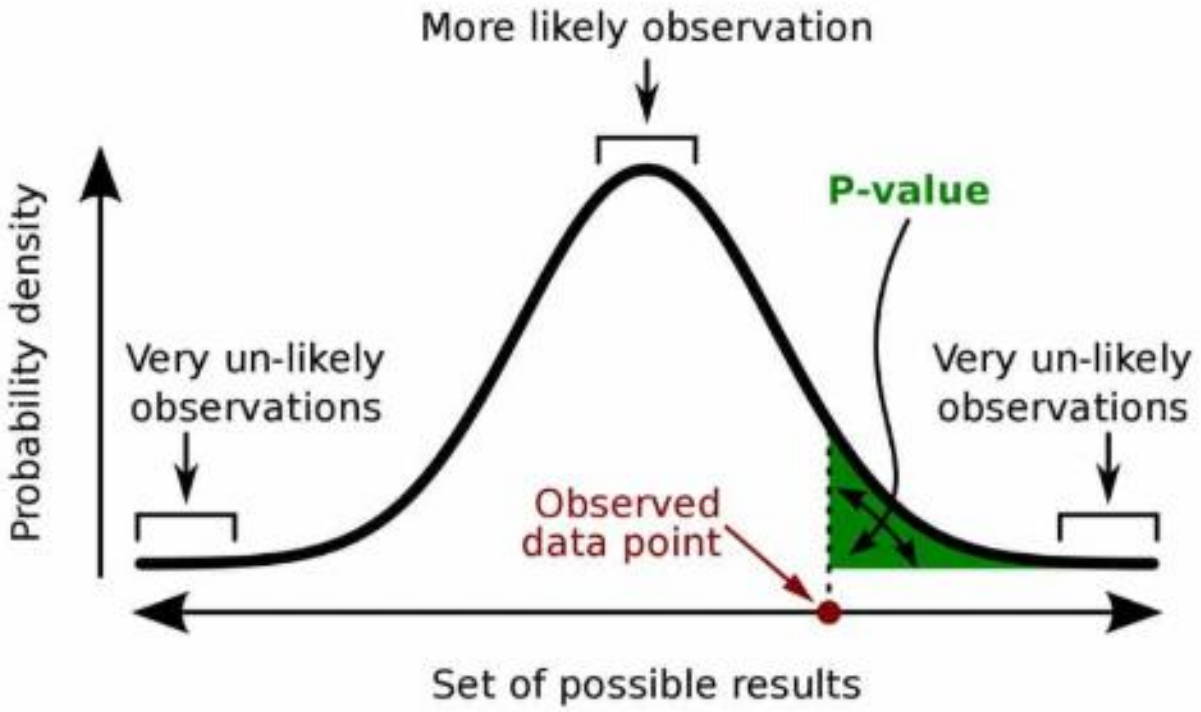
Sample: Subset of the population used for analysis.

Sampling is cost-effective and sufficient for inference.

Hypothesis Testing



Hypothesis Testing P Value



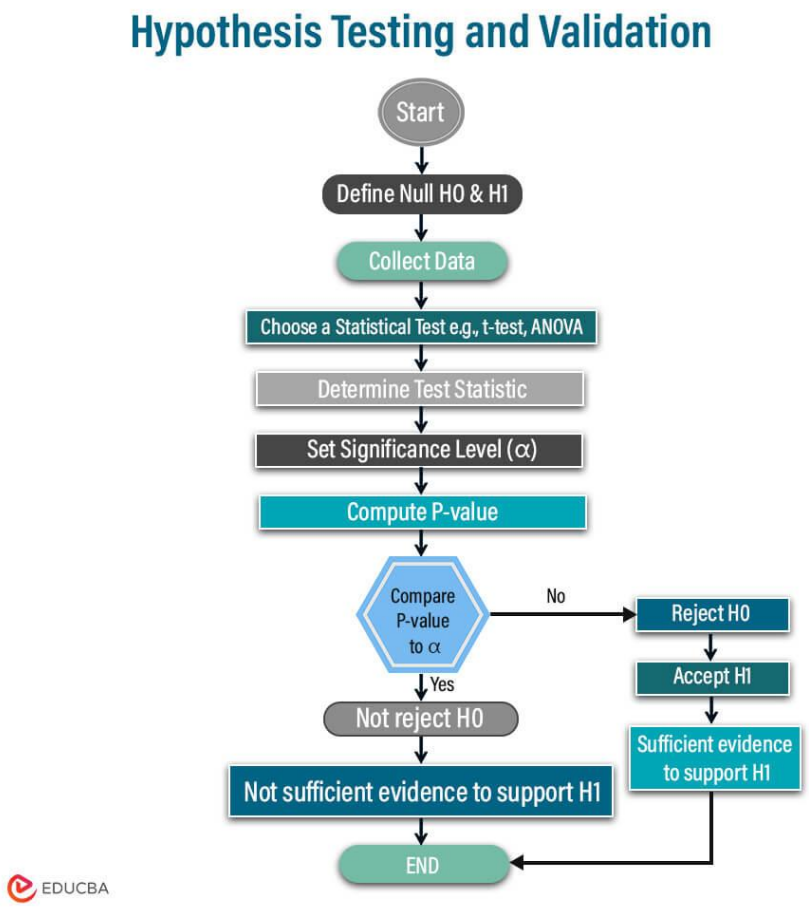
A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

Hypothesis Testing

- Purpose: Make decisions about a population based on sample data.
- Null Hypothesis (H_o): No effect or difference.
- Alternative Hypothesis (H_1): There is an effect or difference.
- P-value: Probability of observing the data if H_o is true.
- Significance Level (α): Commonly set at 0.05.

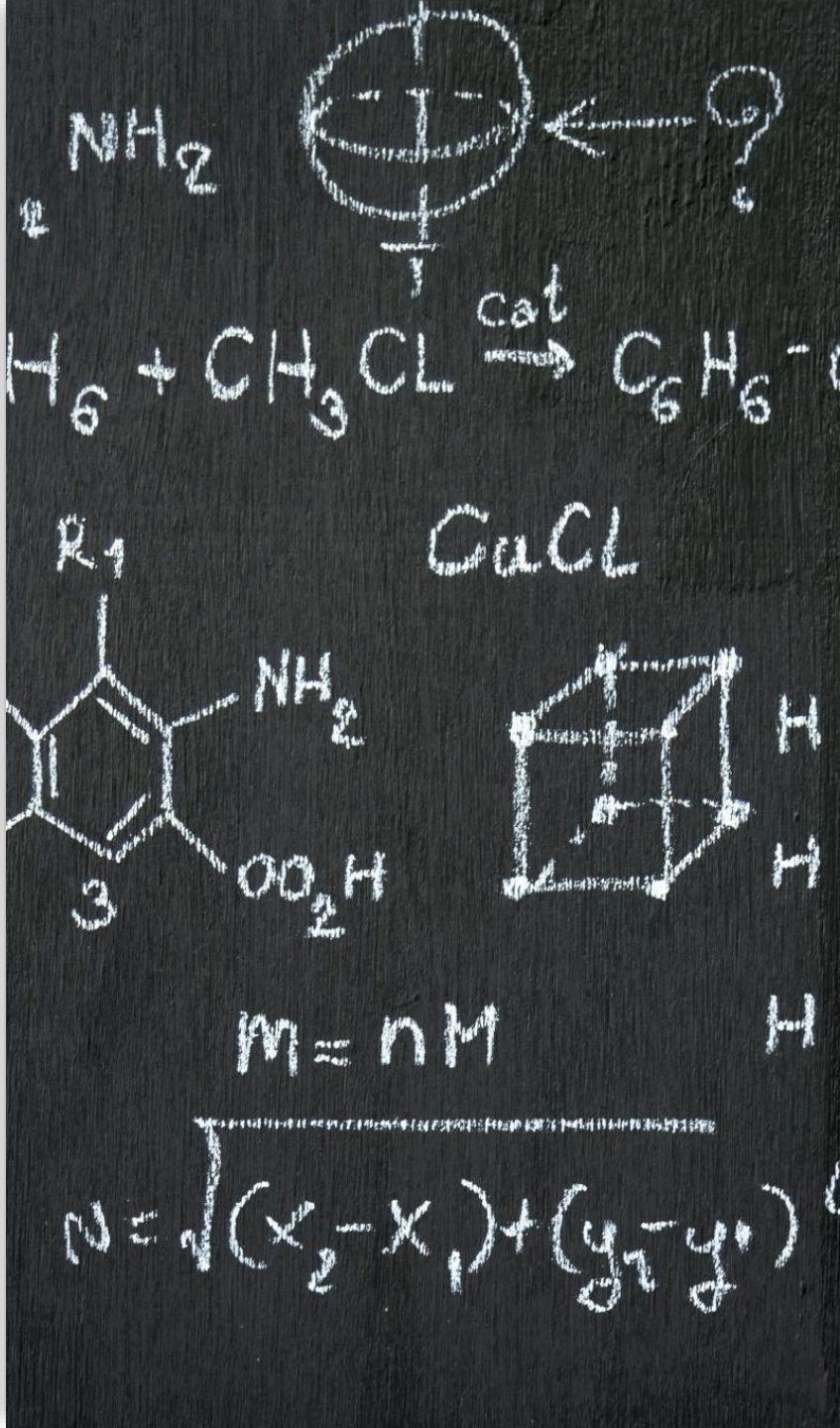


Hypothesis Testing and Validation

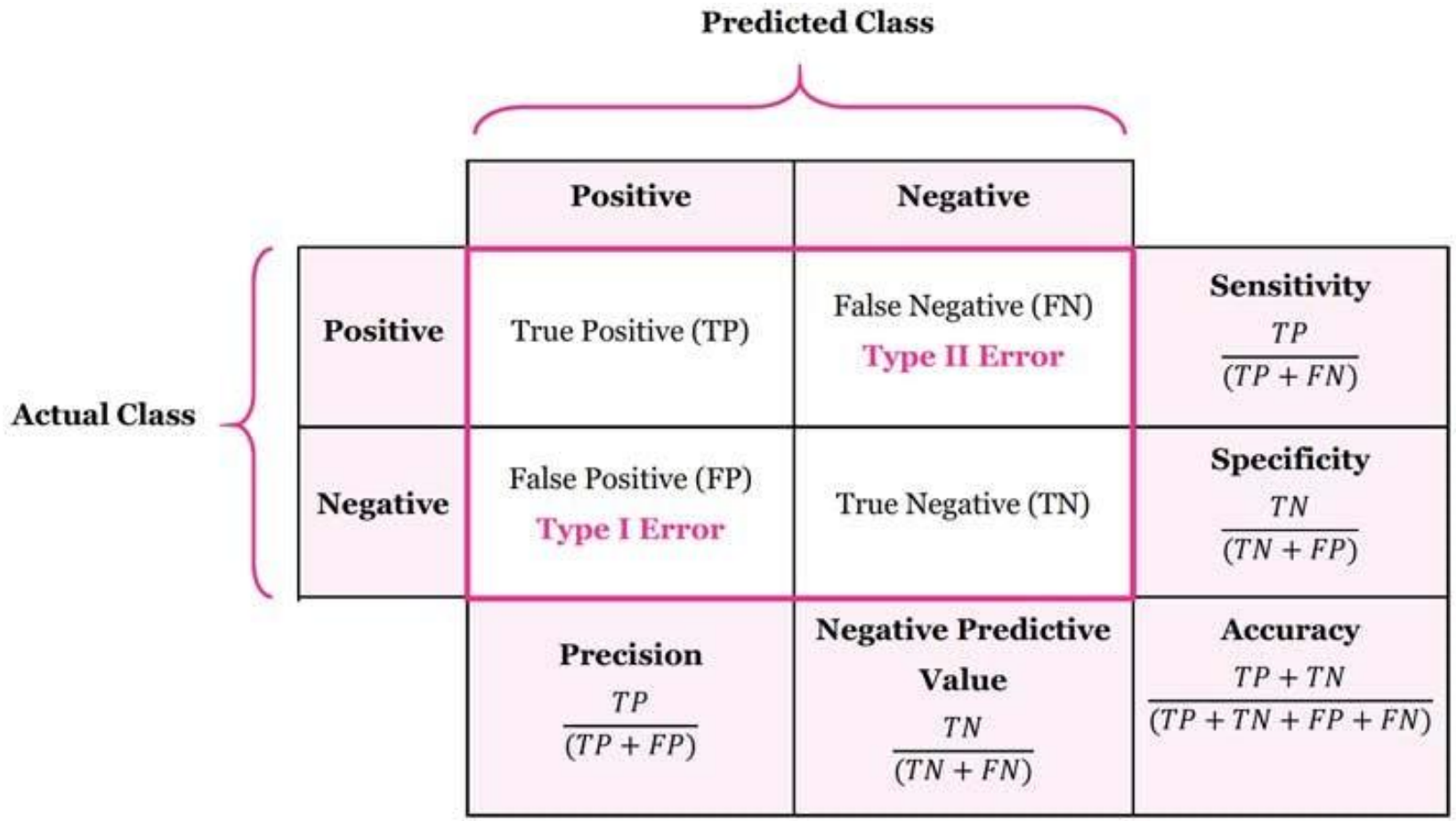


Correlation Matrix

- Definition: Table showing correlation coefficients between variables.
- Values range from -1 to +1.
- +1: Perfect positive correlation, -1: Perfect negative correlation, 0: No correlation.
- Use Case: Identifying relationships between multiple variables.



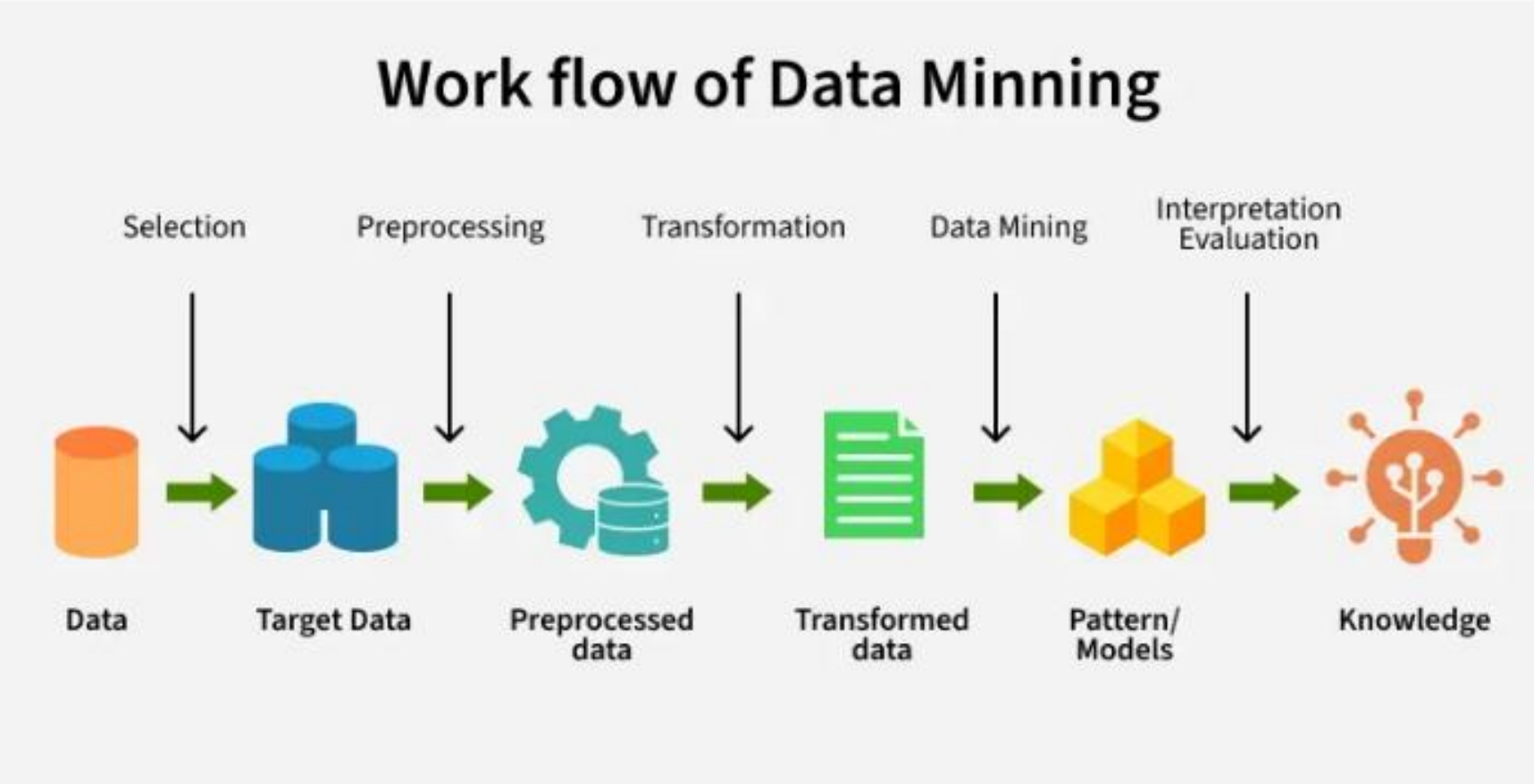
Confusion Matrix



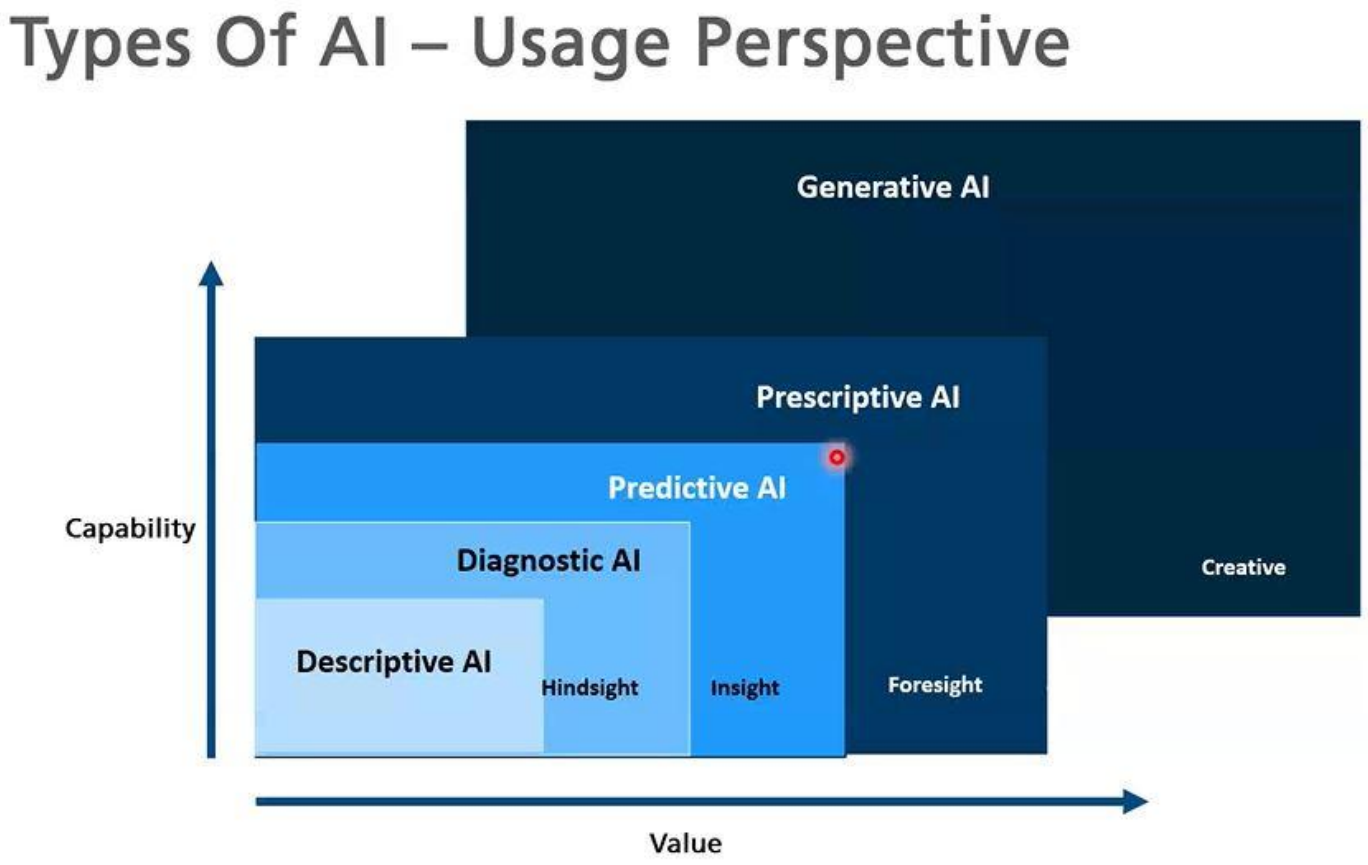
Data Minning



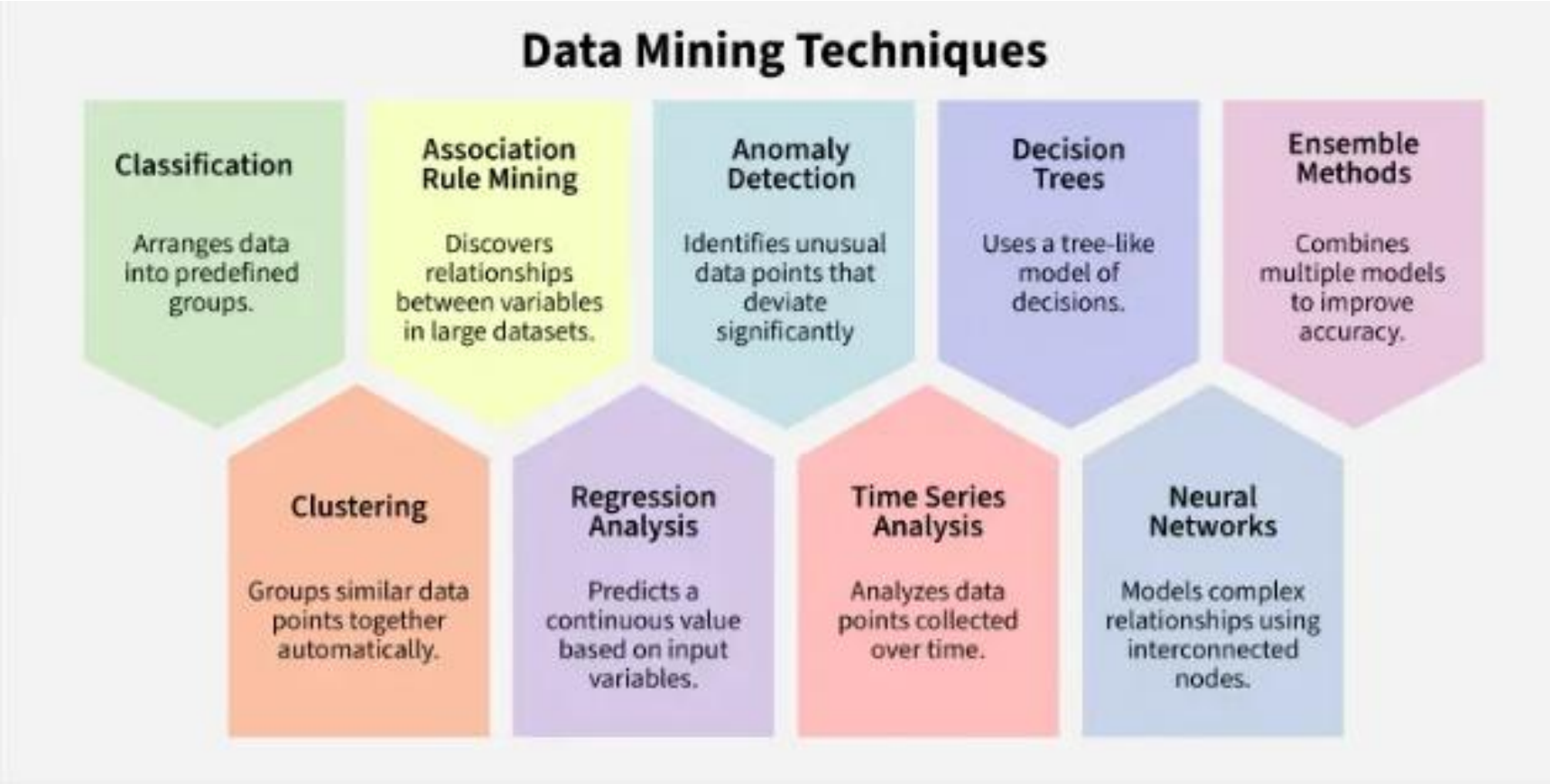
Workflow of Data Minning



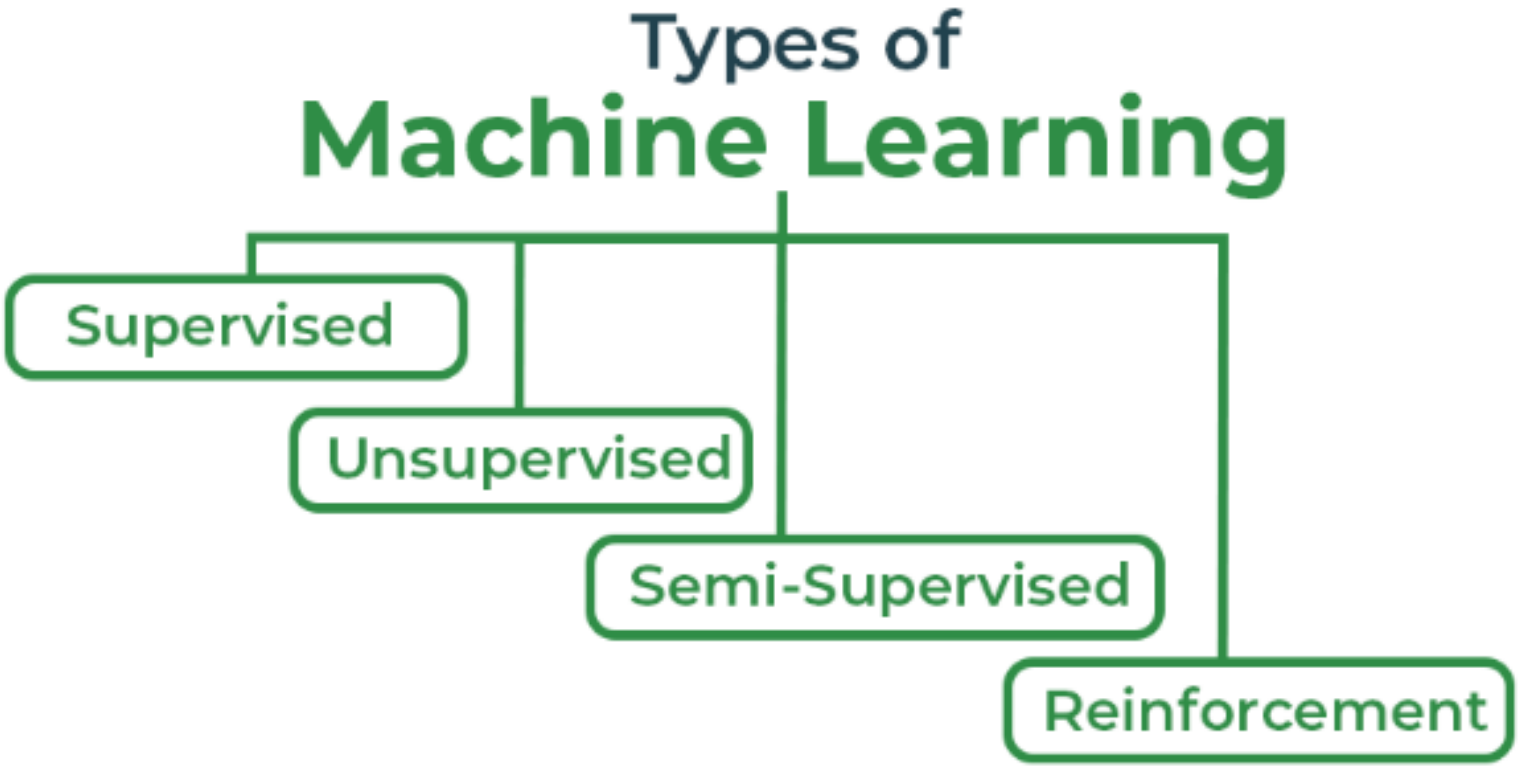
Types of AI – Usage Perspective



Data Minning Techniques



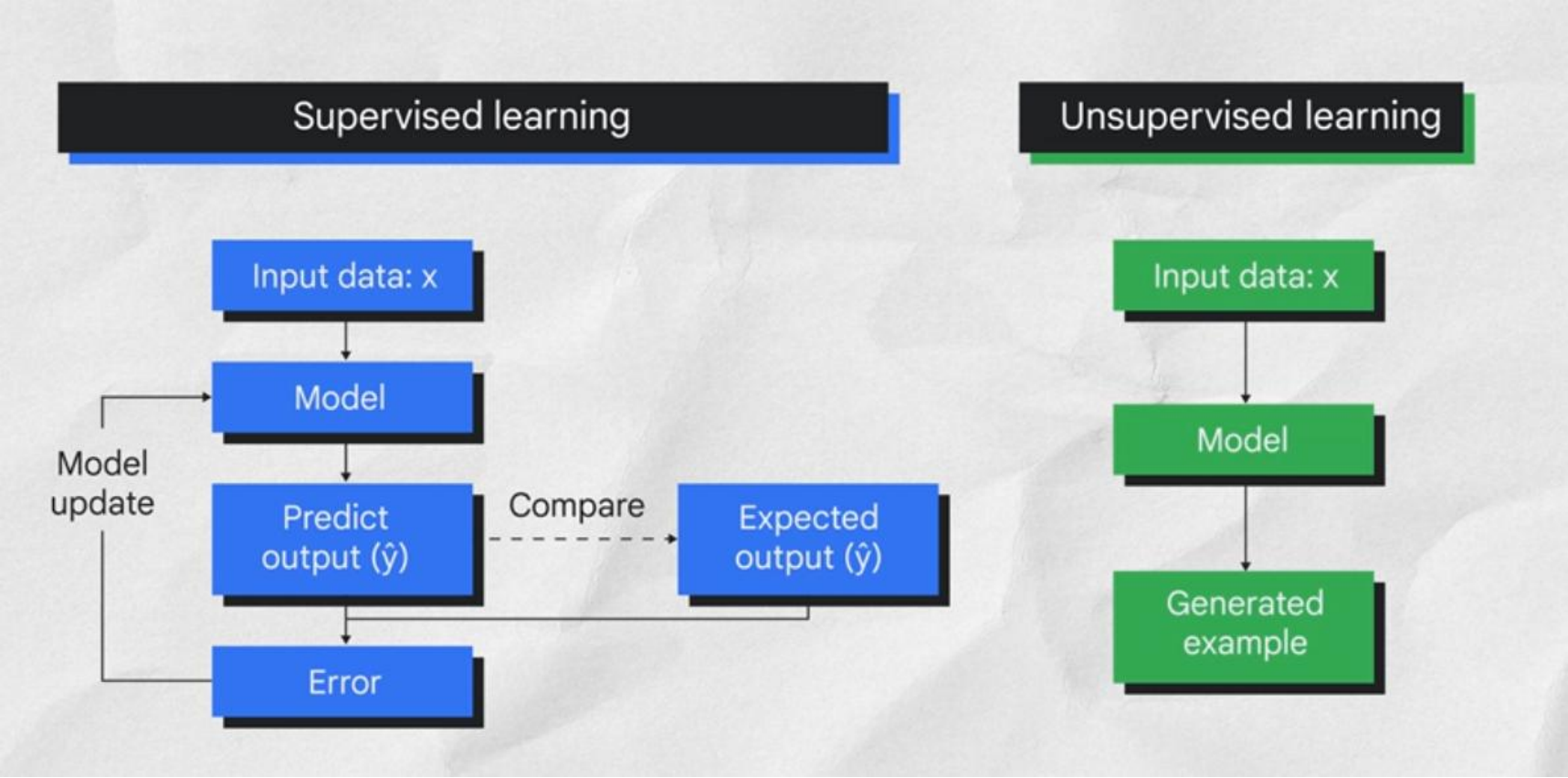
Types of Machine Learning



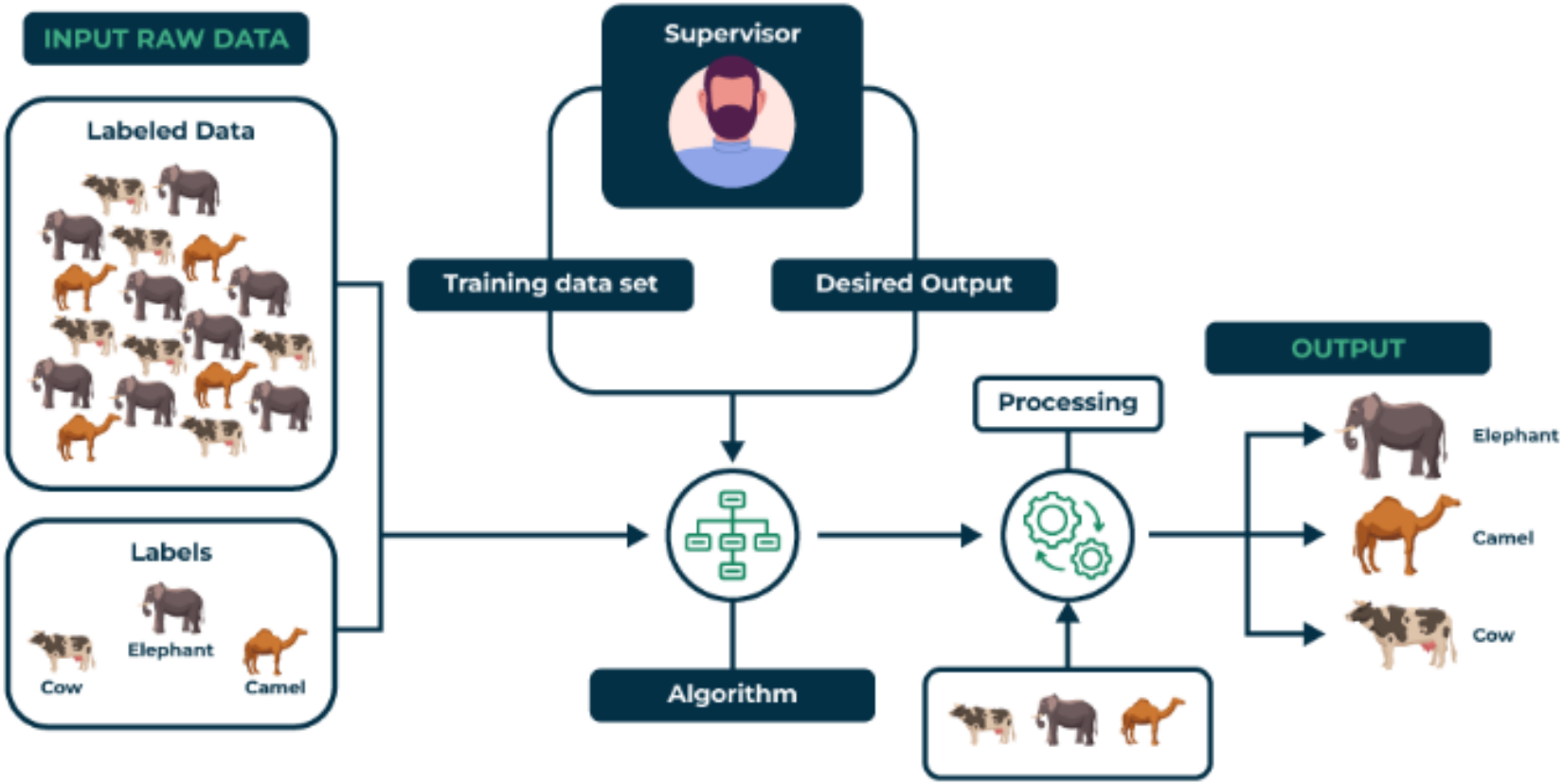
Supervised & Un Supervised learning



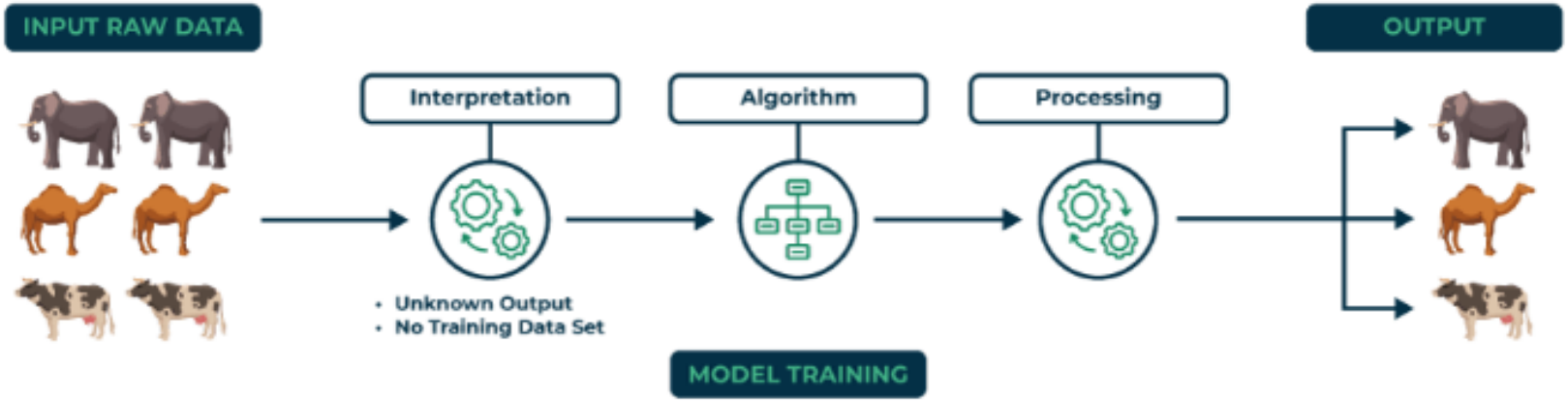
Supervised & Un Supervised learning



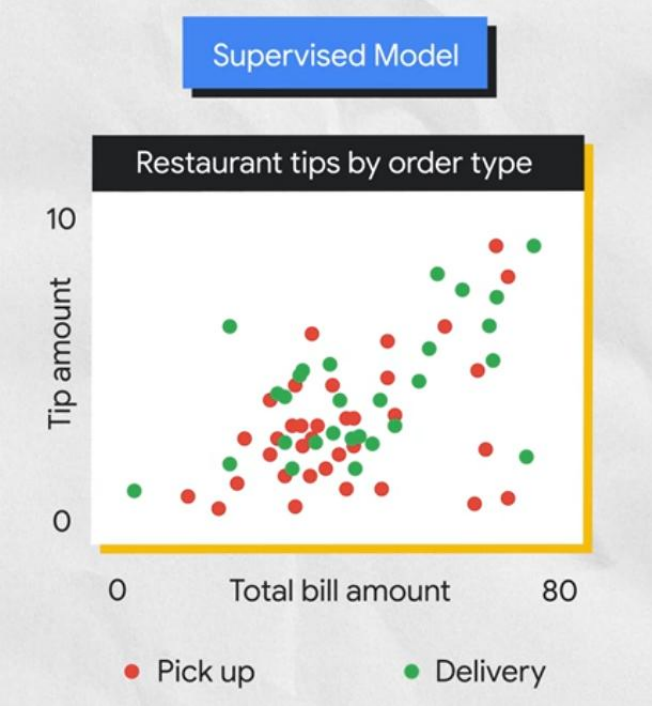
Supervised Machine Learning

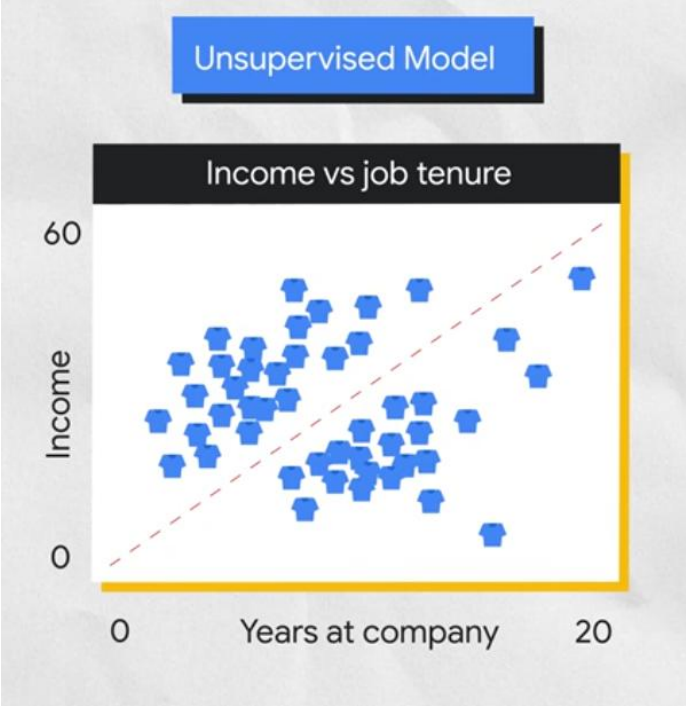


Un Supervised Machine Learning

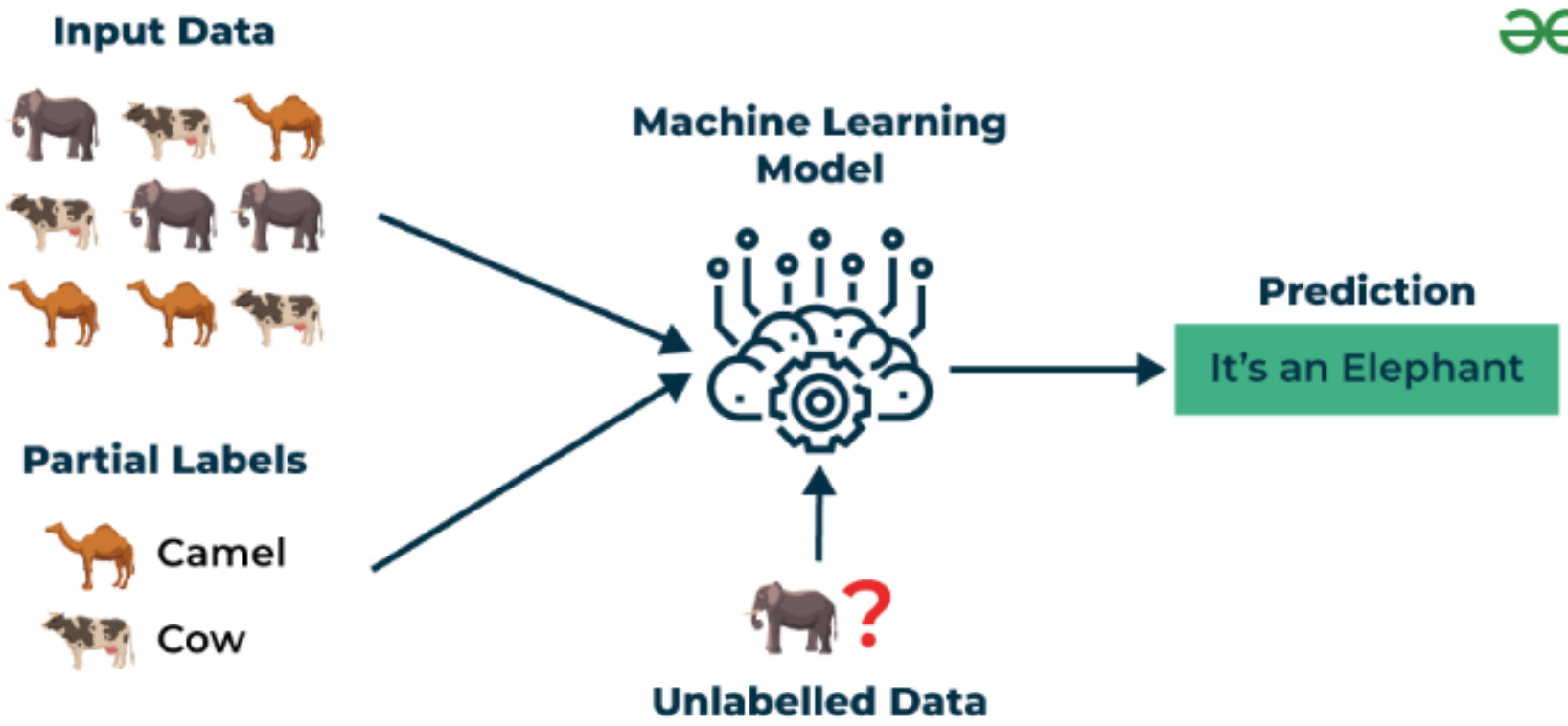


Supervised & Un Supervised learning



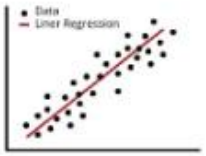


Semi Supervised Machine Learning

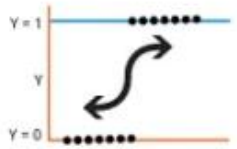


Machine Learning Types

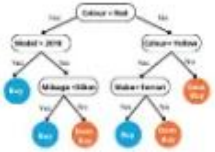
Linear Regression



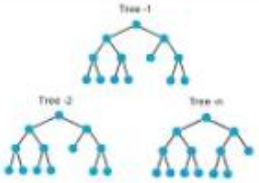
Logistic Regression



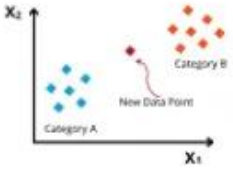
Decision Trees



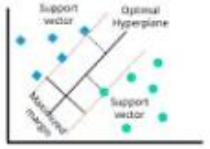
Random Forest



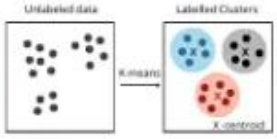
K-Nearest Neighbor



Support Vector Machine



K-Means Clustering

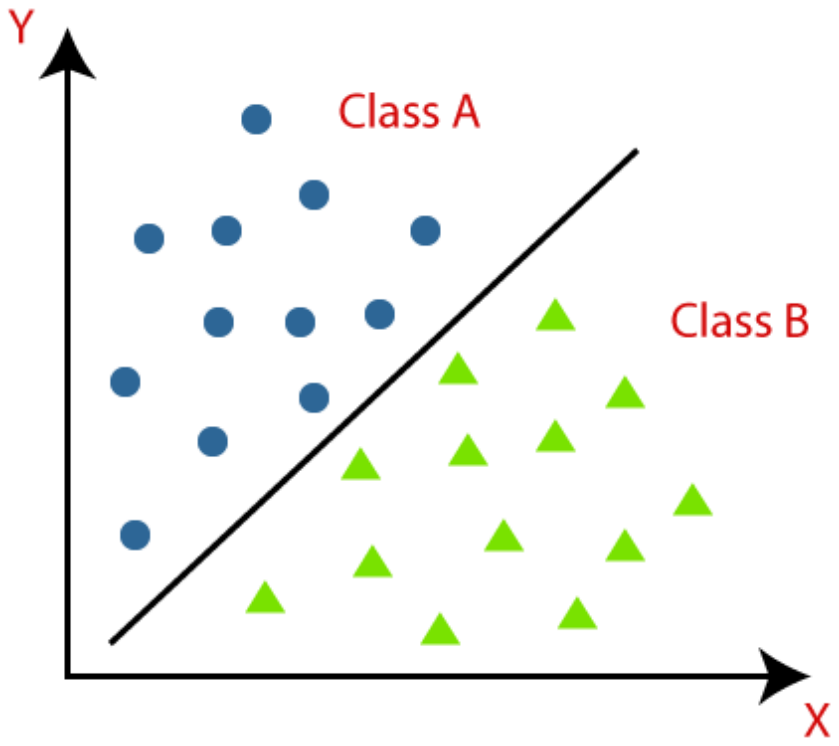


Naïve Bayes

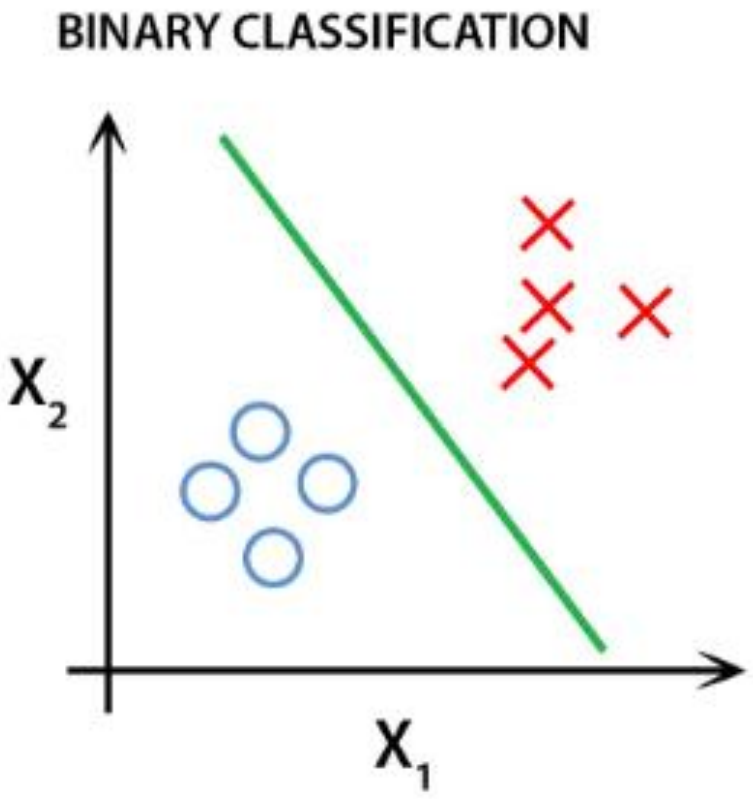


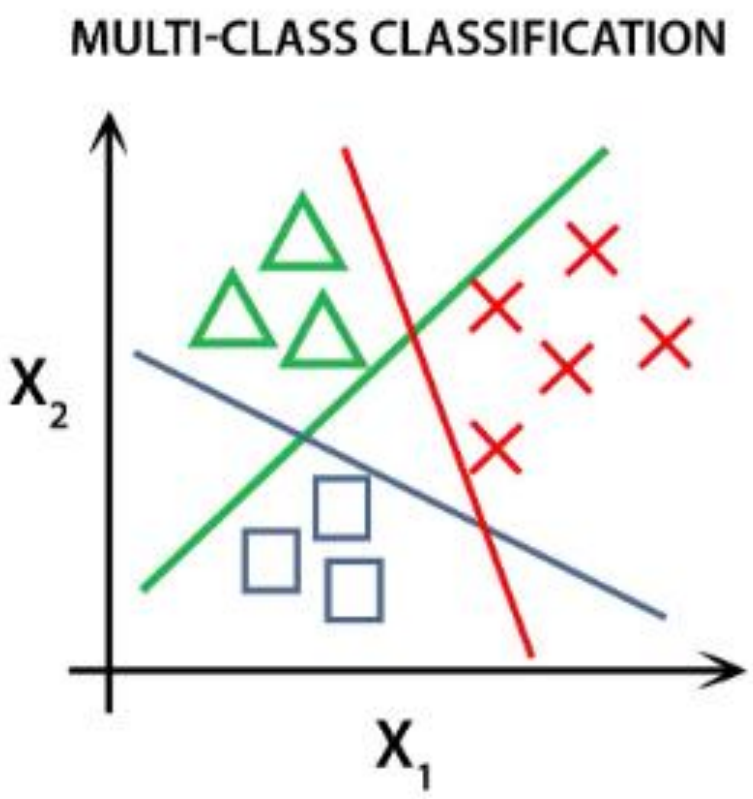
Classification

- Classification is a type of machine learning problem which aims to predict a class label for a given input data point.
- It is one of the most widely used techniques in machine learning and is applied in various domains like image classification, spam detection, sentiment analysis and many more.



Types of Classification





Classification Examples

Email Classification

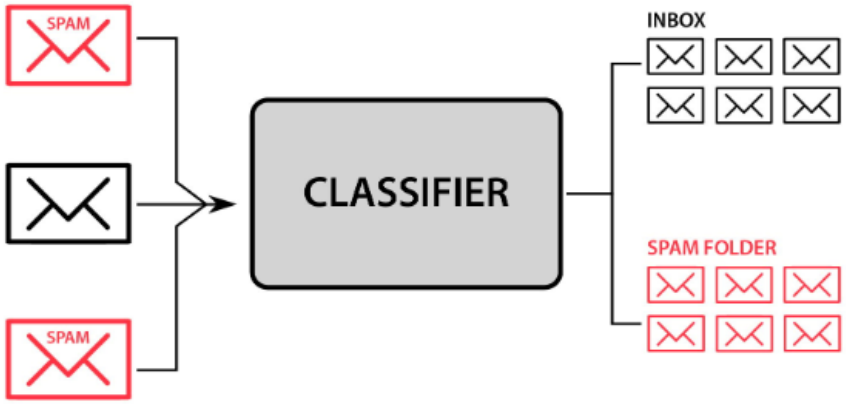
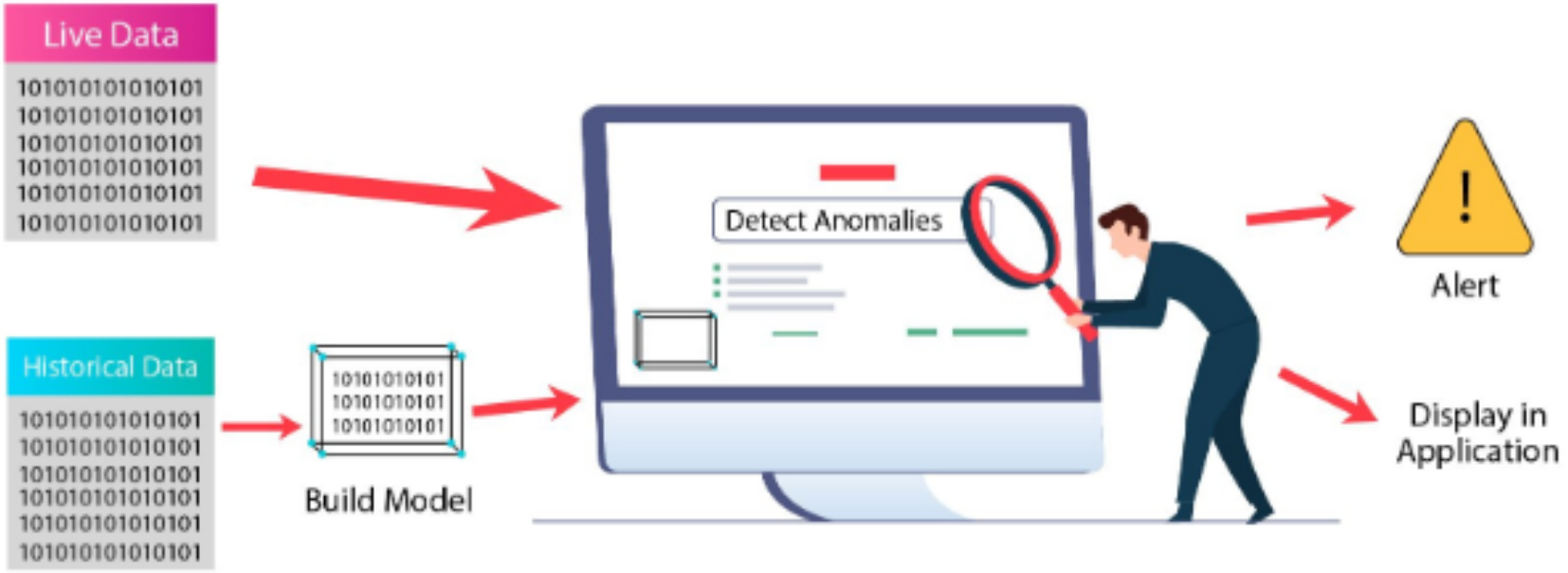


Image Classification



Anomaly Detection

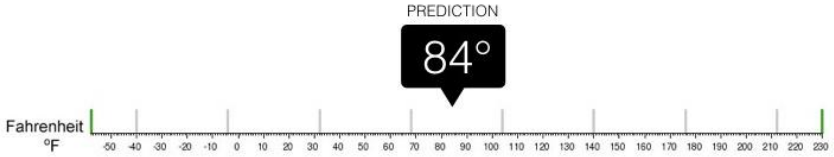


Classification Vs Regression



Regression

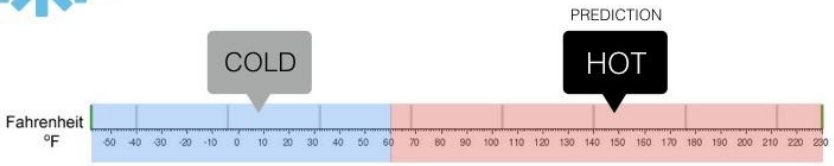
What is the temperature going to be tomorrow?

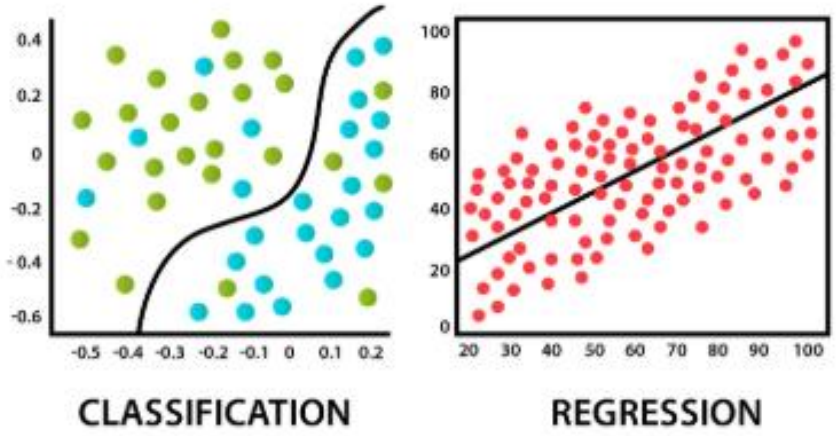




Classification

Will it be Cold or Hot tomorrow?





Linear Regression Analysis

What is Linear Regression?

Linear regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables.

Formula : $\hat{y} = \theta_0 + \theta_1 x$

$\hat{y}$: The predicted value of the dependent variable (output).

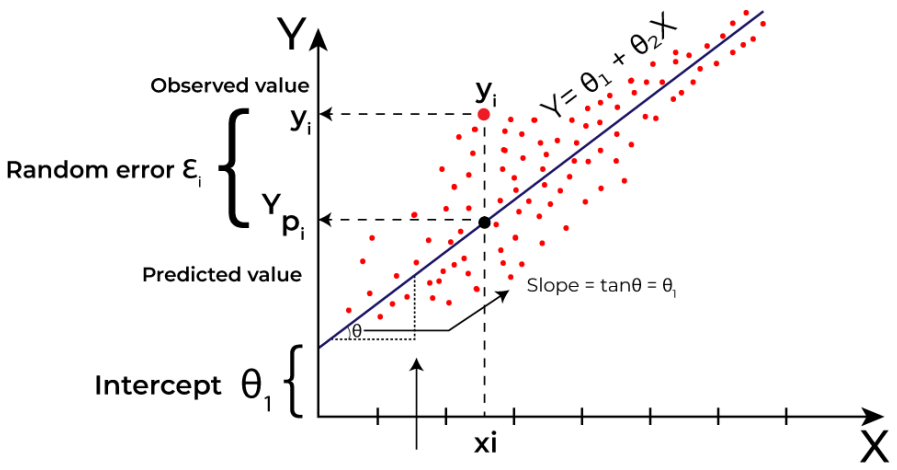
X : Input variable

Example:

Predicting a person's salary based on years of experience.



Linear Regression Analysis



How Linear Regression Works

$$\min \frac{1}{n}\sum_{i=1}^{n} (\bar{Y}_i - Y_i)^2$$

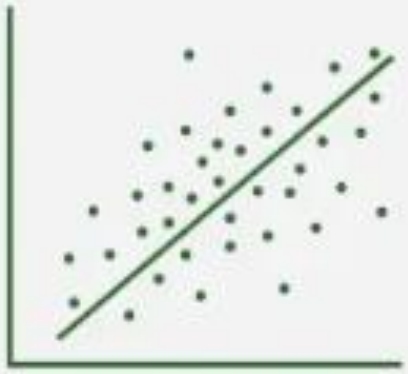
Where:

- $\bar{Y}_i$: Predicted value for the ith observation
- Y_i: Actual observed value for the ith observation
- **n:** Total number of observations
- **Min:** Indicates the goal is to minimize the total error

The model adjusts the parameters 00, 01,..., θη to minimize this error

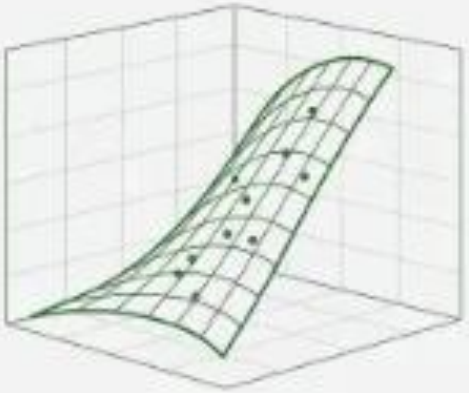
Types of Linear Regression

Simple Linear Regression



Predicts the dependent variable using a single independent variable.

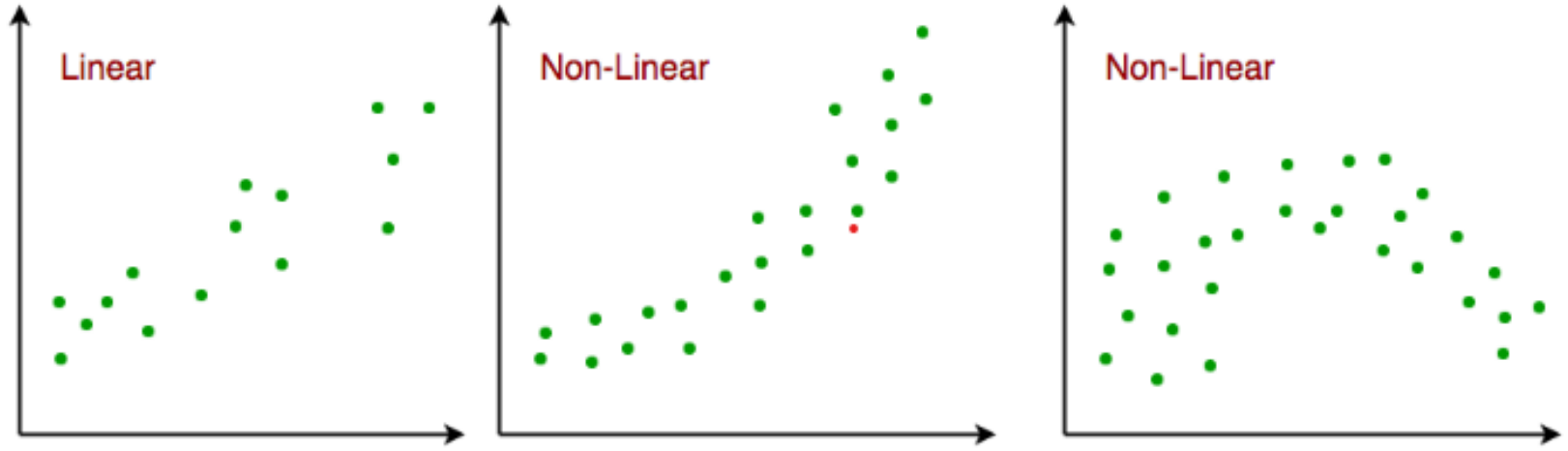
Multiple Linear Regression



Uses two or more independent variables to predict the dependent variable.

Assumptions of the Linear Regression

1. Linearity: The relationship between inputs (X) and the output (Y) is a straight line.

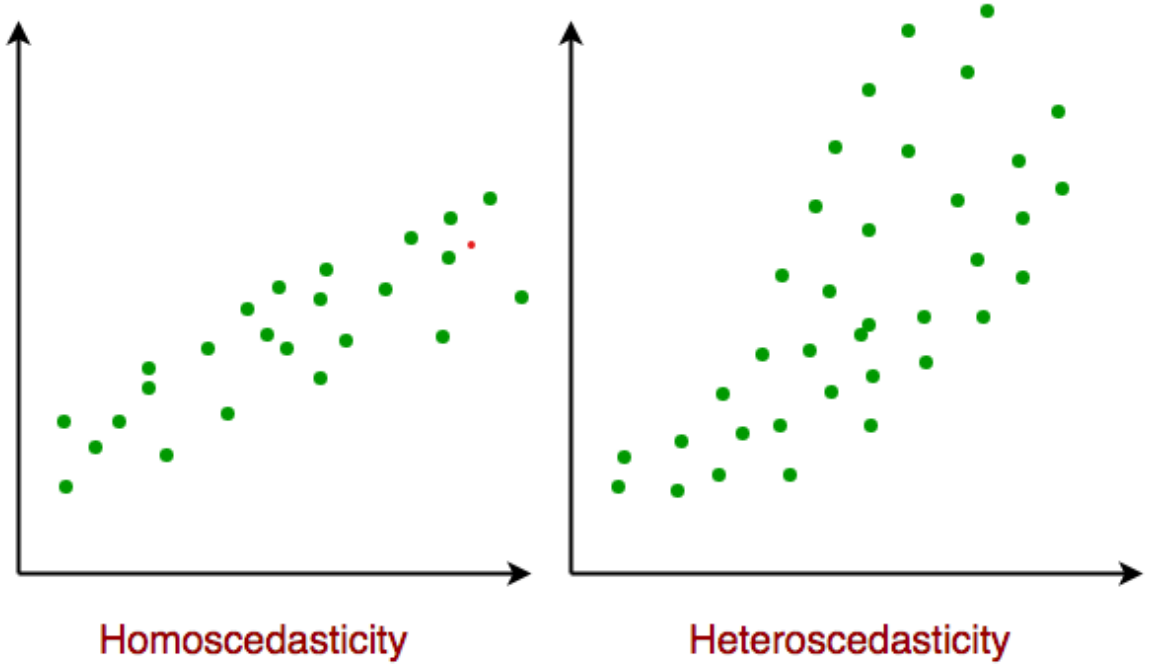


Linearity

2. Independence of Errors: The errors in predictions should not affect each other.

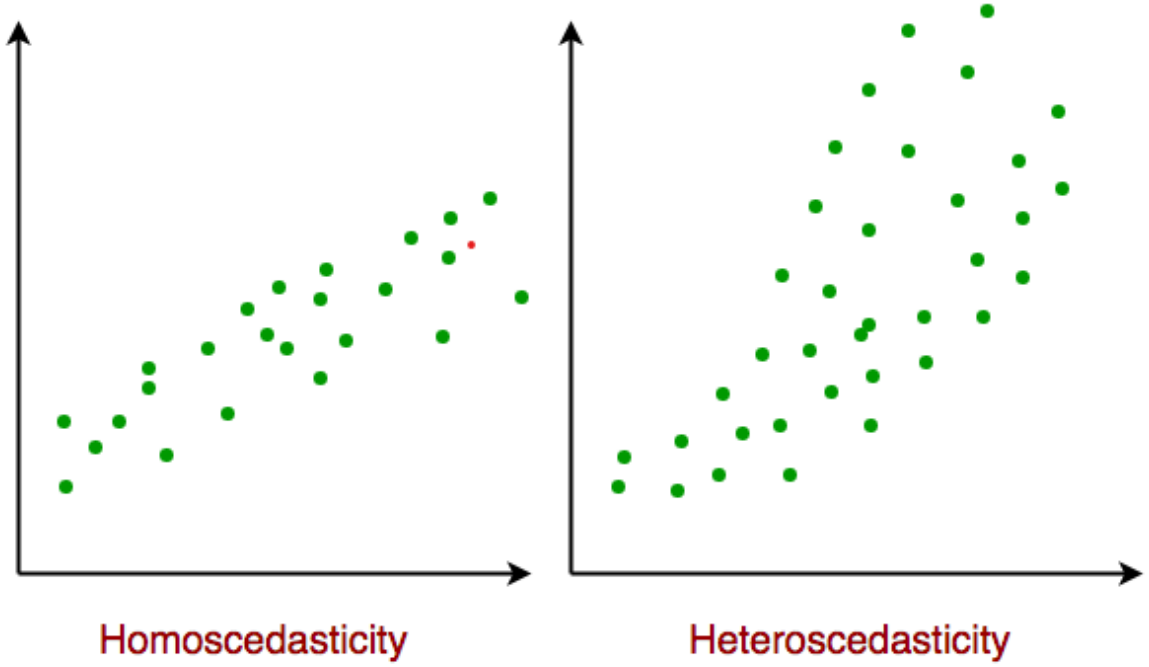
Assumptions of the Linear Regression

3. Constant Variance (Homoscedasticity): The errors should have equal spread across all values of the input. If the spread changes (like fans out or shrinks), it's called heteroscedasticity and it's a problem for the model.



Assumptions of the Linear Regression

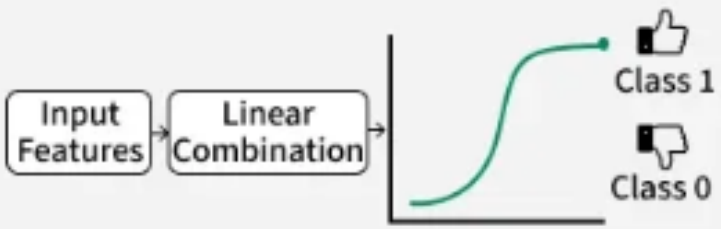
3. Constant Variance (Homoscedasticity): The errors should have equal spread across all values of the input. If the spread changes (like fans out or shrinks), it's called heteroscedasticity and it's a problem for the model.



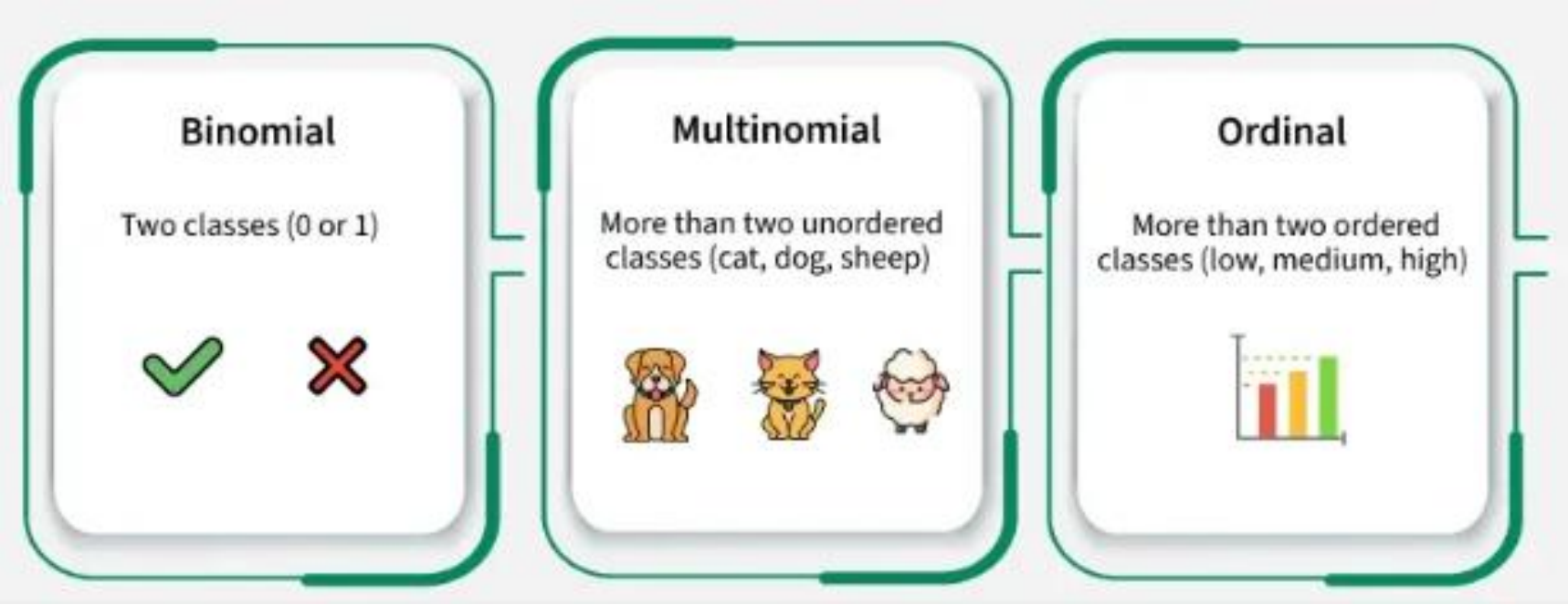
Logistic Regression

What is Logistic Regression?

- Predicts the probability of a binary outcome (Yes/No, 0/1)
- Uses the sigmoid function to map inputs to probabilities (0 to 1)
- Ideal for classification tasks



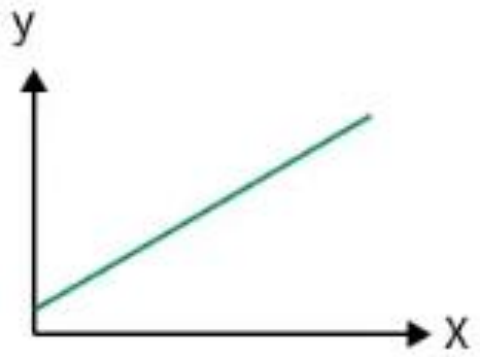
Types of Logistic Regression



Linear Vs Logistic Regression

Linear Regression

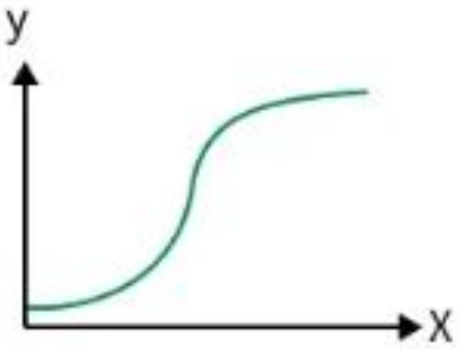
- Predicts continuous values
- Uses best-fit line
- Solves regression problems



vs

Logistic Regression

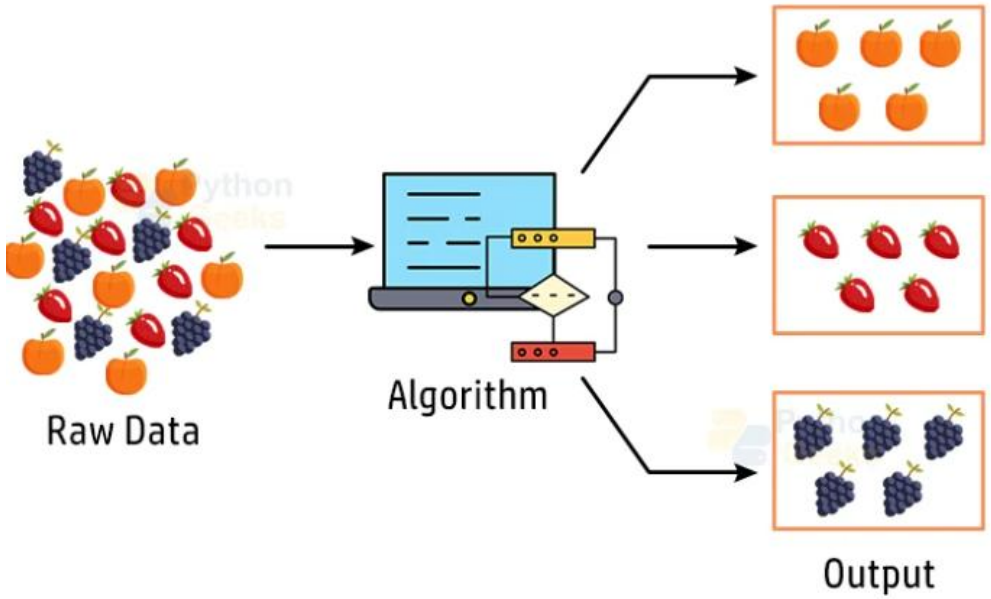
- Predicts categorical classes
- Uses sigmoid S-curve
- Solves classification problems



Clustering

The clustering technique can be widely used in various tasks. Some most common uses of this technique are:

- Market Segmentation
- Statistical data analysis
- Social network analysis
- Image segmentation
- Anomaly detection



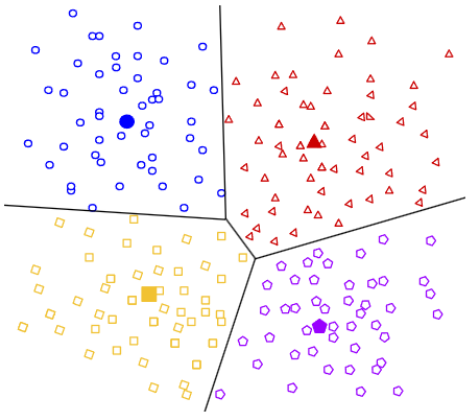


Figure 1: Example of centroid-based clustering.

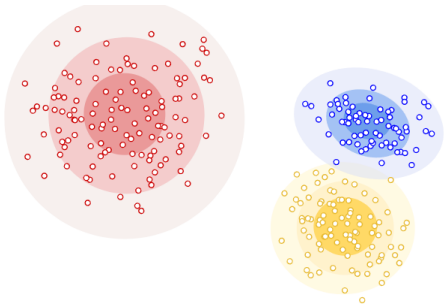


Figure 3: Example of distribution-based clustering.

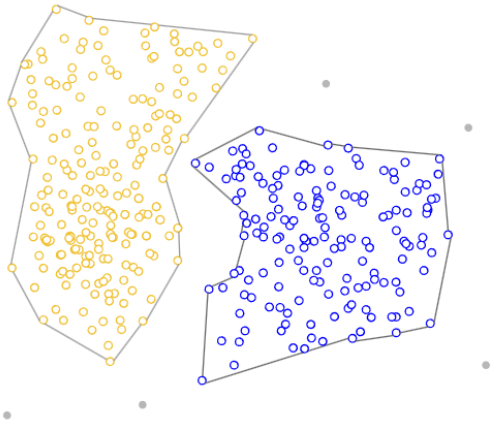
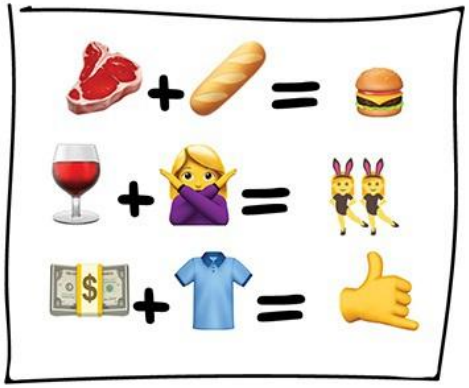


Figure 2: Example of density-based clustering.

Types of Clustering



$$Rule:\ X \Rightarrow Y$$

$$Support = \frac{frq(X,Y)}{N}$$

$$Confidence = \frac{frq(X,Y)}{frq(X)}$$

$$Lift = \frac{Support}{Supp(X) \times Supp(Y)}$$

Association Rules

Clustering VS Association Rules

- Clustering techniques calculate clusters based on similarities whereas Association rule mining finds associations based on co-occurrences.

Let's look at an example to understand this further. Consider the transaction data below.

	Milk	Bread	Butter	Cheese
Customer_1	✅	✅		
Customer_2		✅		✅
Customer_3	✅	✅	✅	
Customer_4		✅		✅

Transaction data

Here the clustering algorithm will make two groups from the samples in the data, i.e., the rows.

- **Cluster_1**: Customer_1 & Customer_3
- **Cluster_2**: Customer_2 & Customer_4

	Milk	Bread	Butter	Cheese
Customer_1	✅	✅		
Customer_2		✅		✅
Customer_3	✅	✅	✅	
Customer_4		✅		✅

Clustering results

On the other hand, association rule mining will suggest associations between products (variables) that are more likely to be bought together in the future.

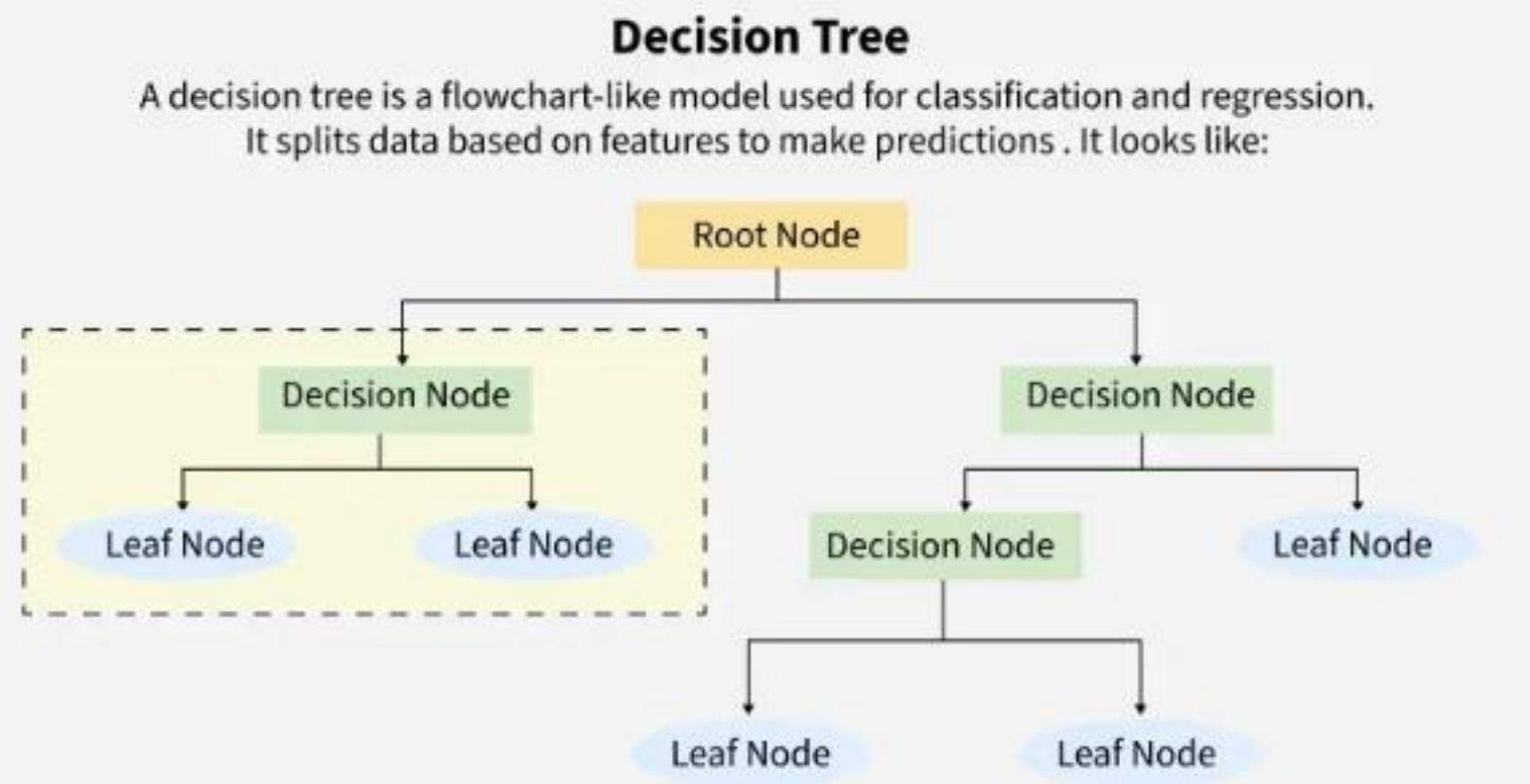
- **Association_1**: Milk & Bread
- **Association_2**: Bread & Cheese

	Milk	Bread	Butter	Cheese
Customer_1	✅	✅		
Customer_2		✅		✅
Customer_3	✅	✅	✅	
Customer_4		✅		✅

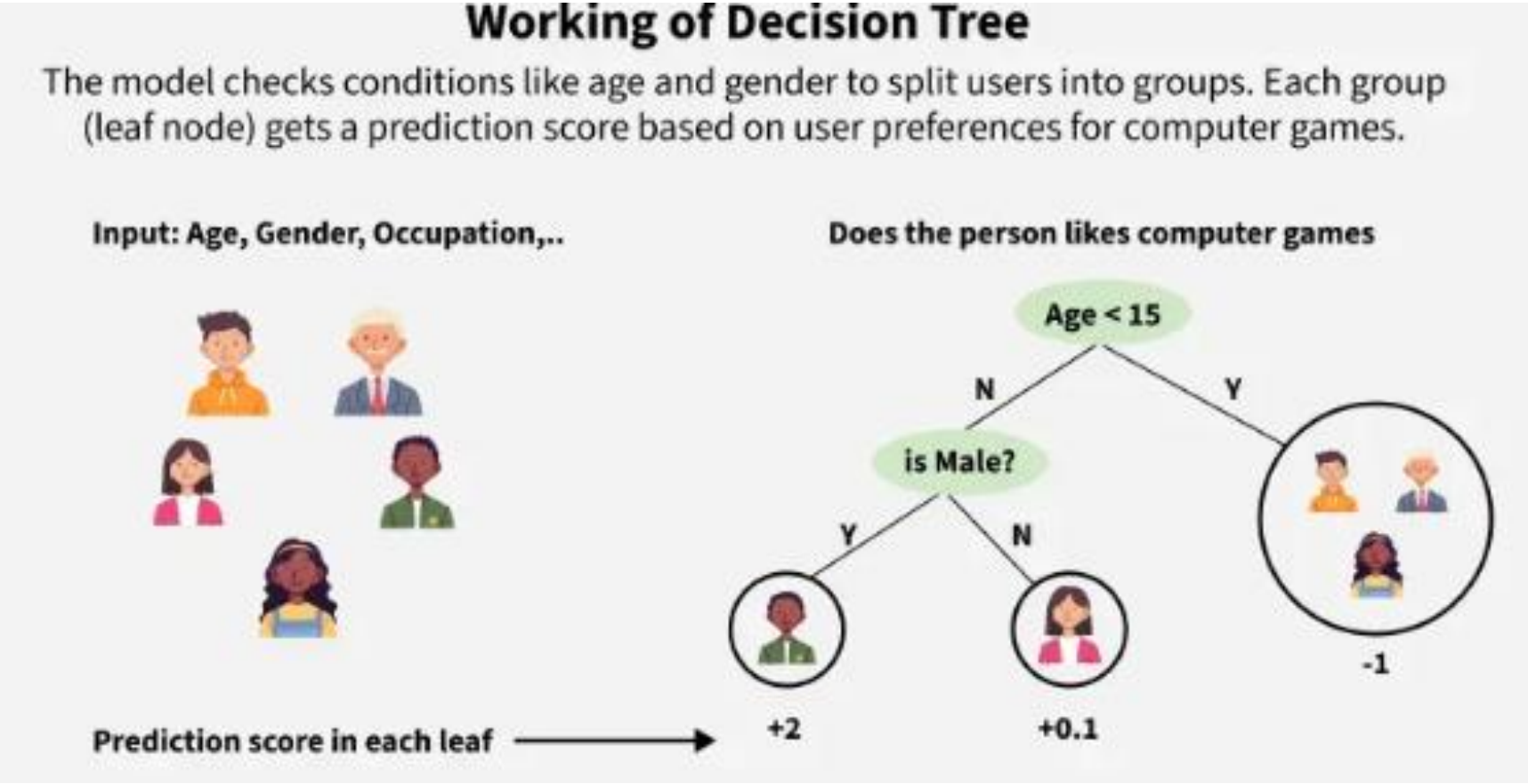
Association Rule Mining results

Clustering is done on customers and association rule mining is performed on the products.

Decision Tree

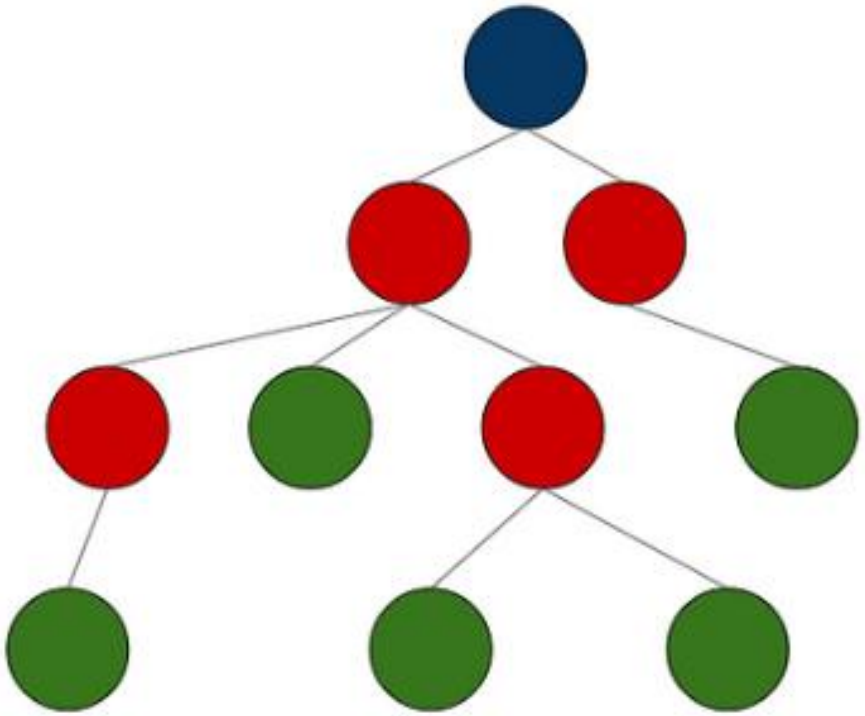


Decision Tree



Decision Tree

Terminologies associated with decision tree





Root Node



Internal Node



Leaf Node

Parent Node

Child Node

Decision node

Branch/Subtree

Splitting

Pruning

Decision Tree

Splitting Criteria In Decision Tree

In decision trees, splitting criteria help decide which feature to split on at each node. The two most common criteria are:

Gini Index

$$I_G = 1 - \sum_{j=1}^{c} p_j^2$$

P_j : proportion of the samples that belongs to class c for a particular node

Entropy

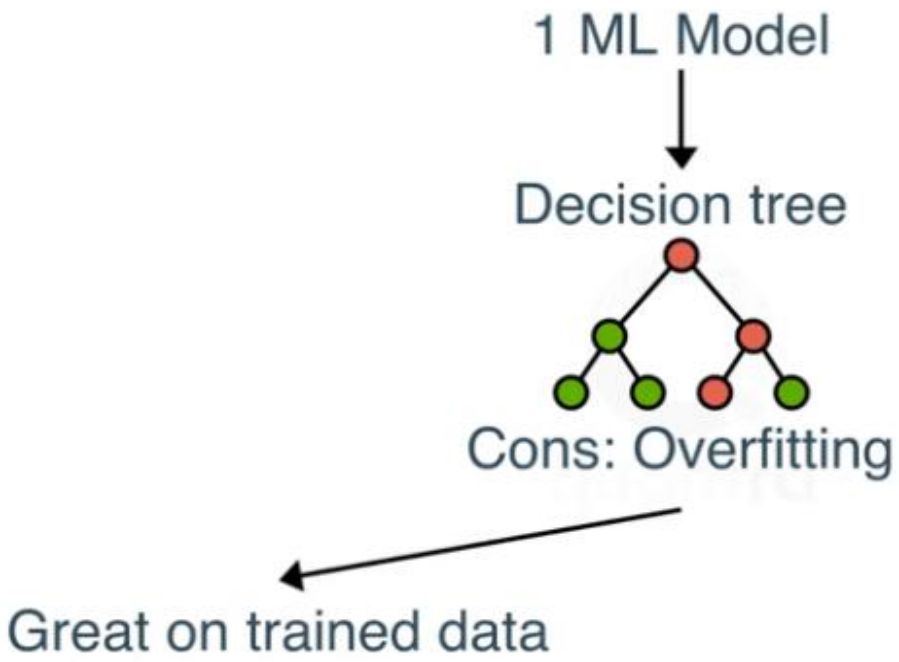
$$I_H = -\sum_{j=1}^{c} p_j log_2(p_j)$$

p_j: proportion of the samples that belongs to class c for a particular node.

*This is the the definition of entropy for all non-empty classes (p ne0) The entropy is 0 if all samples at a node belong to the same class.

Comparison to Decision Tree for Random Forest

Decision trees, characterized by their hierarchical structure, use nodes and branches to guide decision-making based on parameter responses.



In both machine learning, overfitting occurs when the model fits the data too well. Overfitting model learns the detail and noise in the training data to such an extent that it negatively impacts the performance of the model on new data/test data.

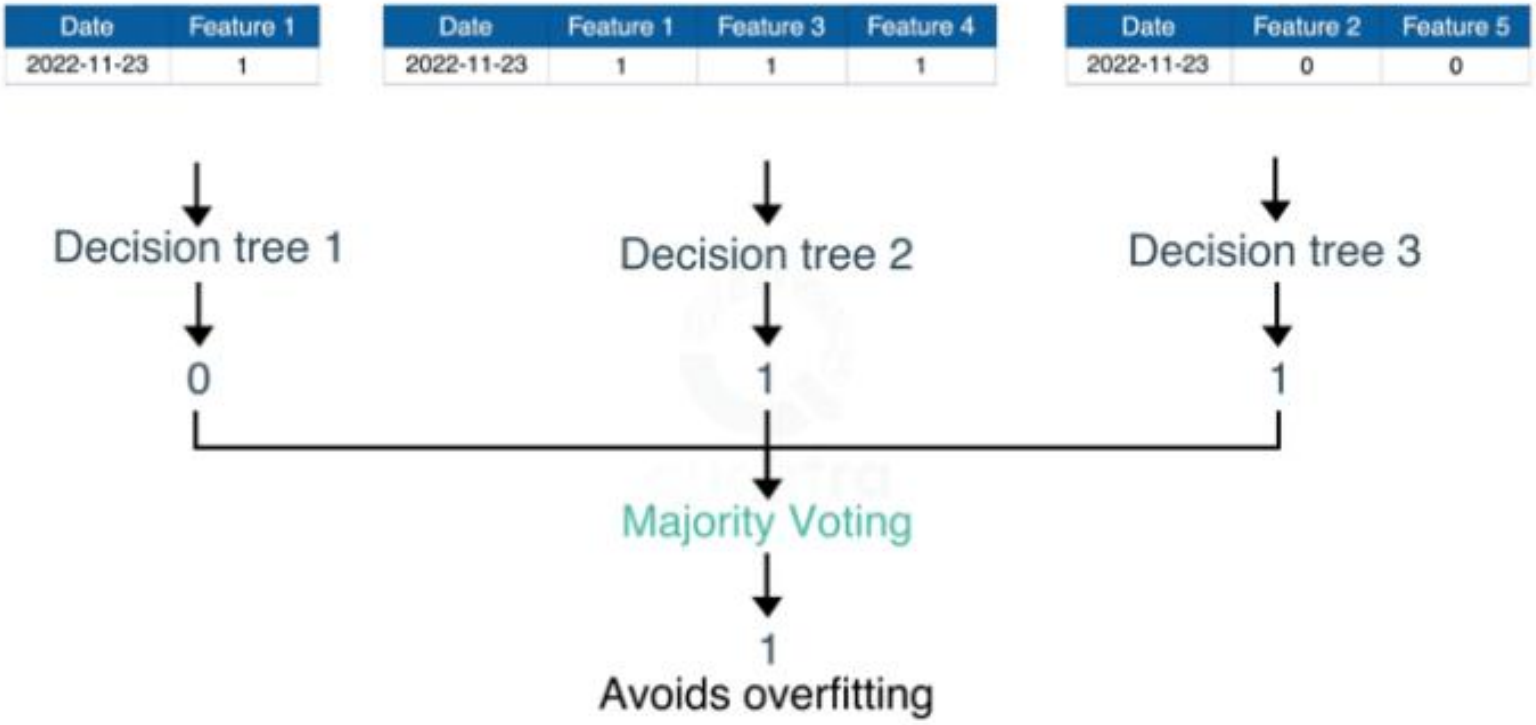
Random Forest

- Random forest algorithm in machine learning is a supervised classification algorithm that addresses the issue of overfitting in decision trees through an ensemble approach. It consists of multiple decision trees constructed randomly by selecting features from the dataset.

- The final prediction of the random forest is determined by aggregating the outcomes from the decision trees, with the most frequent prediction.

- The outcome which is arrived at, for a maximum number of times through the numerous decision trees is considered as the final outcome by the random forest.

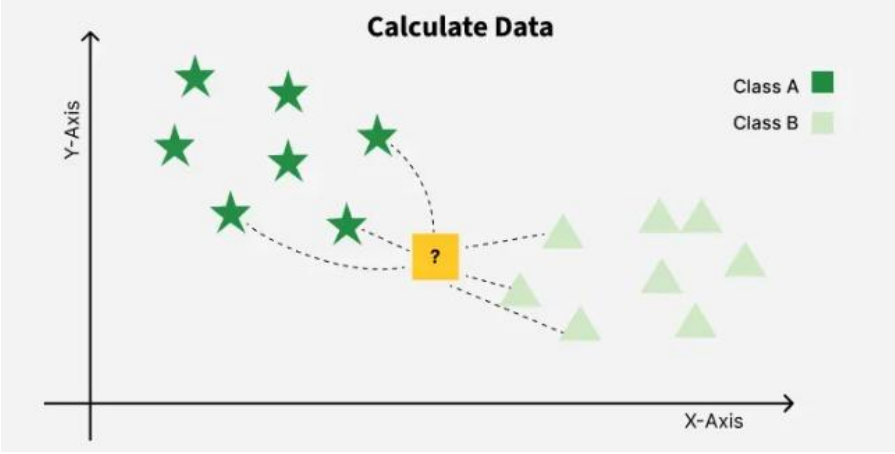
in the diagram below, we can see that there are two trading signals:
1 - is the buy signal
0 - is the sell signal

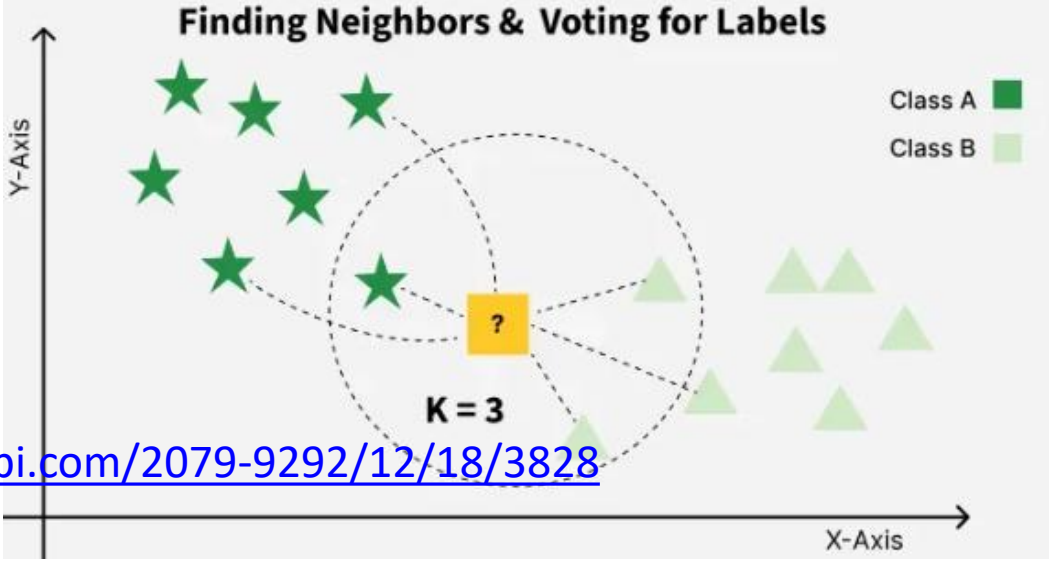


https://blog.quantinsti.com/random-forest-algorithm-in-python

K-Nearest Neighbor (KNN) Algorithm



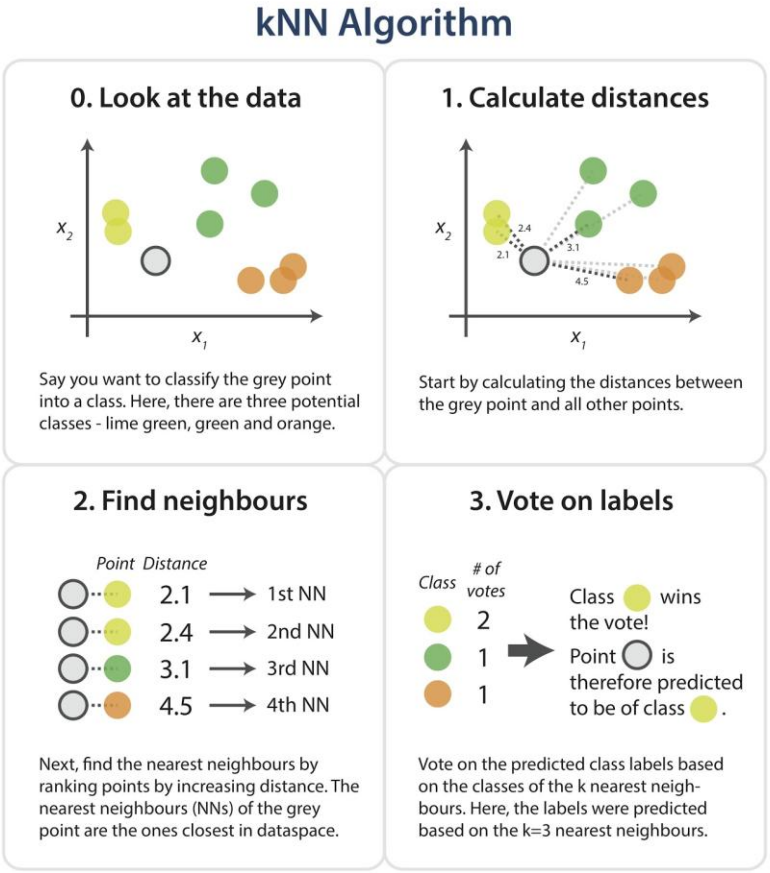




https://www.mdpi.com/2079-9292/12/18/3828

K-Nearest Neighbor (KNN) Algorithm

kNN, or the k-nearest neighbor algorithm, is a machine learning algorithm that uses proximity to compare one data point with a set of data it was trained on and has memorized to make predictions.

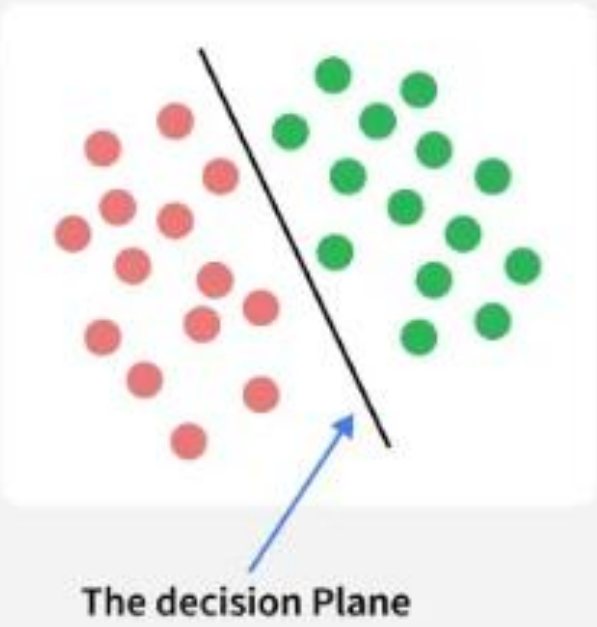


Support Vector Machine (SVM)

SVM (Support Vector Machine) is a supervised machine learning algorithm
used for classification and regression tasks.

SVM is based on the concept of decision planes that define decision boundaries.

A decision plane is one that separates between a set of objects having different class memberships.



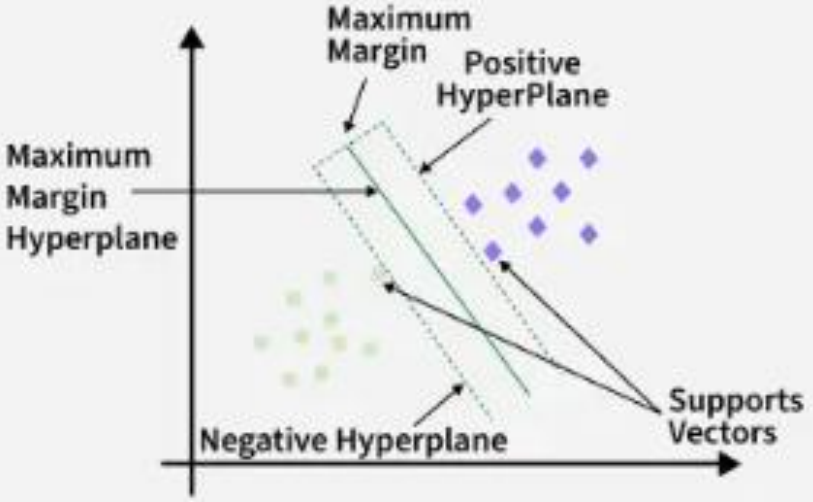
Support Vector Machine (SVM)

Hyperplane:

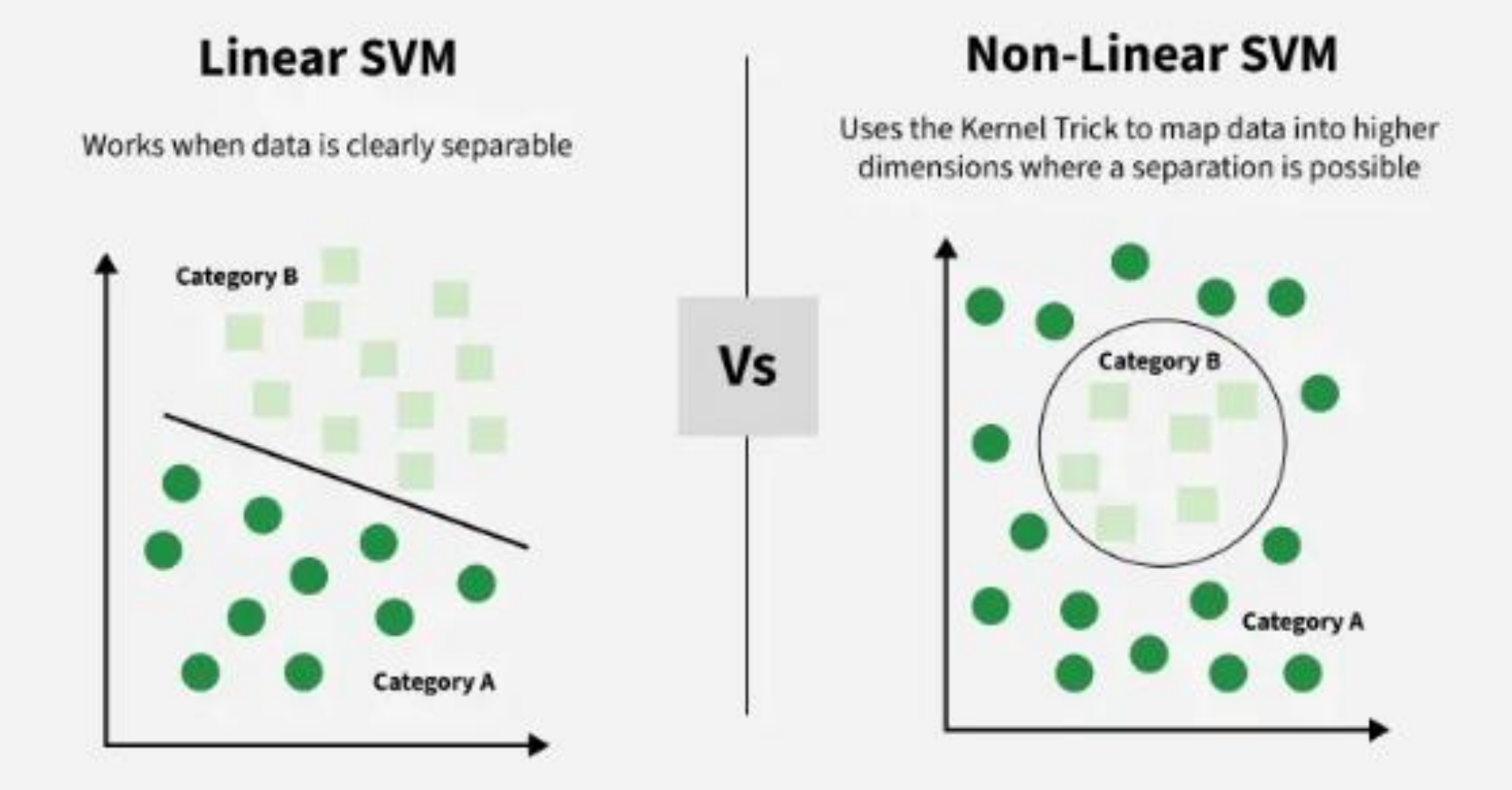
A line or plane that separates classes

Support Vectors:

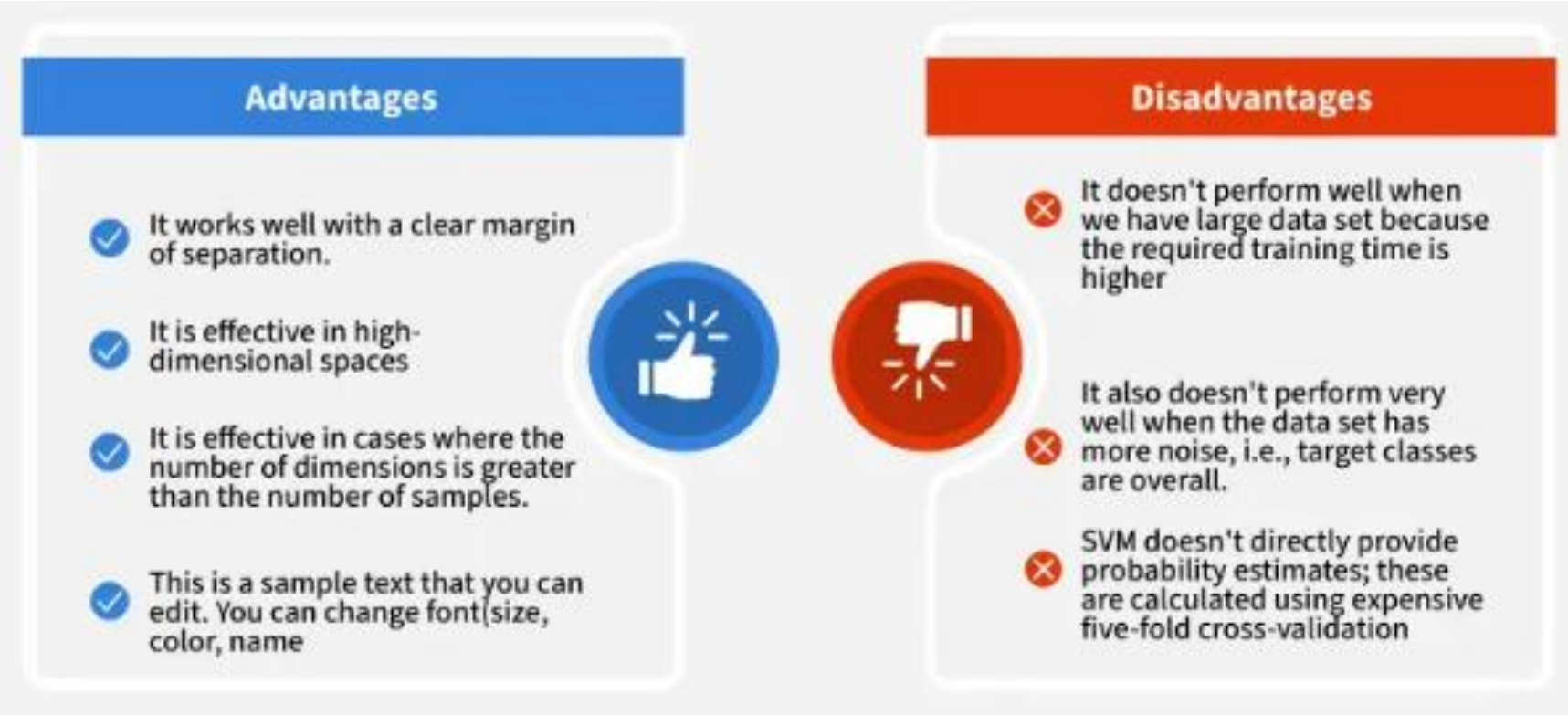
The closest data points to the hyperplane. These points define the boundary and help maximize the margin between classes.



Support Vector Machine (SVM)

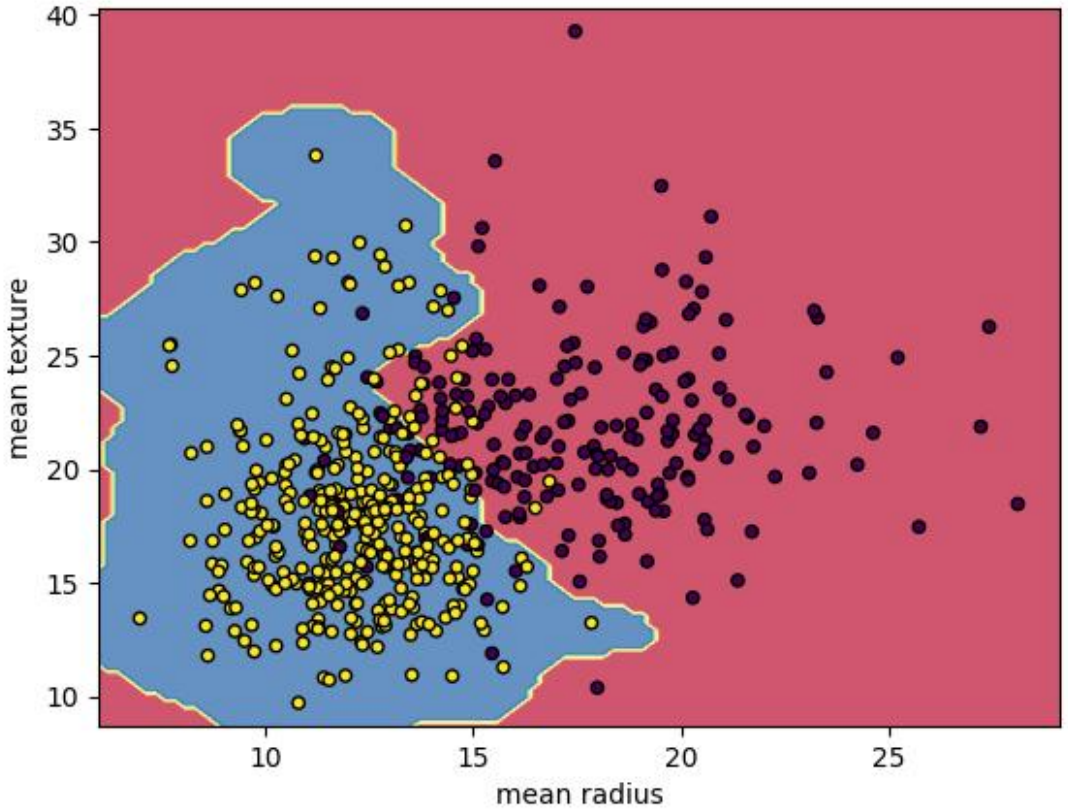


Support Vector Machine (SVM)



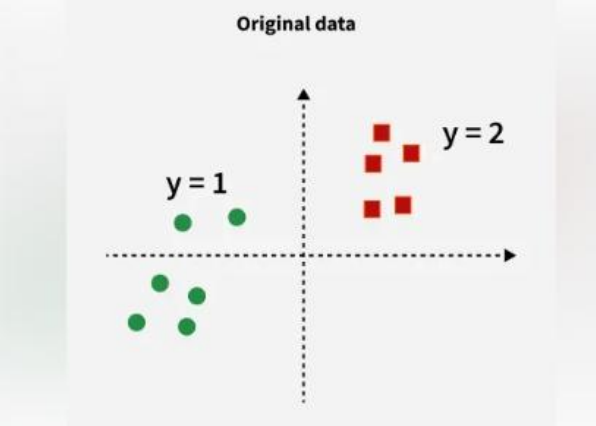
Example(SVM)

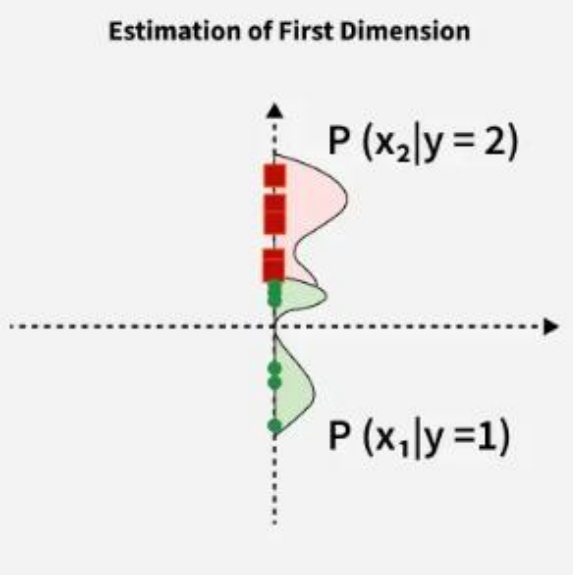
- Predict if cancer is Benign or malignant. Using historical data about patients diagnosed with cancer enables doctors to differentiate malignant cases and benign ones are given independent attributes.
- The **colored background** (red and blue regions) shows how the SVM model **divides the space** based on the first two features (mean radius and mean texture of the tumor).
- The **dots** represent actual patient data:
    - **Red dots**: Patients with **malignant (cancerous)** tumors.
    - **Blue dots**: Patients with **benign (non-cancerous)** tumors.
- The **black edges** around the dots help distinguish individual data points.

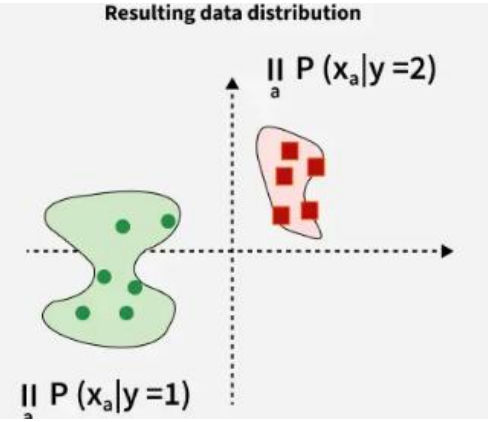


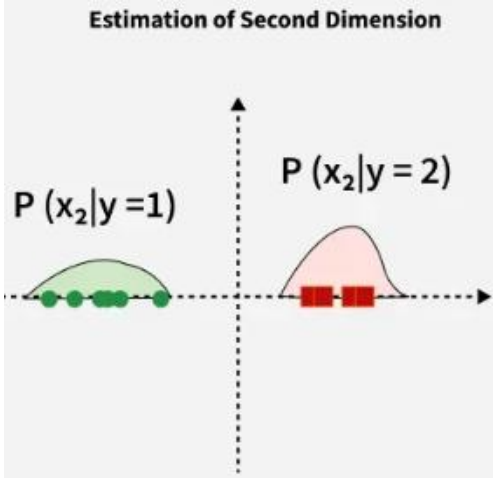
Navie Bayes Classifers

- Naive Bayes is a classification algorithm that uses probability to predict which category a data point belongs to, assuming that all features are unrelated.









Example: Weather Dataset

Let's take a dataset used for predicting if golf is played based on:
Outlook: Sunny, Rainy, Overcast
Temperature: Hot, Mild, Cool
Humidity: High, Normal
Wind: True, False.

X=(Sunny, Hot, Normal, False)
Goal: Predict if golf will be played (Yes or No).

Outlook

	Yes	No	P(yes)	P(no)
Sunny	3	2	3/10	2/4
Overcast	4	0	4/10	0/4
Rainy	3	2	3/10	2/4
Total	10	4	100%	100%

Temperature

	Yes	No	P(yes)	P(no)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
Total	9	5	100%	100%

Humidity

	Yes	No	P(yes)	P(no)
High	3	4	3/9	4/5
Normal	6	1	6/9	1/5
Total	9	5	100%	100%

Wind

	Yes	No	P(yes)	P(no)
False	6	2	6/9	2/5
True	3	3	3/9	3/5
Total	9	5	100%	100%

Play		P(Yes)/P(No)
Yes	9	9/14
No	5	5/14
Total	14	100%

6. Calculate Posterior Probabilities

For Class = Yes:

$$P(\text{Yes} \mid \text{today}) \propto \frac{2}{9} \cdot \frac{2}{9} \cdot \frac{6}{9} \cdot \frac{6}{9} \cdot \frac{9}{14}$$

$$P(\text{Yes} \mid \text{today}) \approx 0.02116$$

For Class = No:

$$P(\text{No} \mid \text{today}) \propto \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} \cdot \frac{2}{5} \cdot \frac{5}{14}$$

$$P(\text{No} \mid \text{today}) \approx 0.0068$$

7. Normalize Probabilities

To compare:

$$P(\text{Yes} \mid \text{today}) = \frac{0.02116}{0.02116 + 0.0068} \approx 0.756$$

$$P(\text{No} \mid \text{today}) = \frac{0.0068}{0.02116 + 0.0068} \approx 0.244$$

Class Probabilities:

From dataset of 14 rows:

- $P(\text{Yes}) = \frac{9}{14}$
- $P(\text{No}) = \frac{5}{14}$

Conditional Probabilities (Tables 1–4):

Feature	Value	P (Value | Yes)	P (Value | No)
Outlook	Sunny	2/9	3/5
Temperature	Hot	2/9	2/5
Humidity	Normal	6/9	1/5
Wind	False	6/9	2/5

8. Final Prediction

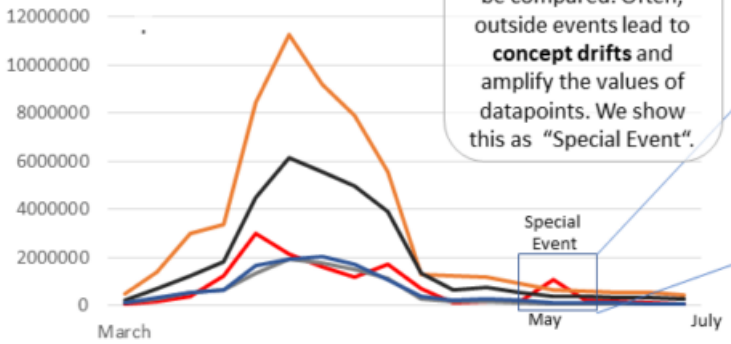
Since:

$$P(\text{Yes} \mid \text{today}) > P(\text{No} \mid \text{today})$$

The model predicts: ***Yes (Play Golf)***

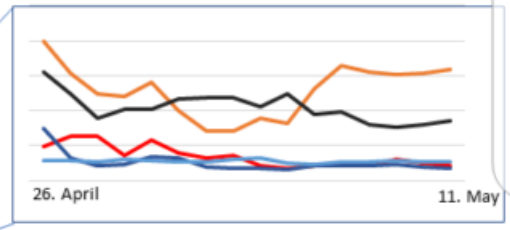
Time Series Analysis

Example:
Tweets with different Hashtags



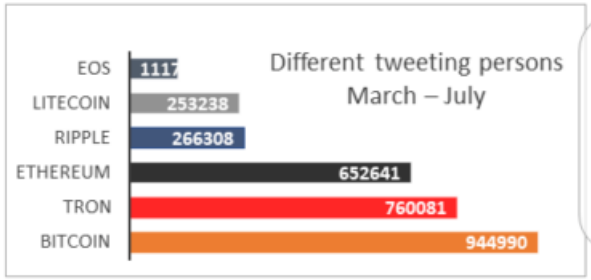
Typcially, periods of data transmissions can be compared. Often, outside events lead to **concept drifts** and amplify the values of datapoints. We show this as "Special Event".

Drill down and indicator change



Regular for Time Series Data is to have **coarse- and fine-grained viewpoints** while allowing different **time changing indicators** to be looked at.

Roll up and aggregate



Time Series Databases often support to **store aggregated** data for certain **time periods**.



The nature of

Time Series Data

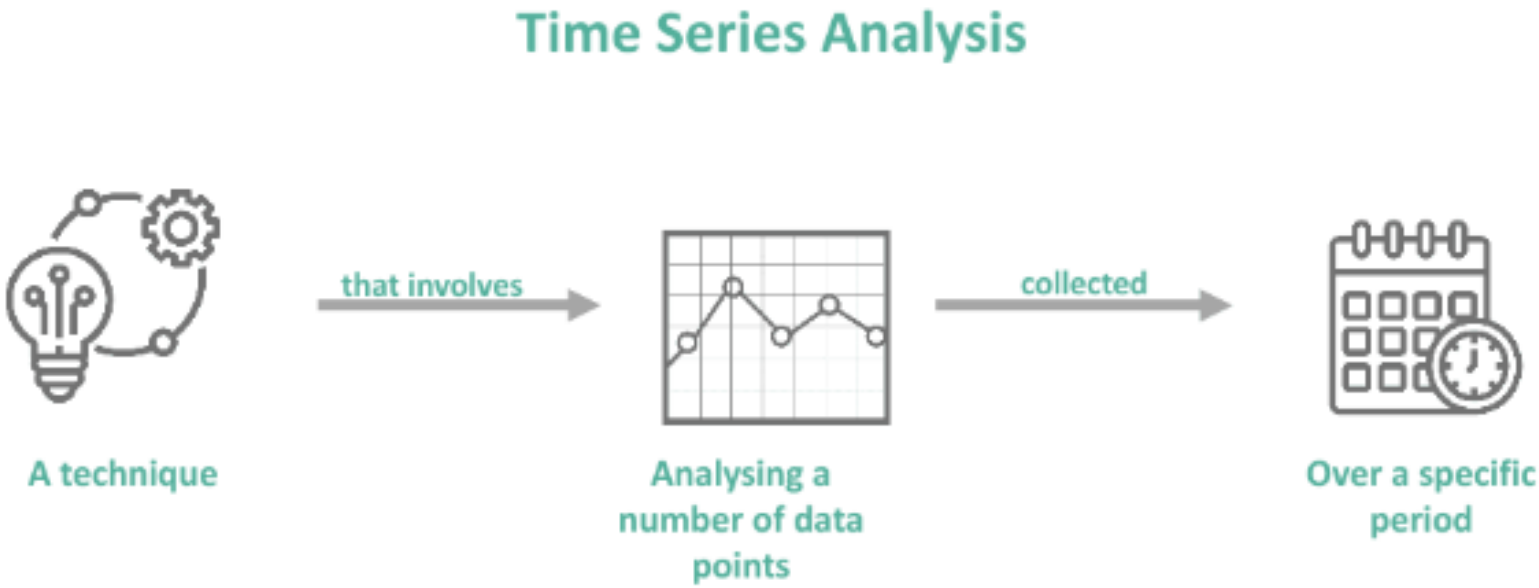
Data which arrives in a subsequent manner. Each new record is normally regarded as new dataset.

iunera

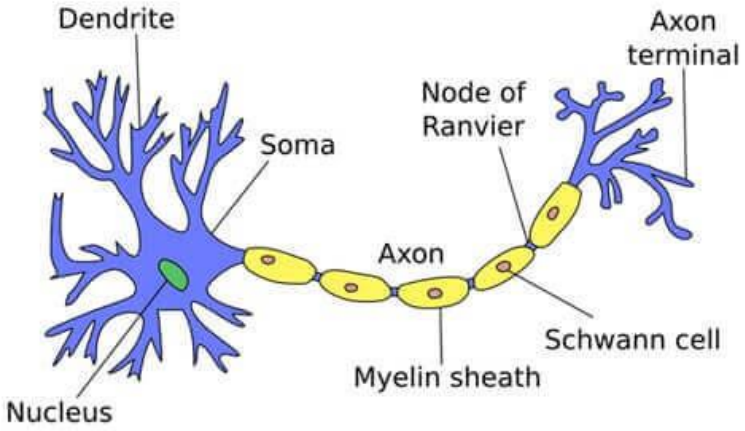
Time Series Analysis

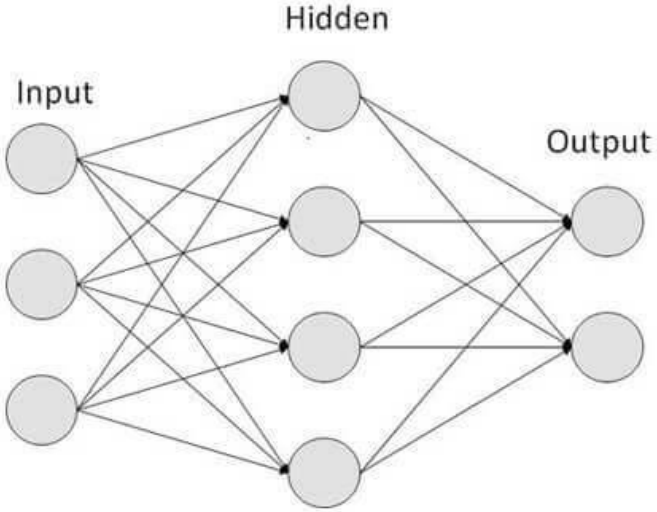
Key concepts in time series analysis include:

- Trend: The long-term movement or direction of a stock's price over time. Trends can be upward (bullish), downward (bearish), or sideways (range-bound).
- Seasonality: Regular patterns or cycles that repeat at fixed intervals within the data, such as daily, weekly, monthly, or yearly.
- Autocorrelation: The correlation between a data point and its lagged values. Autocorrelation allows us to understand how past values influence future values.
- Stationarity: A desirable property of time series data where statistical properties, such as mean and variance, remain constant over time. Stationary data simplifies the analysis and enhances prediction accuracy.



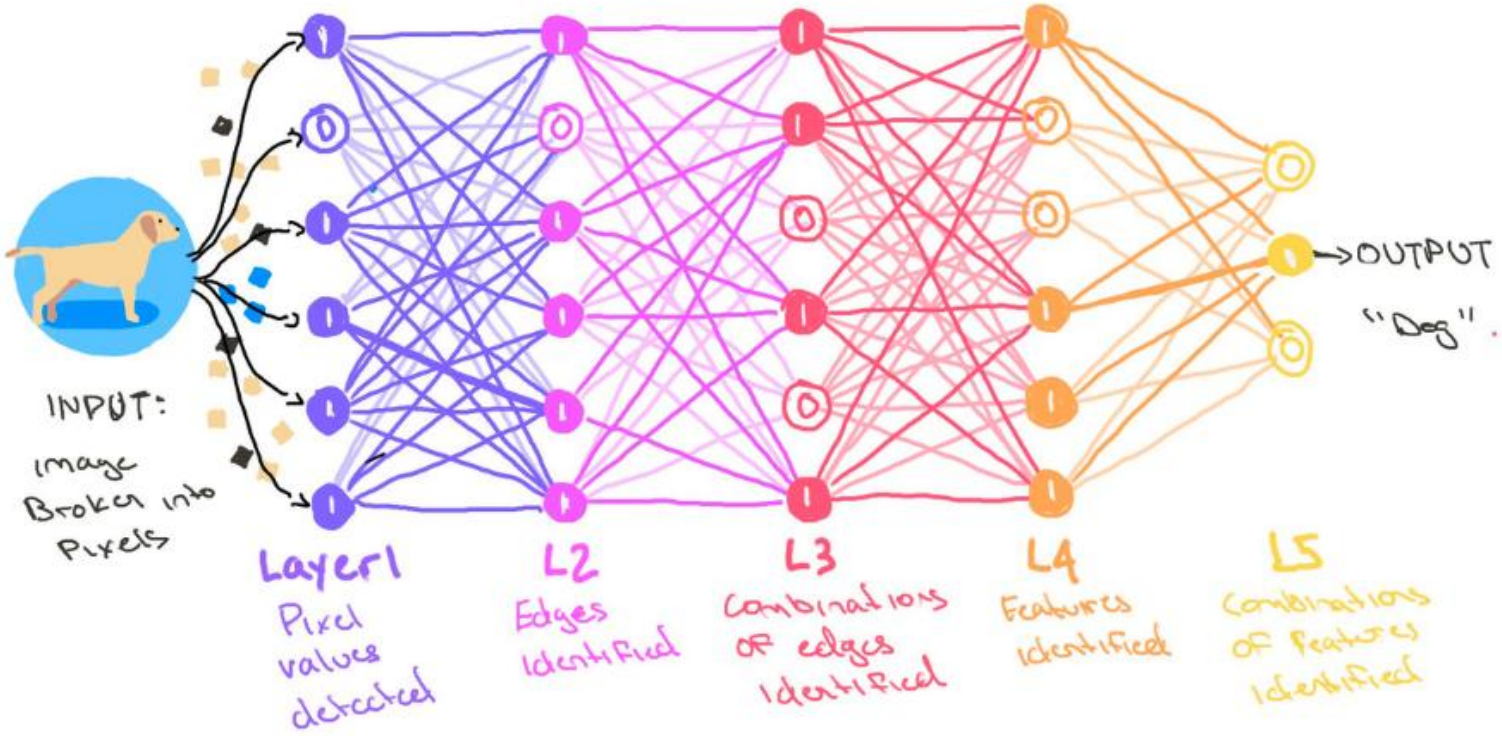
Neural Networks





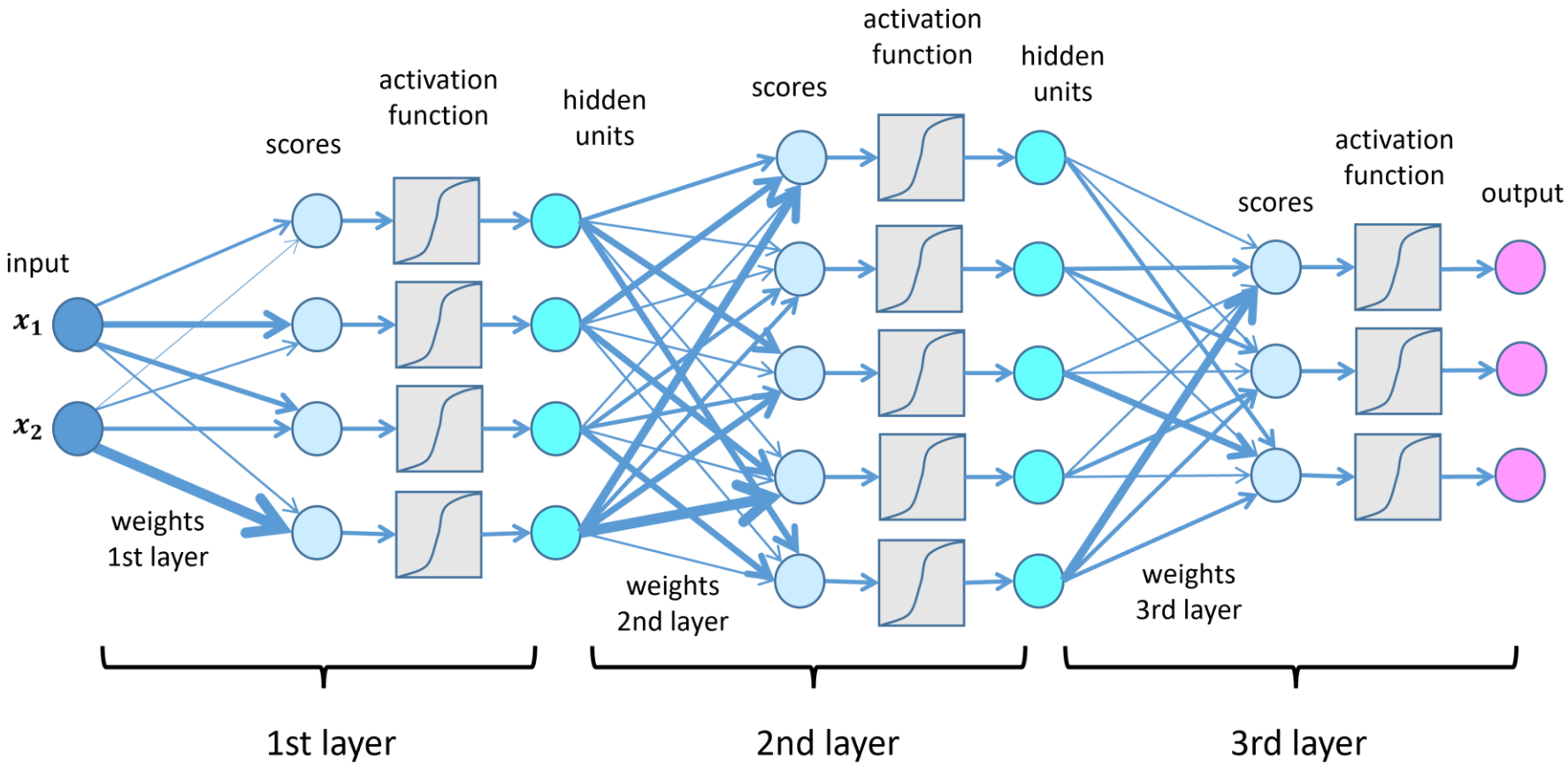
ON the left: The biological neuron graph & on the right: the artificial neural network

Neural Networks



https://www.notablecap.com/blog/the-anatomy-of-a-neural-network

Neural Networks



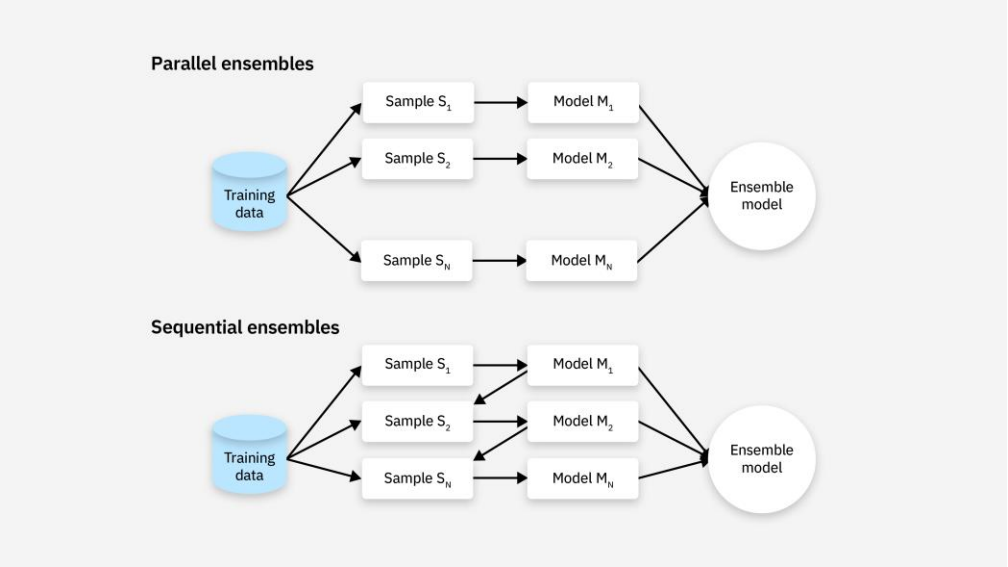
https://lamarr-institute.org/blog/deep-neural-networks/

Ensemble Methods

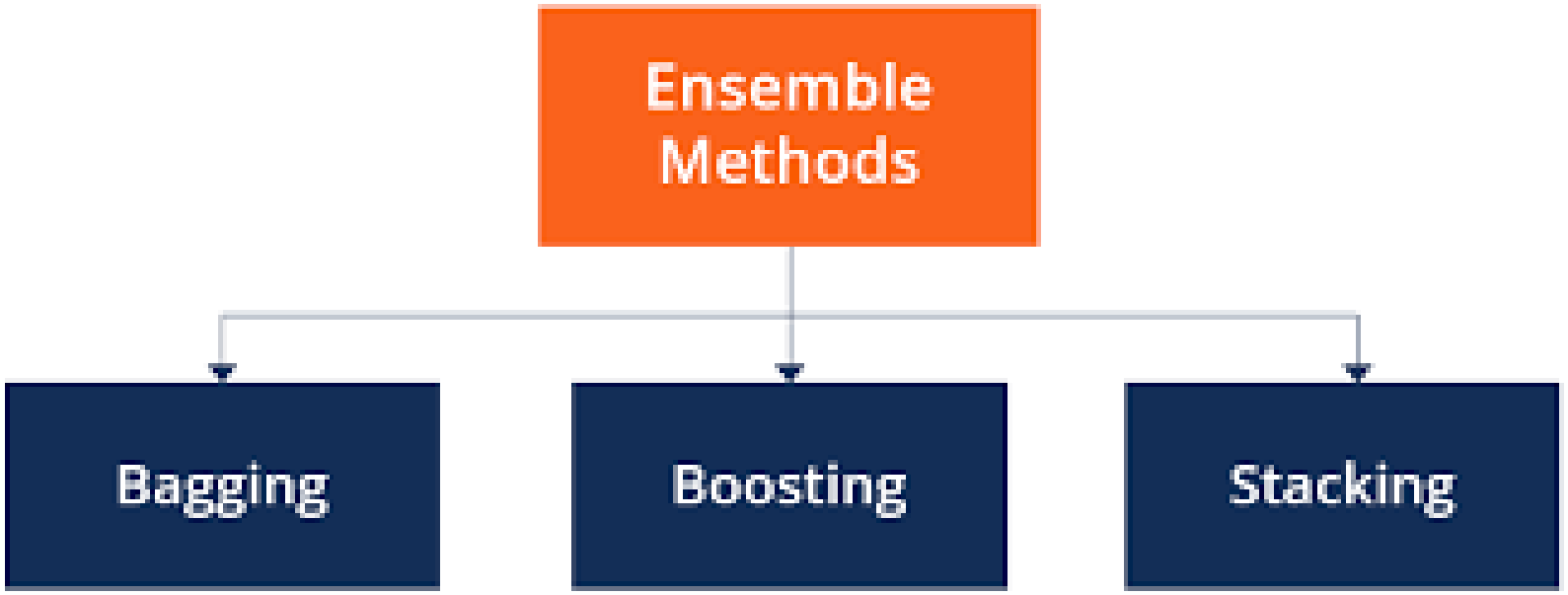
Ensemble methods work by aggregating predictions from several base models (often called **weak learners**) to produce a **stronger overall model**. The idea is that while individual models may make errors, combining them can reduce variance, bias, or improve predictions.

Types of ensemble models

- **Parallel** methods train each base learner apart from the others of the others
- **Sequential** methods train a new base learner so that it minimizes errors made by the previous model trained in the preceding step.

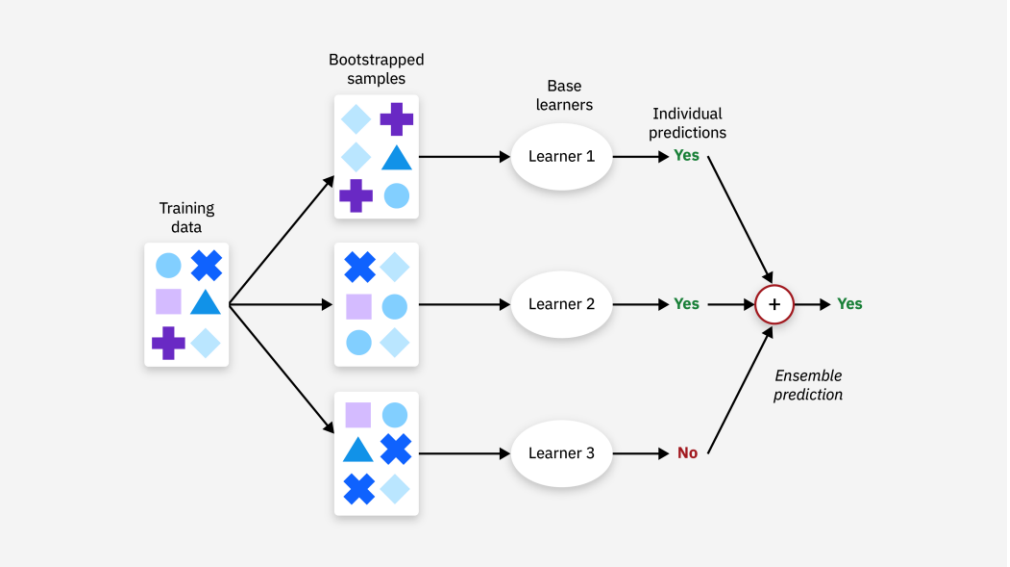


Ensemble Methods



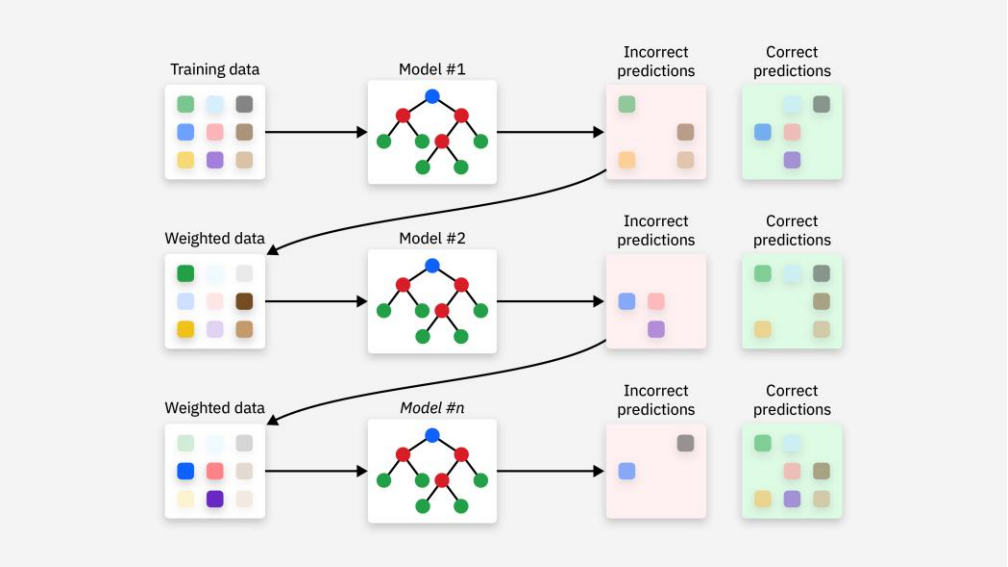
Bagging

- **Bagging (Bootstrap Aggregating)**
- **Goal:** Reduce variance.
- **How it works:** Multiple models are trained on different subsets of the data (sampled with replacement).
- **Popular Algorithm: Random Forest**
- **Use Case:** Improves stability and accuracy in classification and regression tasks.



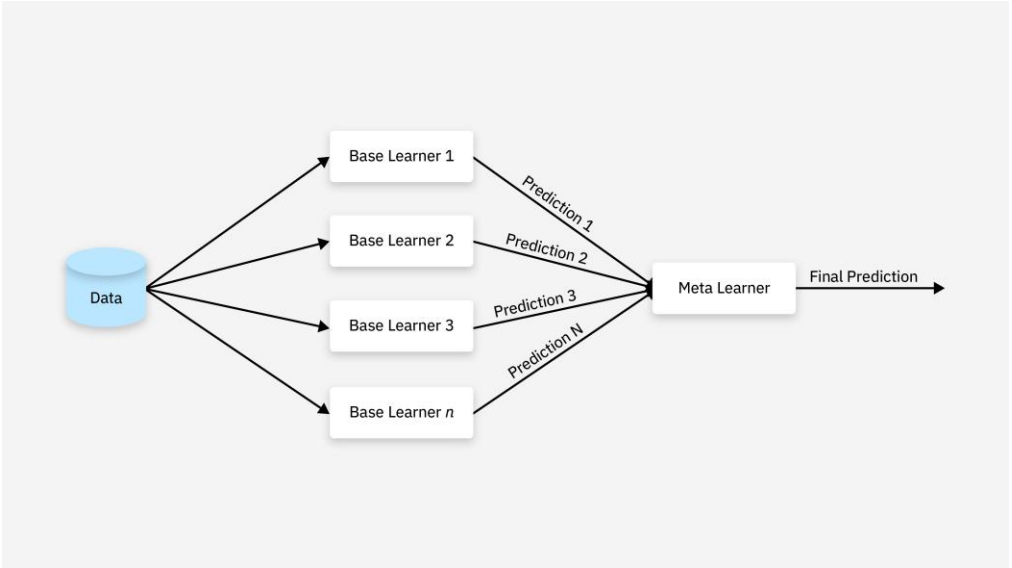
Boosting

- **Boosting**
- **Goal:** Reduce bias.
- **How it works:** Models are trained sequentially, each trying to correct the errors of the previous one.
- **Popular Algorithms:**
  - **AdaBoost**
  - **Gradient Boosting Machines (GBM)**
  - **XGBoost**, **LightGBM**, **CatBoost**
- **Use Case:** Excellent for structured/tabular data with high predictive power.

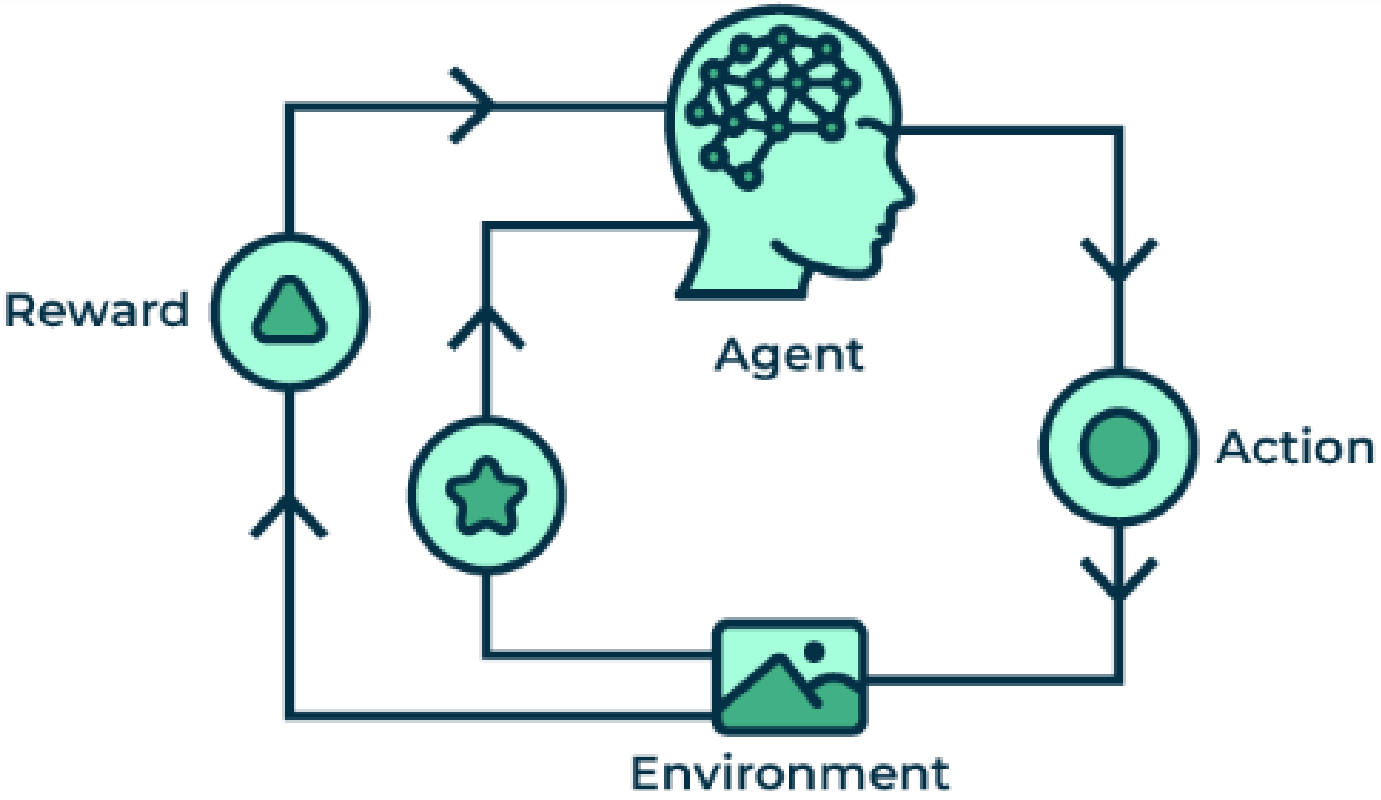


Stacking

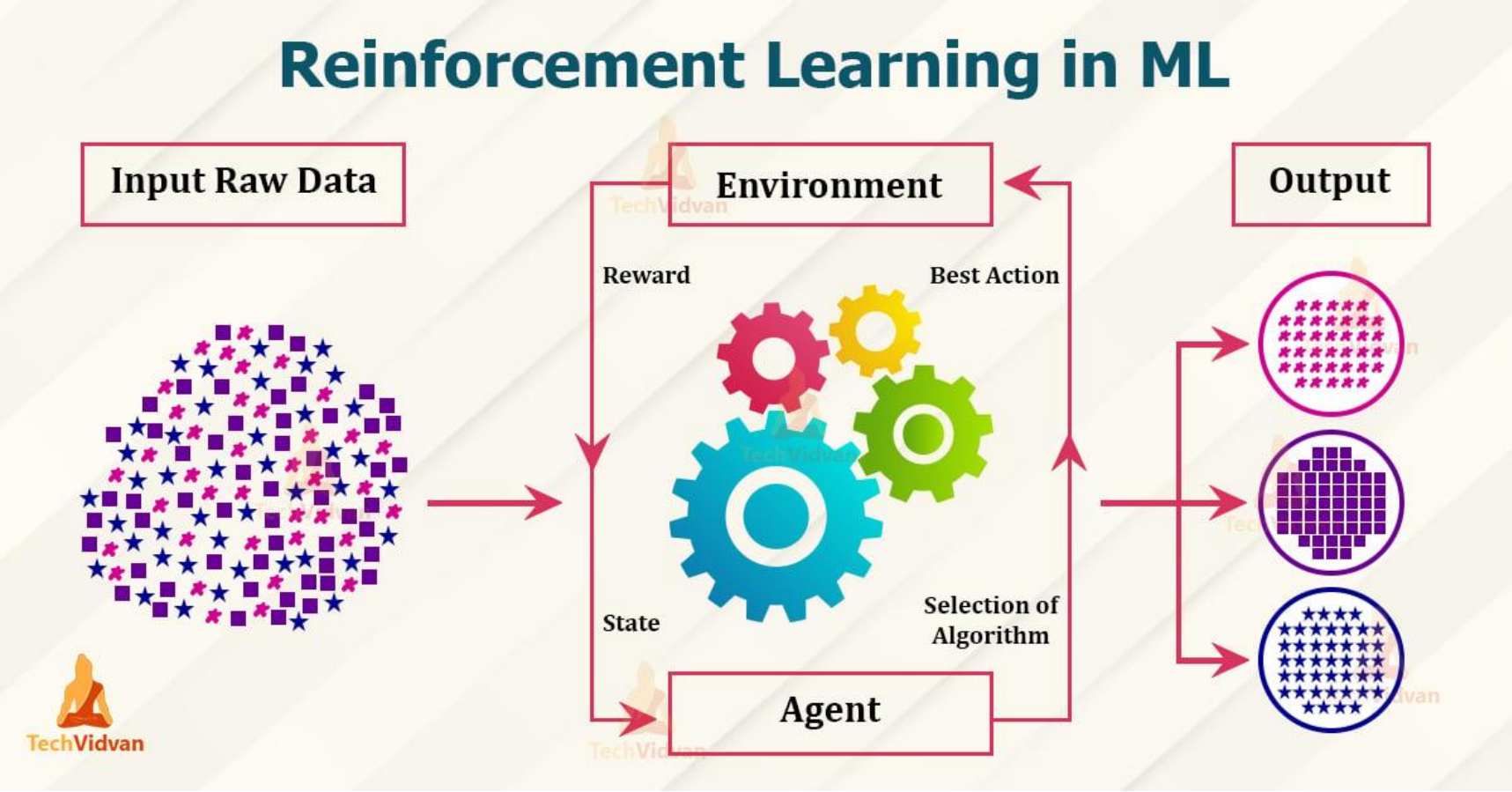
- **Stacking (Stacked Generalization)**
- **Goal:** Combine different types of models.
- **How it works:** Multiple models are trained, and their outputs are used as inputs to a **meta-model** that makes the final prediction.
- **Use Case:** Useful when combining models of different types (e.g., decision trees, neural networks, SVMs).



Reinforcement Machine Learning

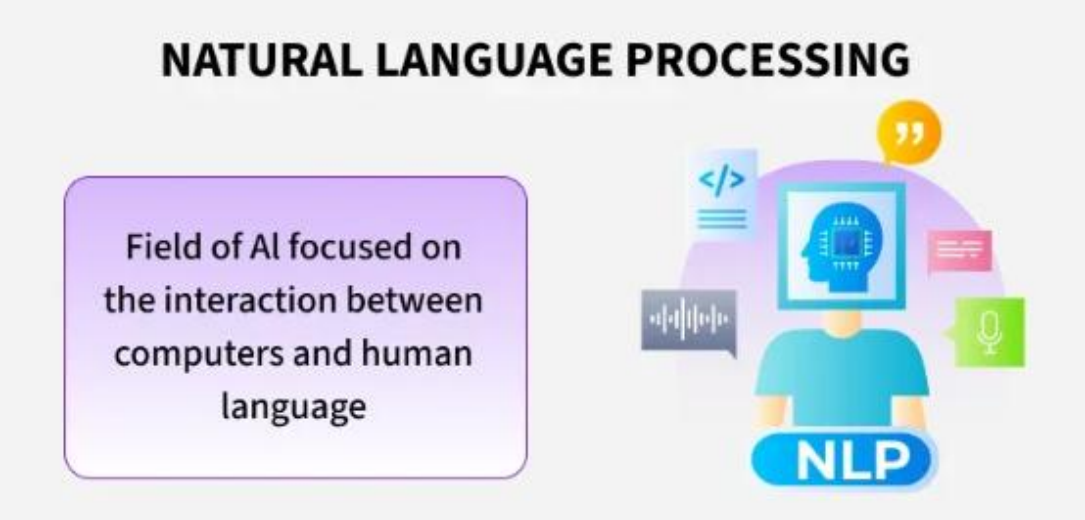


Reinforcement Machine Learning



https://techvidvan.com/tutorials/reinforcement-learning/

Natural Language Processing (NLP)



Natural Language Processing (NLP) is a field that combines computer science, artificial intelligence and language studies. It helps computers understand, process and create human language in a way that makes sense and is useful. With the growing amount of text data from social media, websites and other sources, NLP is becoming a key tool to gain insights and automate tasks like analyzing text or translating languages.

NLP Techniques



1. Text Processing and Preprocessing

Tokenization: Dividing text into smaller units, such as words or sentences.

Stemming and Lemmatization: Reducing words to their base or root forms.

Stopword Removal: Removing common words (like "and", "the", "is") that may not carry significant meaning.

Text Normalization: Standardizing text, including case normalization, removing punctuation and correcting spelling errors.



2. Syntax and Parsing

Part-of-Speech (POS) Tagging: Assigning parts of speech to each word in a sentence (e.g., noun, verb, adjective).

Dependency Parsing: Analyzing the grammatical structure of a sentence to identify relationships between words.

Constituency Parsing: Breaking down a sentence into its constituent parts or phrases (e.g., noun phrases, verb phrases).



3. Semantic Analysis

Named Entity Recognition (NER): Identifying and classifying entities in text, such as names of people organizations, locations, dates, etc.

Word Sense Disambiguation (WSD): Determining which meaning of a word is used in a given context.

Coreference Resolution: Identifying when different words refer to the same entity in a text (e.g., "he" refers to "John").



4. Information Extraction

Entity Extraction: Identifying specific entities and their relationships within the text.

Relation Extraction: Identifying and categorizing the relationships between entities in a text.



5. Text Classification in NLP

Sentiment Analysis: Determining the sentiment or emotional tone expressed in a text (e.g., positive, negative, neutral).

Topic Modeling: Identifying topics or themes within a large collection of documents.

Spam Detection: Classifying text as spam or not spam.

NLP Techniques



6. Language Generation

Machine Translation: Translating text from one language to another.

Text Summarization: Producing a concise summary of a larger text.

Text Generation: Automatically generating coherent and contextually relevant text.



7. Speech Processing

Speech Recognition: Converting spoken language into text.

Text-to-Speech (TTS) Synthesis: Converting written text into spoken language.



8. Question Answering

Retrieval-Based QA: Finding and returning the most relevant text passage in response to a query.

Generative QA: Generating an answer based on the information available in a text corpus.



9. Dialogue Systems

Chatbots and Virtual Assistants: Enabling systems to engage in conversations with users, providing responses and performing tasks based on user input.



10. Sentiment and Emotion Analysis in NLP

Emotion Detection: Identifying and categorizing emotions expressed in text.

Opinion Mining: Analyzing opinions or reviews to understand public sentiment toward products, services or topics.

How Natural Language Processing (NLP) Works



How Natural Language Processing (NLP) Works

1. Text Input and Data Collection

Data Collection: Gathering text data from various sources such as websites, books, social media or proprietary databases.

Data Storage: Storing the collected text data in a structured format, such as a database or a collection of documents.

2. Text Preprocessing

Preprocessing is crucial to clean and prepare the raw text data for analysis. Common preprocessing steps include:

Tokenization: Splitting text into smaller units like words or sentences.

Lowercasing: Converting all text to lowercase to ensure uniformity.

Stopword Removal: Removing common words that do not contribute significant meaning, such as "and," "the," "is."

Punctuation Removal: Removing punctuation marks.

Stemming and Lemmatization: Reducing words to their base or root forms. Stemming cuts off suffixes, while lemmatization considers the context and converts words to their meaningful base form.

Text Normalization: Standardizing text format, including correcting spelling errors, expanding contractions and handling special characters.

How Natural Language Processing (NLP) Works



3. Text Representation

Bag of Words (BoW): Representing text as a collection of words, ignoring grammar and word order but keeping track of word frequency.

Term Frequency-Inverse Document Frequency (TF-IDF): A statistic that reflects the importance of a word in a document relative to a collection of documents.

Word Embeddings: Using dense vector representations of words where semantically similar words are closer together in the vector space (e.g., Word2Vec, GloVe).



4. Feature Extraction

Extracting meaningful features from the text data that can be used for various NLP tasks.

N-grams: Capturing sequences of N words to preserve some context and word order.

Syntactic Features: Using parts of speech tags, syntactic dependencies and parse trees.

Semantic Features: Leveraging word embeddings and other representations to capture word meaning and context.



5. Model Selection and Training

Selecting and training a machine learning or deep learning model to perform specific NLP tasks.

Supervised Learning: Using labeled data to train models like Support Vector Machines (SVM), Random Forests or deep learning models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs).

Unsupervised Learning: Applying techniques like clustering or topic modeling (e.g., Latent Dirichlet Allocation) on unlabeled data.

Pre-trained Models: Utilizing pre-trained language models such as BERT, GPT or transformer-based models that have been trained on large corpora.

How Natural Language Processing (NLP) Works



6. Model Deployment and Inference

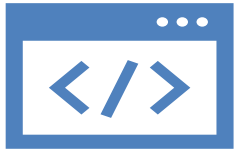
Deploying the trained model and using it to make predictions or extract insights from new text data.

Text Classification: Categorizing text into predefined classes (e.g., spam detection, sentiment analysis).

Named Entity Recognition (NER): Identifying and classifying entities in the text.

Machine Translation: Translating text from one language to another.

Question Answering: Providing answers to questions based on the context provided by text data.



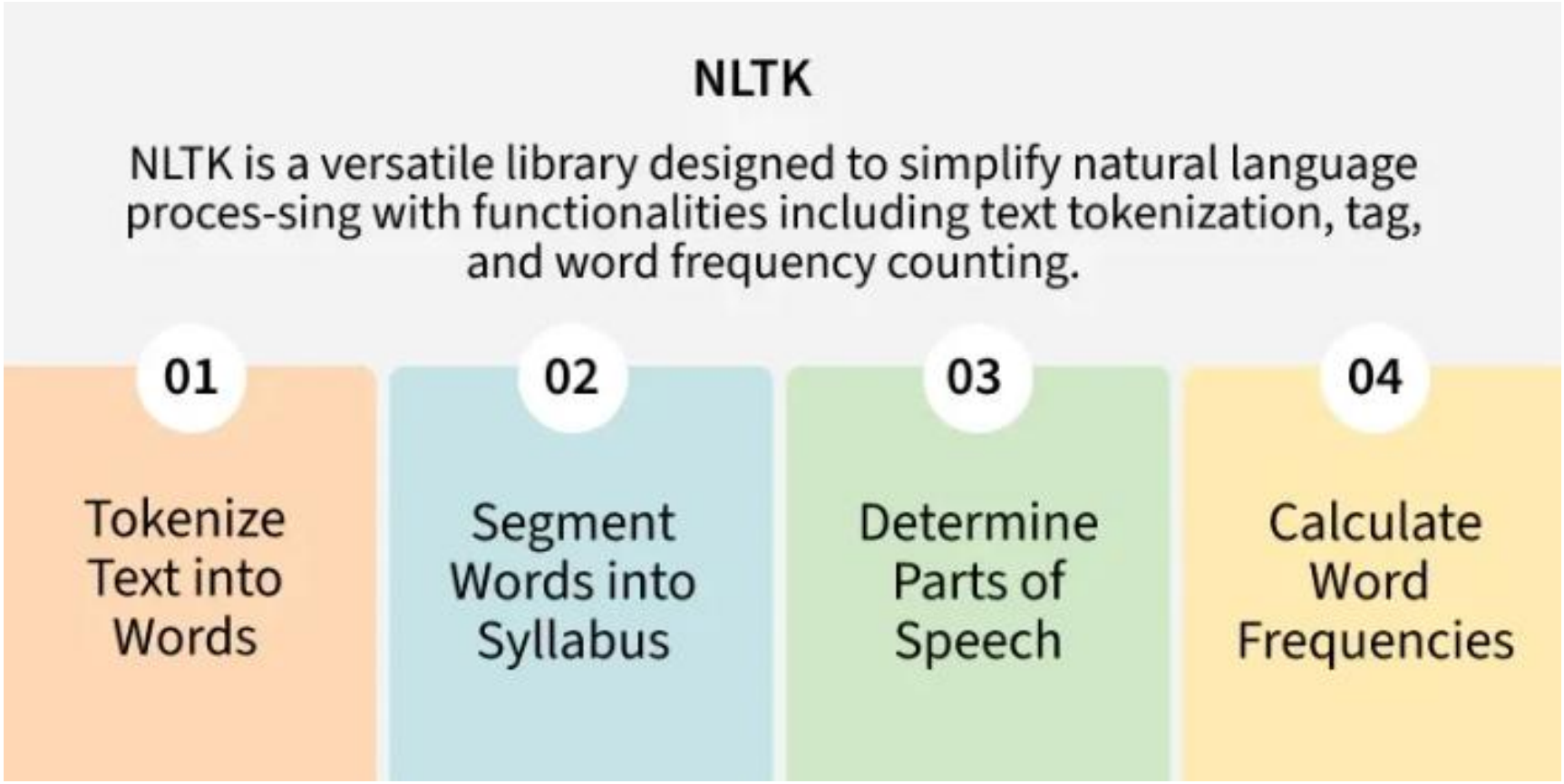
7. Evaluation and Optimization

Evaluating the performance of the NLP algorithm using metrics such as accuracy, precision, recall, F1-score and others.

Hyperparameter Tuning: Adjusting model parameters to improve performance.

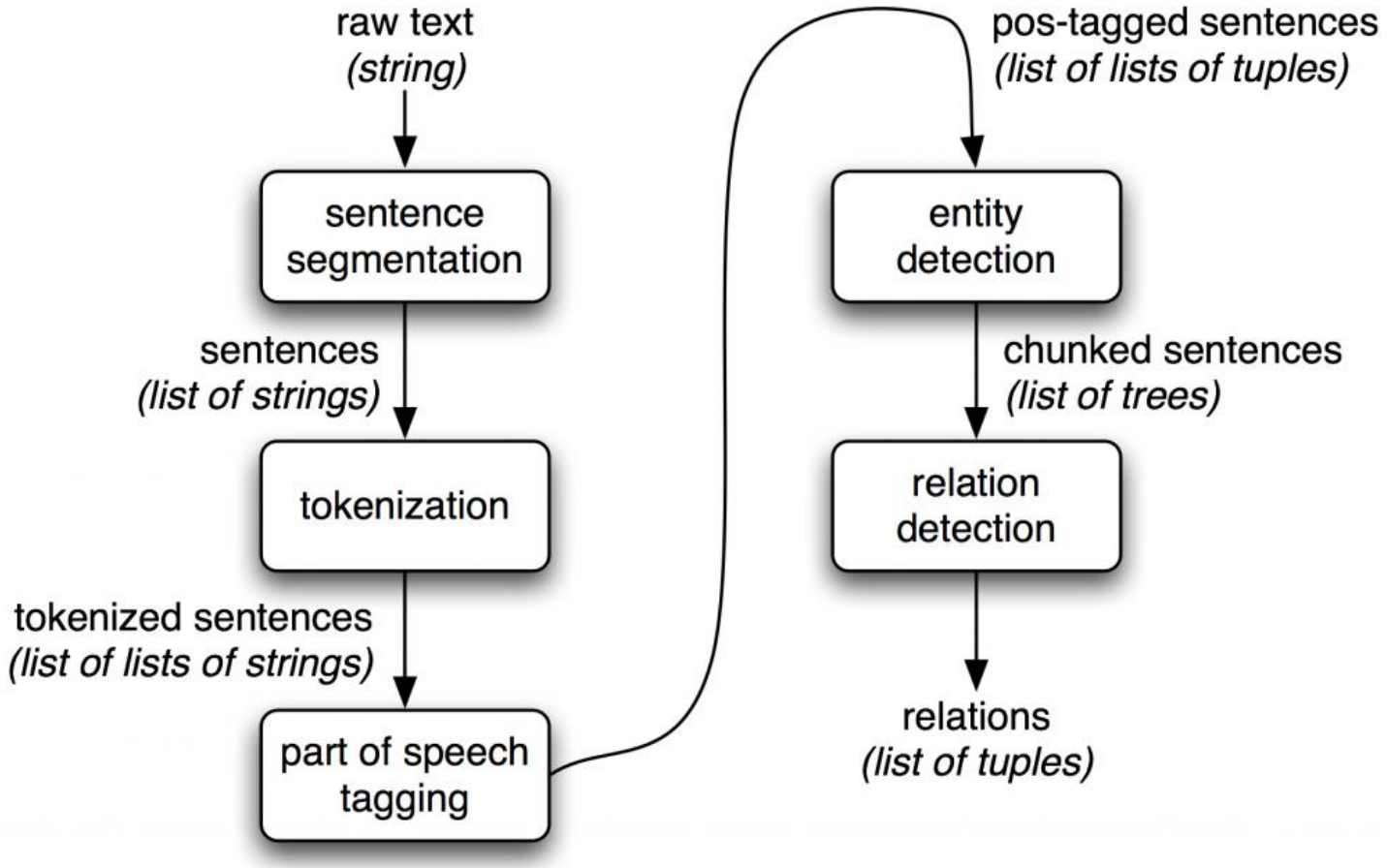
Error Analysis: Analyzing errors to understand model weaknesses and improve robustness.

NLTK Package



https://www.nltk.org/

What are the feature of NLTK?

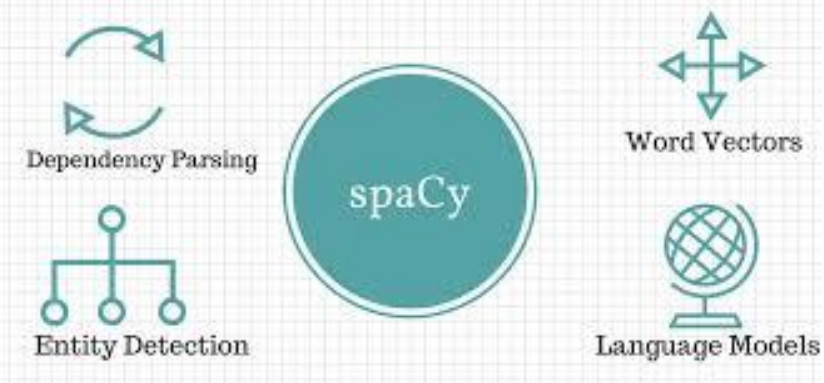


https://www.devopsschool.com/blog/what-is-nltk-and-use-cases-of-nltk/

Spacy Package

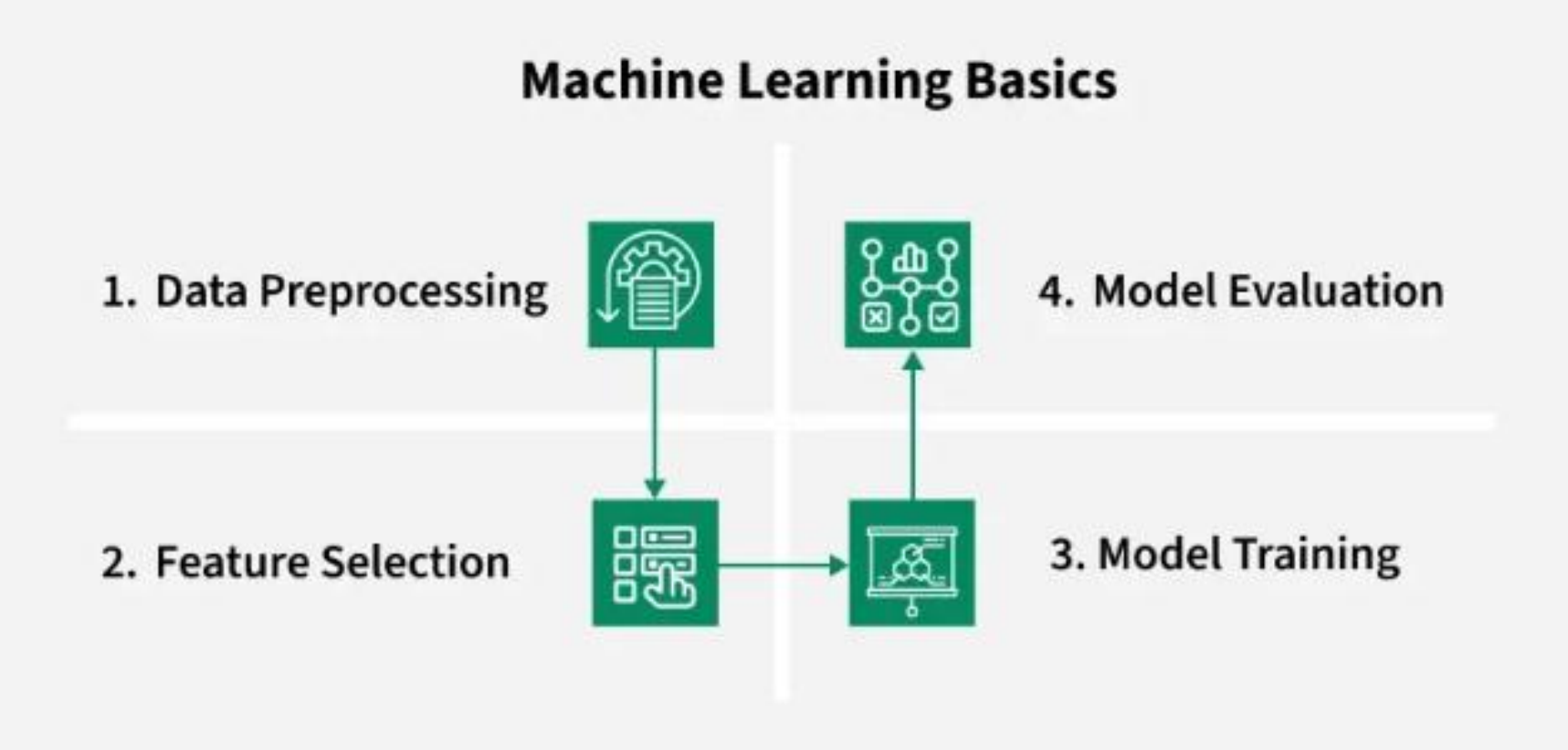
https://spacy.io/

The **spaCy** package is a powerful and fast open-source library for **Natural Language Processing (NLP)** in Python. It's widely used for tasks like text preprocessing, named entity recognition, part-of-speech tagging, and syntactic parsing.

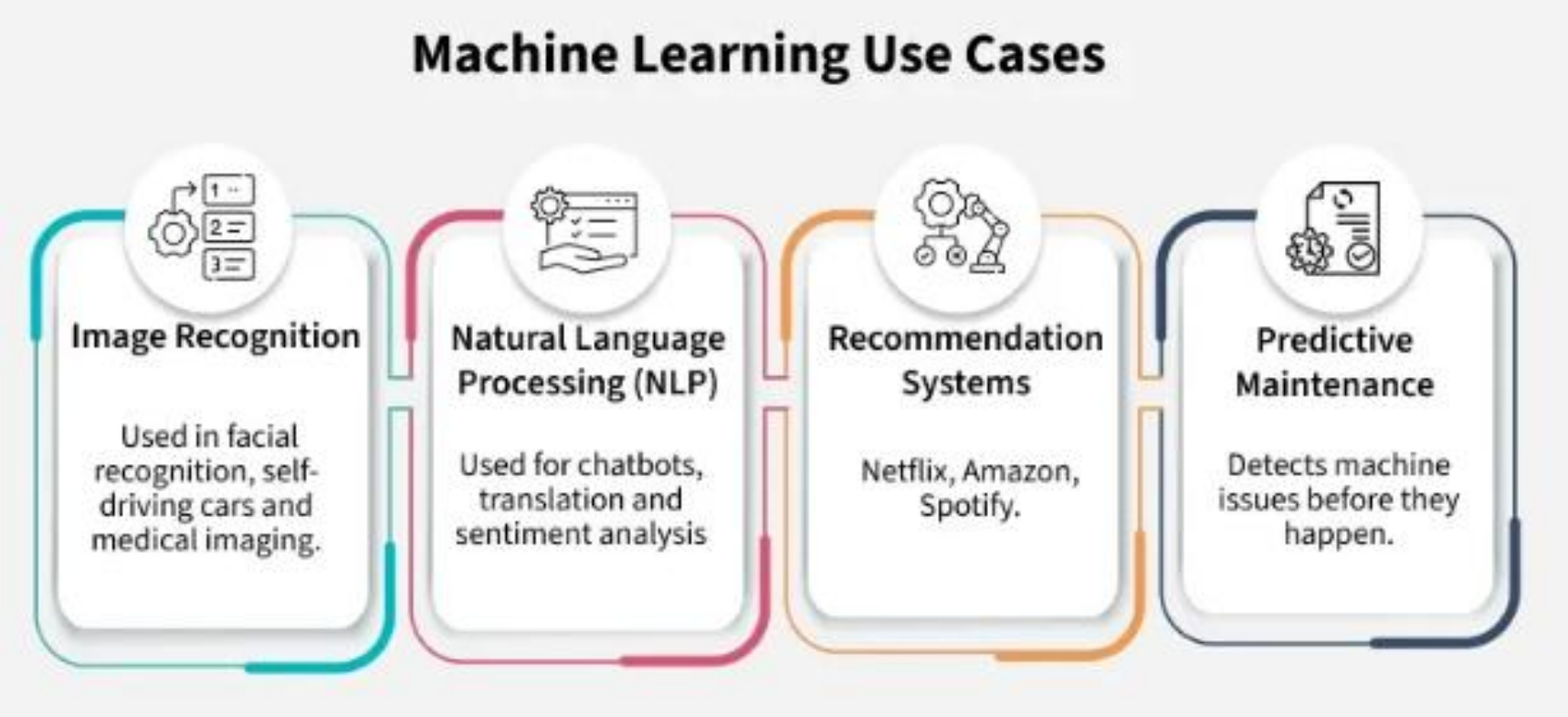




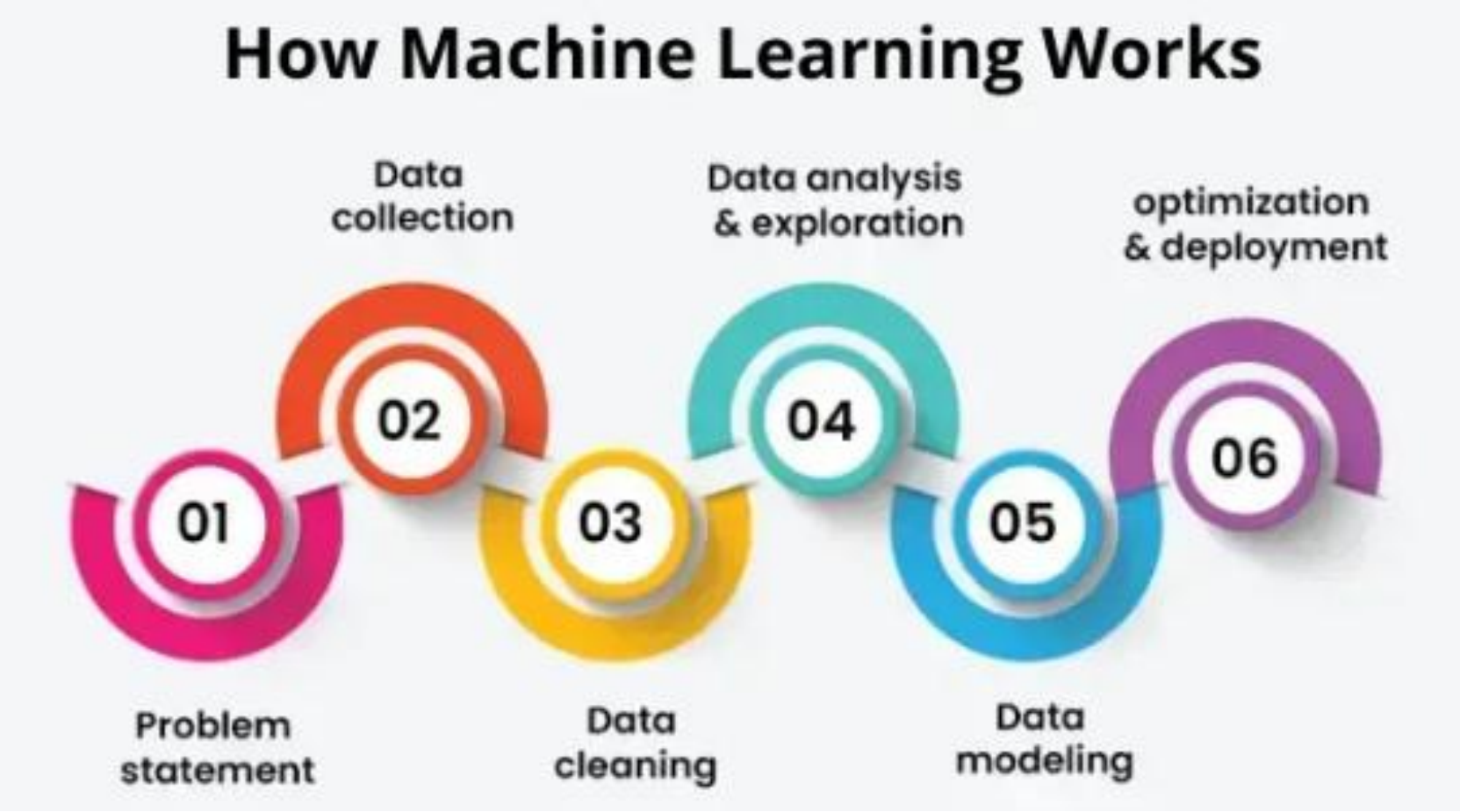
Machine Learning Basics



Machine Learning Use Cases



Machine Learning Works



Feature Engineering



Feature Creation



Feature Transformation



Feature Extraction



Feature Selection



Feature Scaling

Feature Engineering

Vary	Feature engineering can vary depending on the specific problem but the general steps are:
Cleansing	Data Cleansing: Identify and correct errors or inconsistencies in the dataset to ensure data quality and reliability.
Transform	Data Transformation: Transform raw data into a format suitable for modeling including scaling, normalization and encoding.
Feature	Feature Extraction: Create new features by combining or deriving information from existing ones to provide more meaningful input to the model.
Feature	Feature Selection: Choose the most relevant features for the model using techniques like correlation analysis, mutual information and stepwise regression.
Feature	Feature Iteration: Continuously refine features based on model performance by adding, removing or modifying features for improvement.

Common Techniques in Feature Engineering

One-Hot Encoding: One-Hot Encoding converts categorical variables into binary indicators, allowing them to be used by machine learning models.

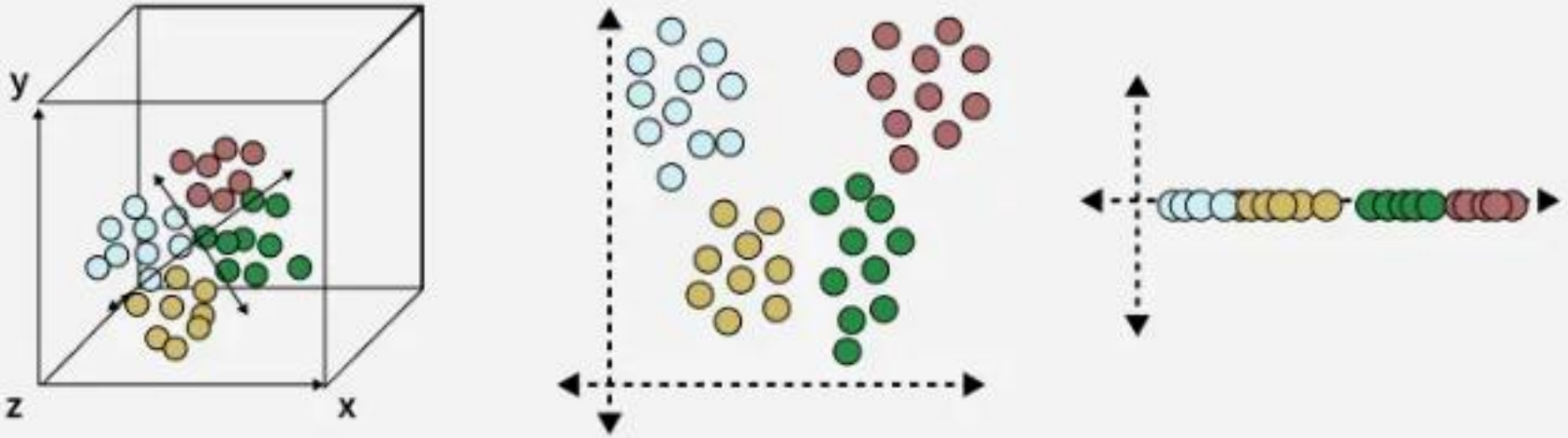
Binning: Binning transforms continuous variables into discrete bins, making them categorical for easier analysis.

Text Data Preprocessing: Involves removing stop-words, stemming and vectorizing text data to prepare it for machine learning models.

Feature Splitting: Divides a single feature into multiple sub-features, uncovering valuable insights and improving model performance.

Dimensionality Reduction

Dimensionality Reduction is the process of reducing the number of input variables (features) in a dataset while preserving as much important information as possible.



Feature extraction

Feature extraction involves creating new features by combining or transforming the original features. These new features retain most of the dataset's important information in fewer dimensions. Common feature extraction methods are:

Principal Component Analysis (PCA): Converts correlated variables into uncorrelated 'principal components, reducing dimensionality while maintaining as much variance as possible enabling more efficient analysis.

Missing Value Ratio: Variables with missing data beyond a set threshold are removed, improving dataset reliability.

Backward Feature Elimination: Starts with all features and removes the least significant ones in each iteration. The process continues until only the most impactful features remain, optimizing model performance.

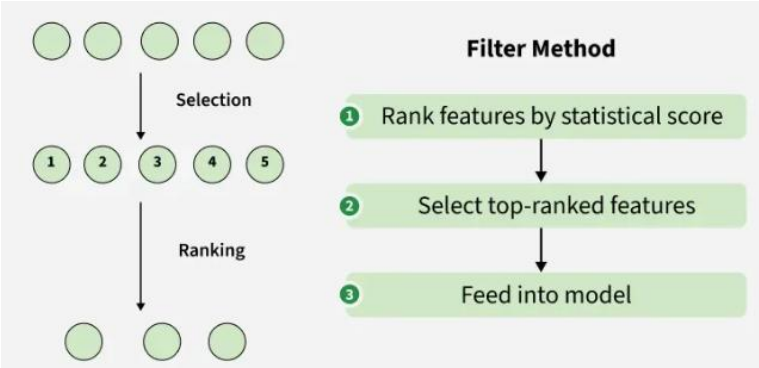
Forward Feature Selection: Forward Feature Selection Begins with one feature, adds others incrementally and keeps those improving model performance.

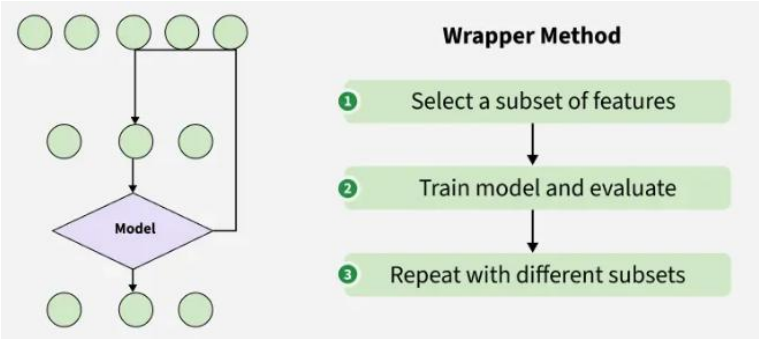
Random Forest: Random forest Uses decision trees to evaluate feature importance, automatically selecting the most relevant features without the need for manual coding, enhancing model accuracy.

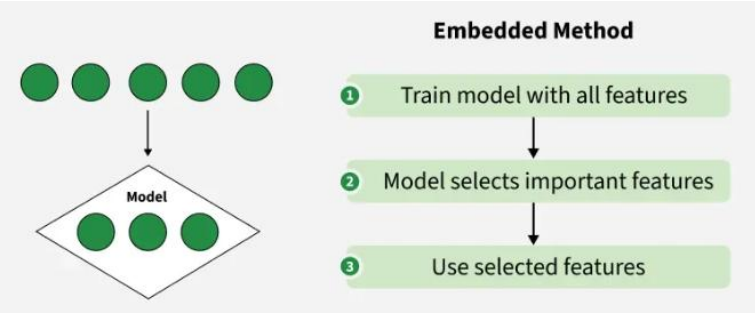
Factor Analysis: Groups variables by correlation and keeps the most relevant ones for further analysis.

Independent Component Analysis (ICA): Identifies statistically independent components, ideal for applications like 'blind source separation' where traditional correlation-based methods fall short.

Feature Selection







Feature Engineering: Scaling, Normalization, and Standardization

1. Absolute Maximum Scaling
This method of scaling requires two-step:

We should first select the maximum absolute value out of all the entries of a particular measure.

Then after this we divide each entry of the column by this maximum value.

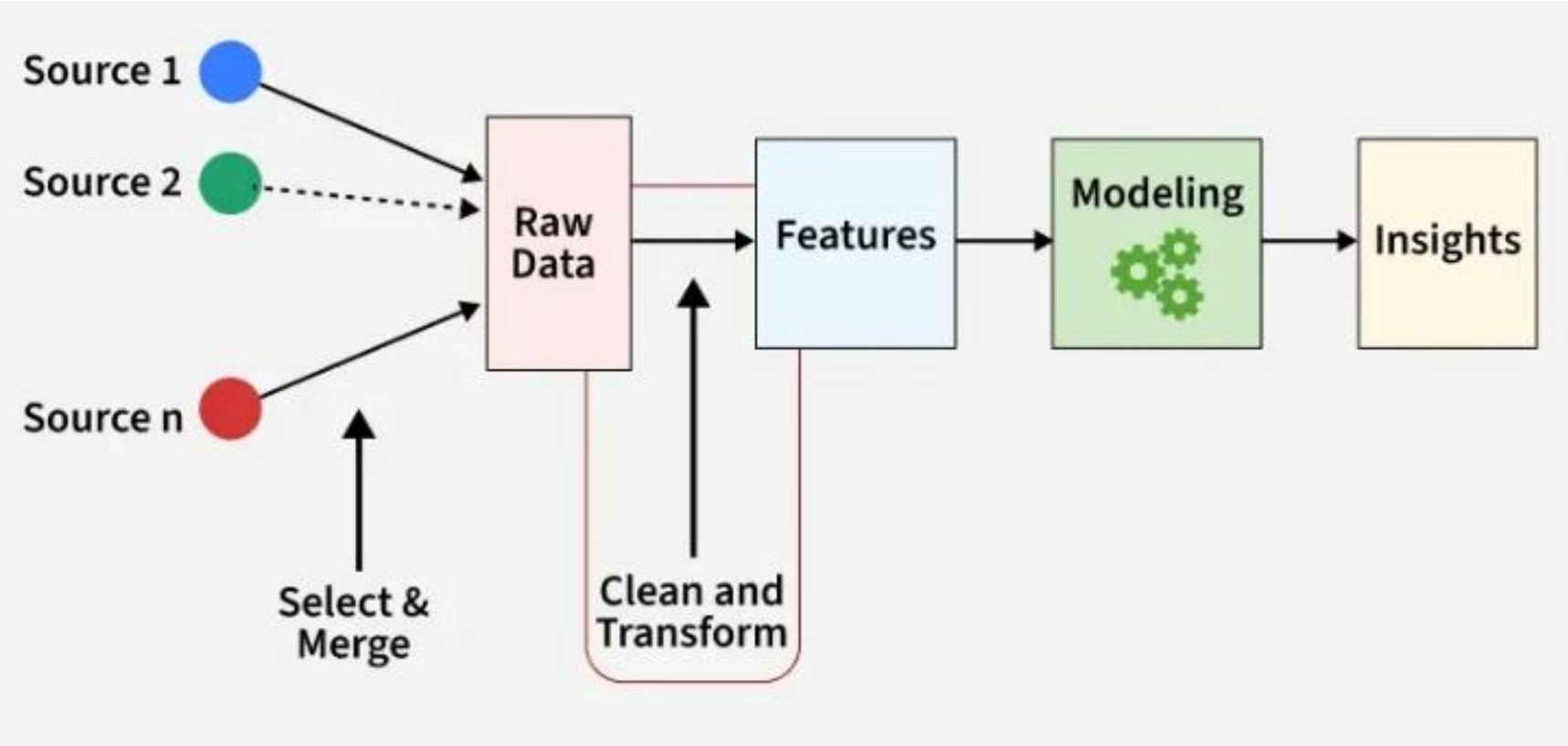
$X_{scaled} = \max(|X|)X_i - \max(|X|)$

2. Min-Max Scaling

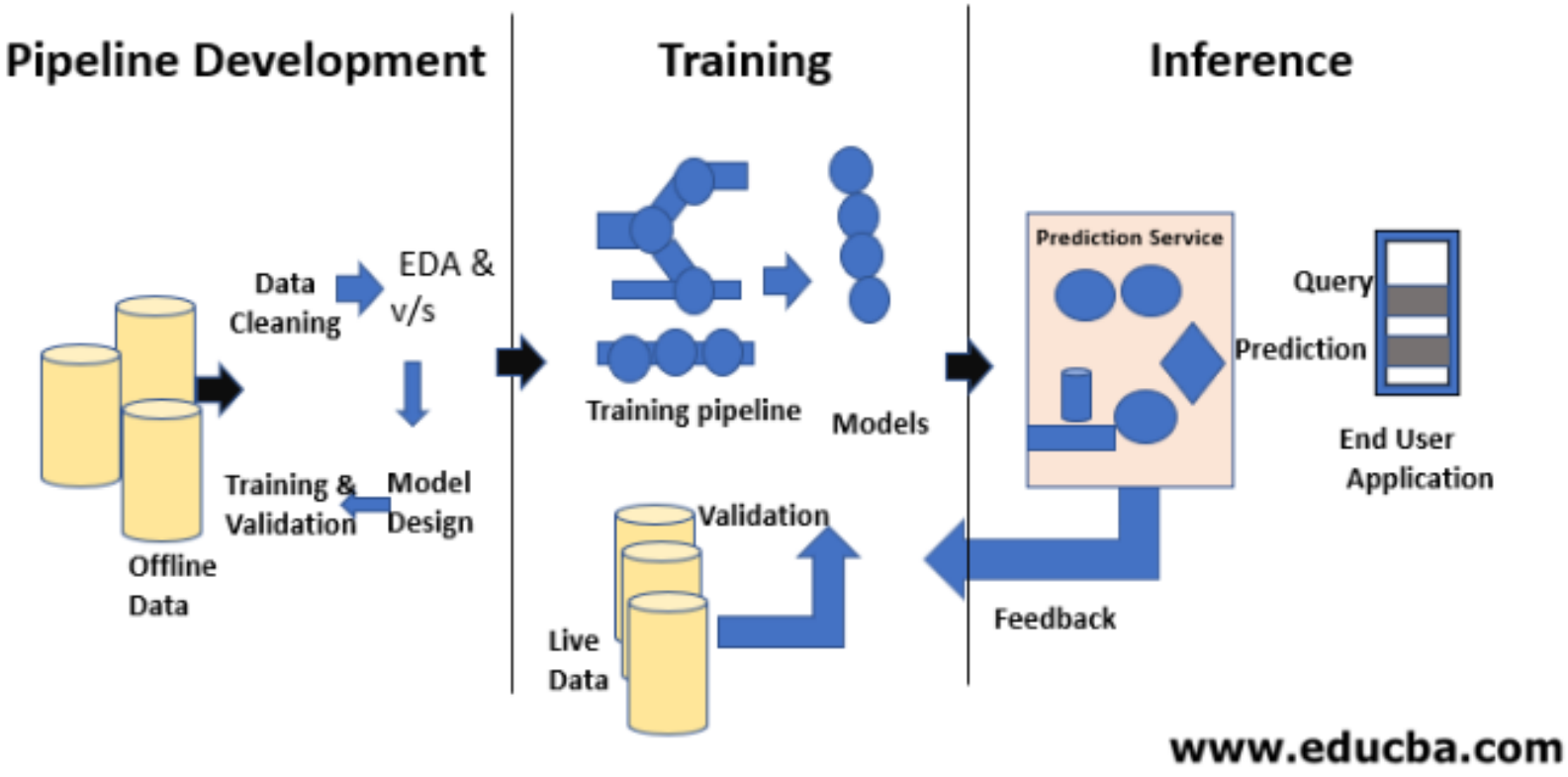
This method of scaling requires below two-step:
First we are supposed to find the minimum and the maximum value of the column.
Then we will subtract the minimum value from the entry and divide the result by the difference between the maximum and the minimum value.
$X_{scaled} = X_i - X_{min} X_{max} - X_{min}$

3. Normalization
Normalization is the process of adjusting the values of data points so that they all have the same length or size, specifically a length of 1.
This is done by dividing each data point by the "length" (called as Euclidean norm) of that data point.
Think of it like adjusting the size of a vector so that it fits within a standard size of 1.
The formula for Normalization looks like this:
$X_{scaled} = \|X\| X_i$
Where:
X_i is each individual value.
$\|X\|$ represents the Euclidean norm (or length) of the vector X.

Feature Engineering

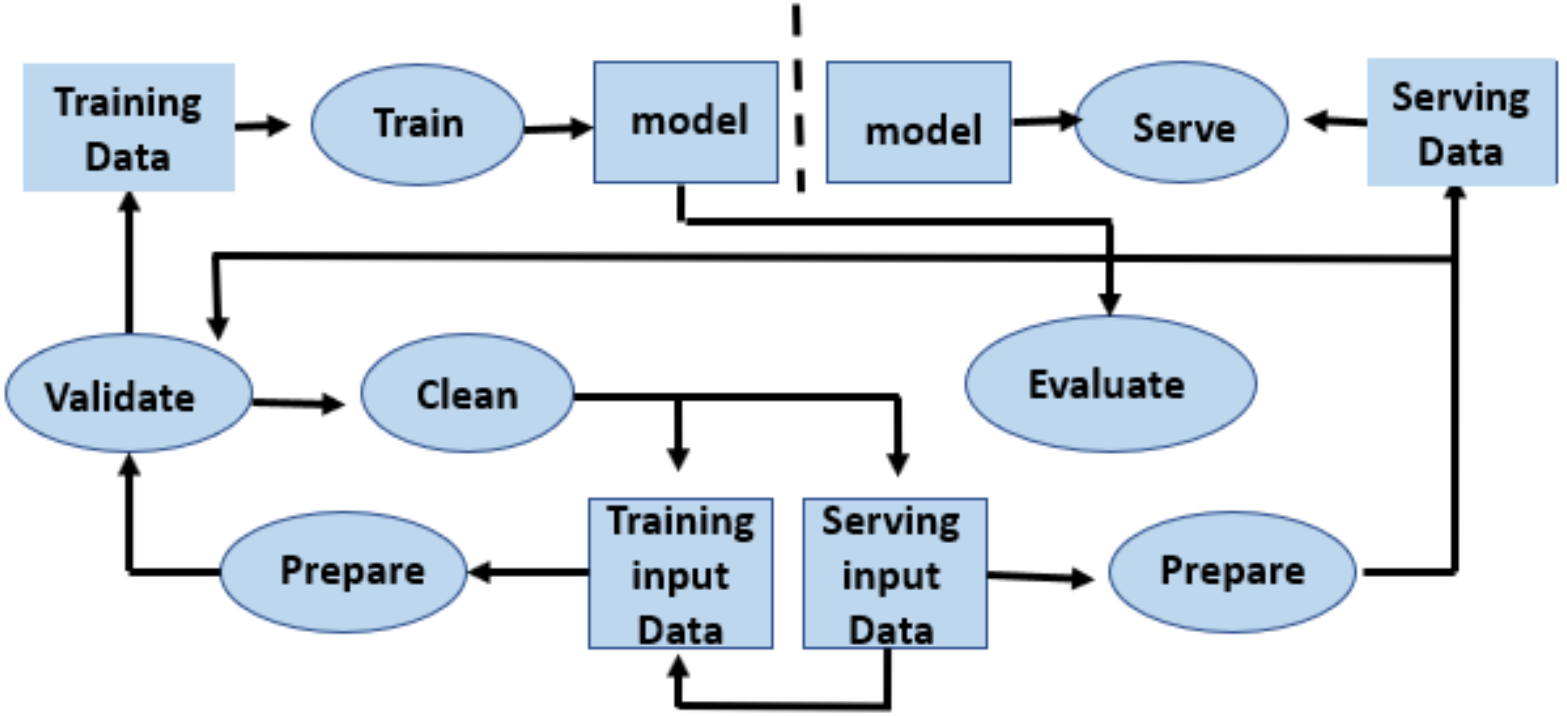


Detailed representation of the Machine Learning life cycle



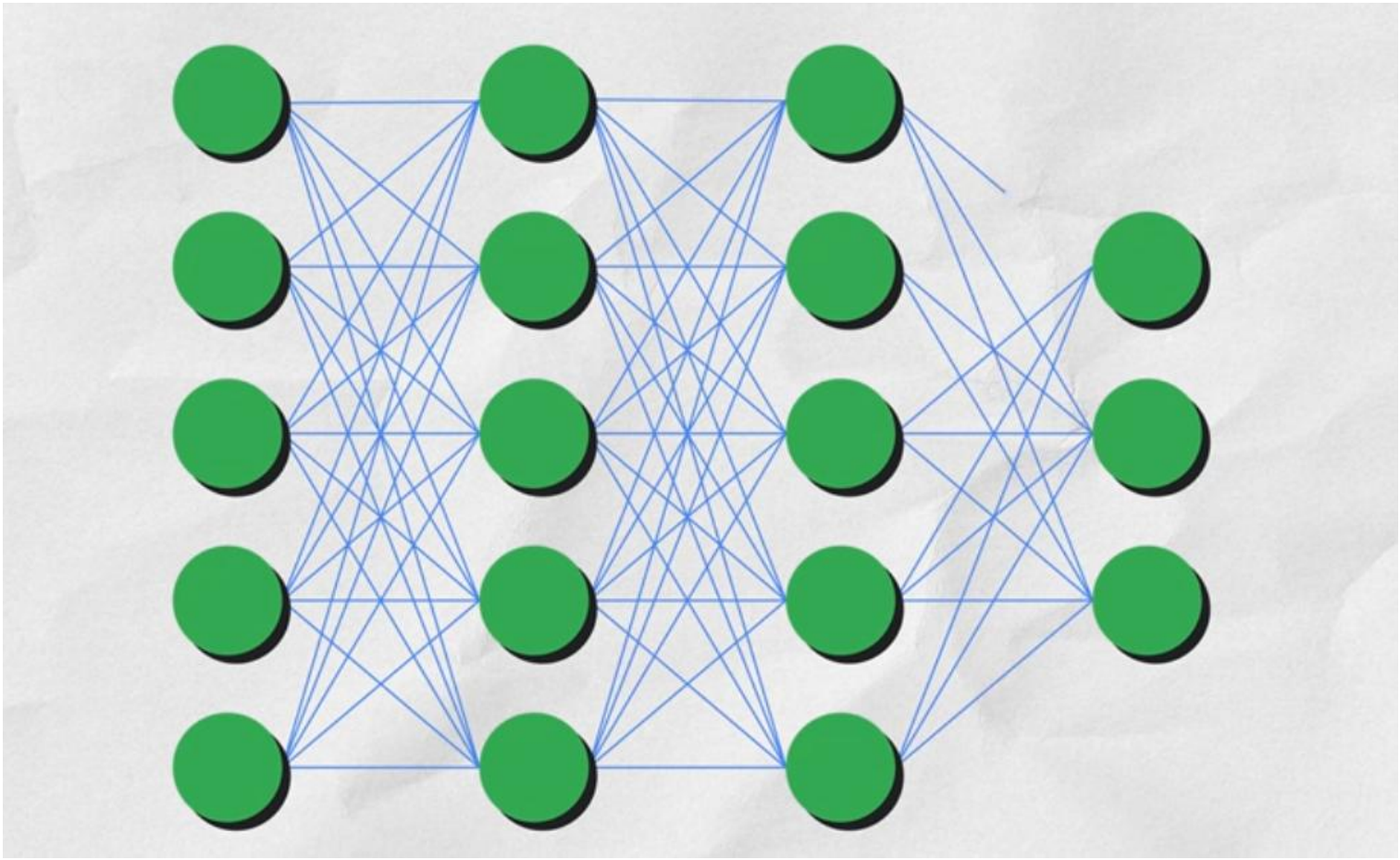
https://www.educba.com/machine-learning-life-cycle/

Pipeline with a data point of view



https://www.educba.com/machine-learning-life-cycle/

Deep learning

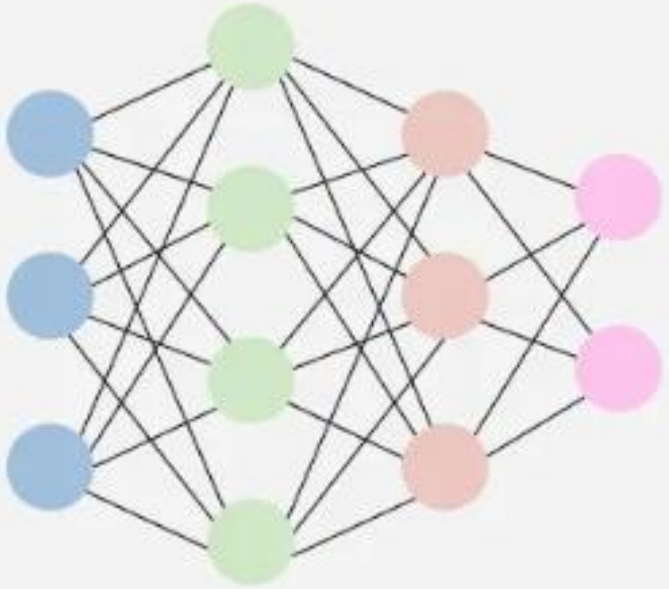


Deep Learning

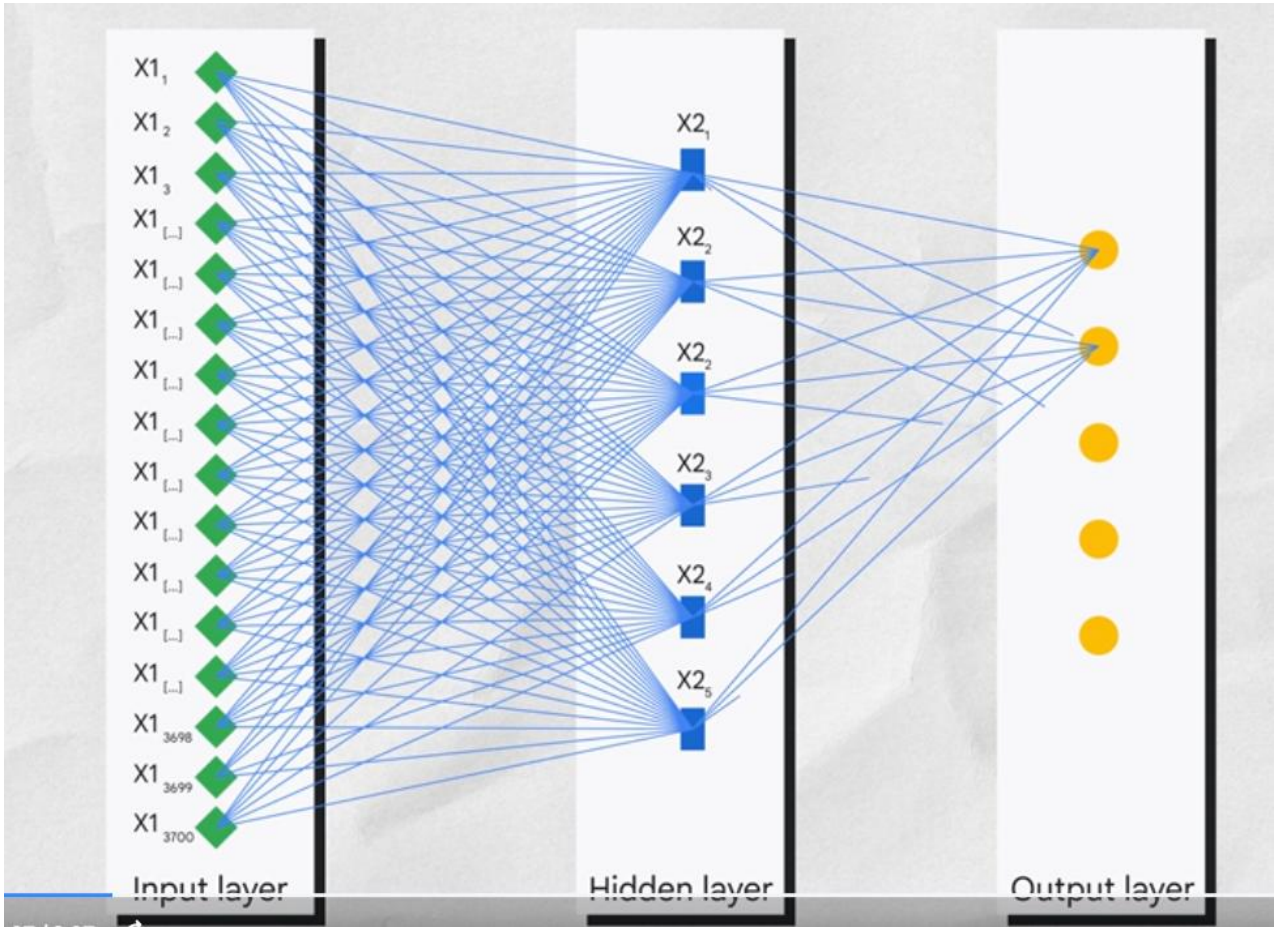
What is Deep Learning?

1. Deep learning uses multi-layered neural networks.

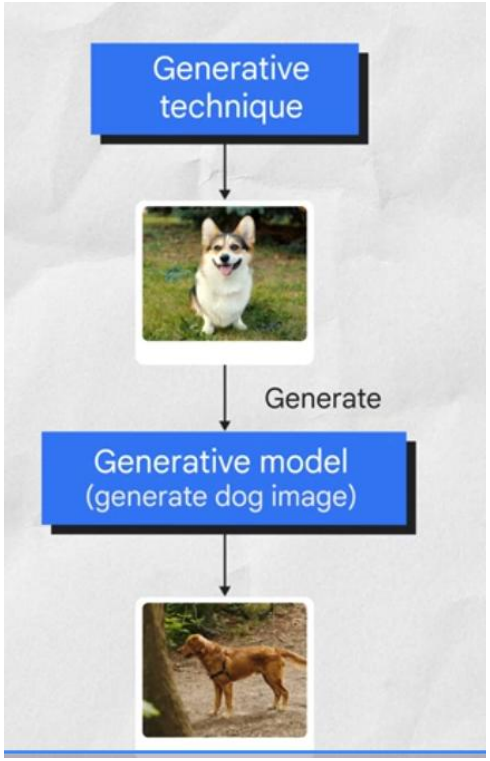
2. Learns from large datasets to make predictions automatically.

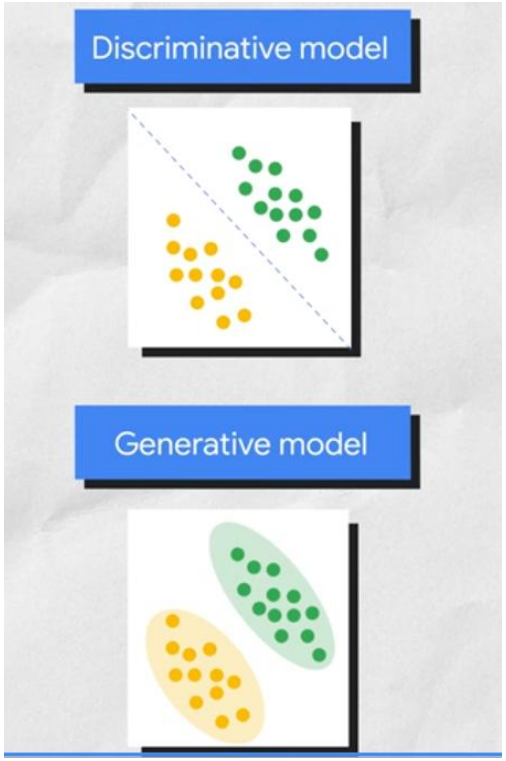


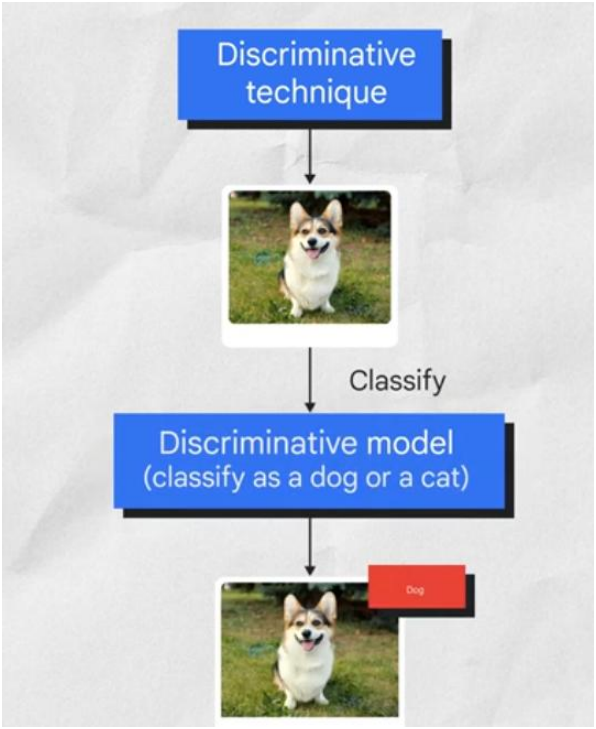
Deep learning



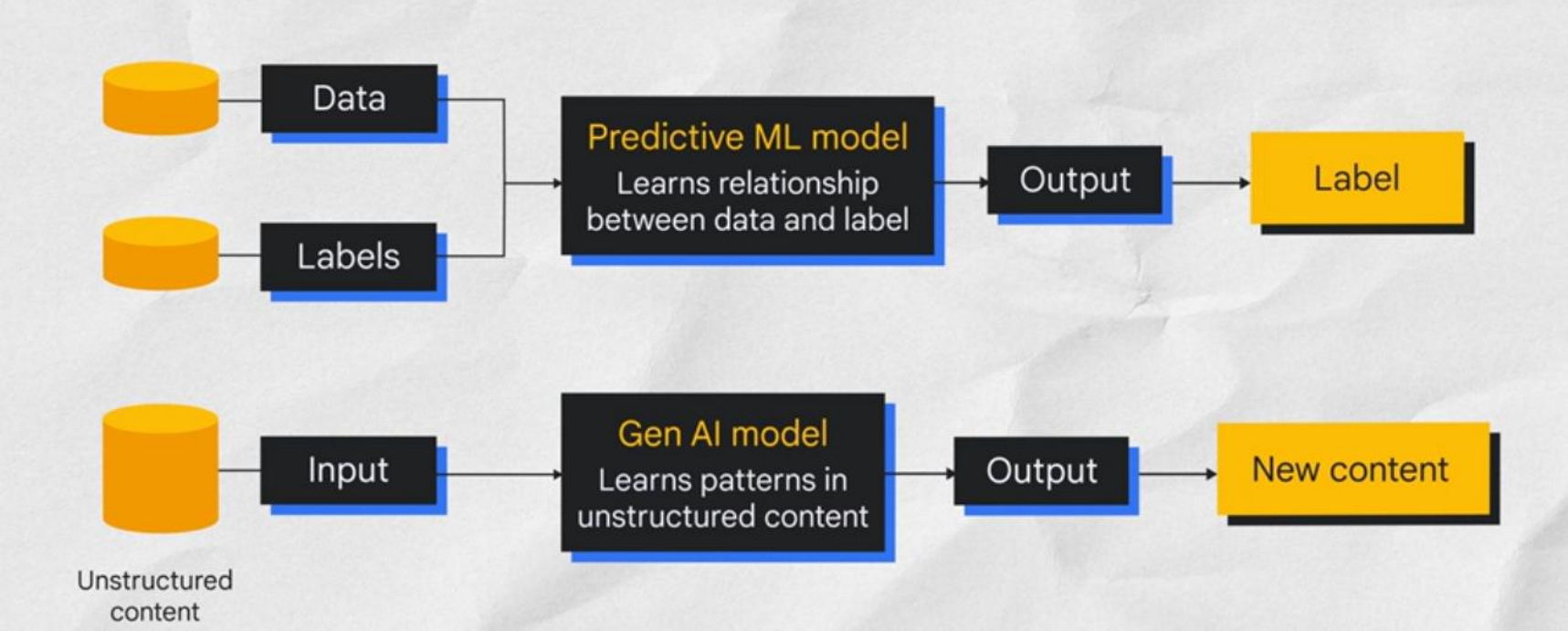
Discriminative Model & Generative Model



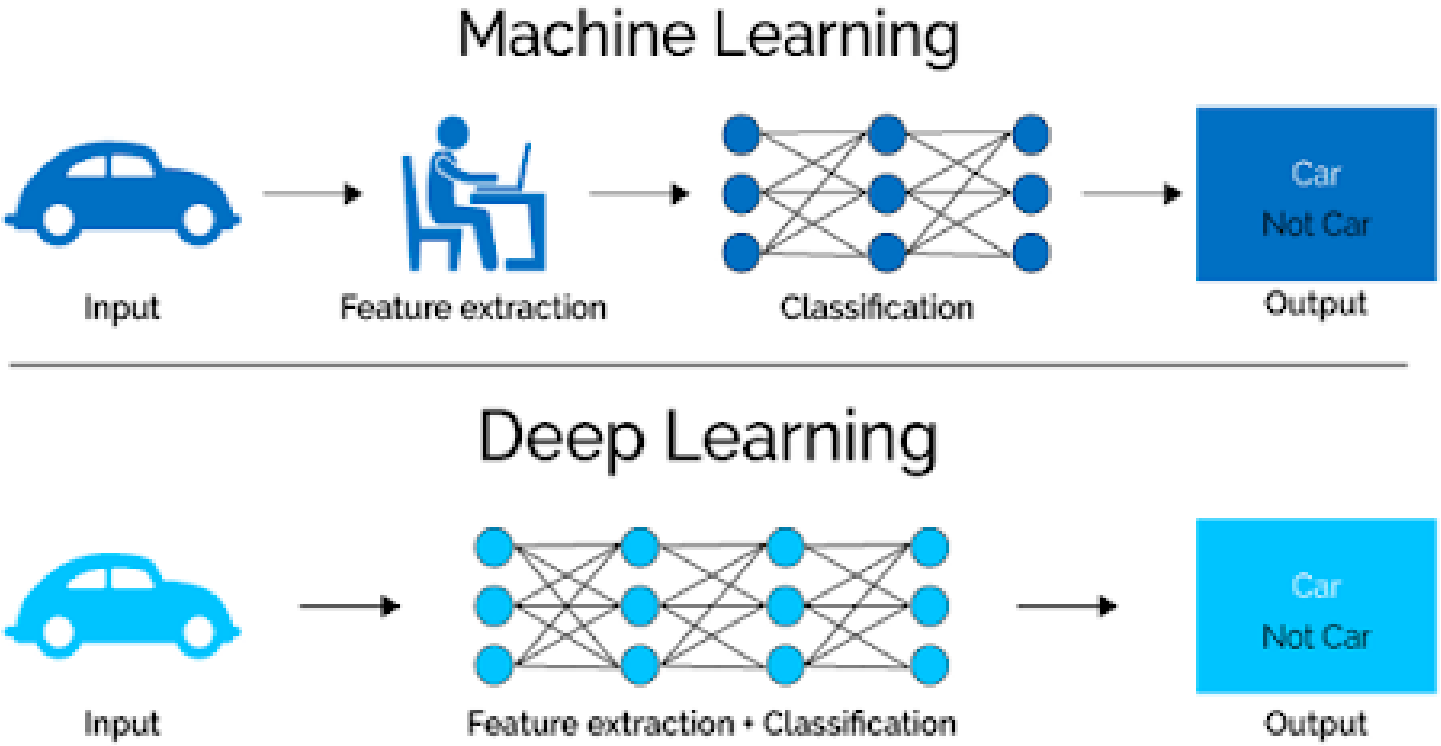




Discriminative Model & Generative Model

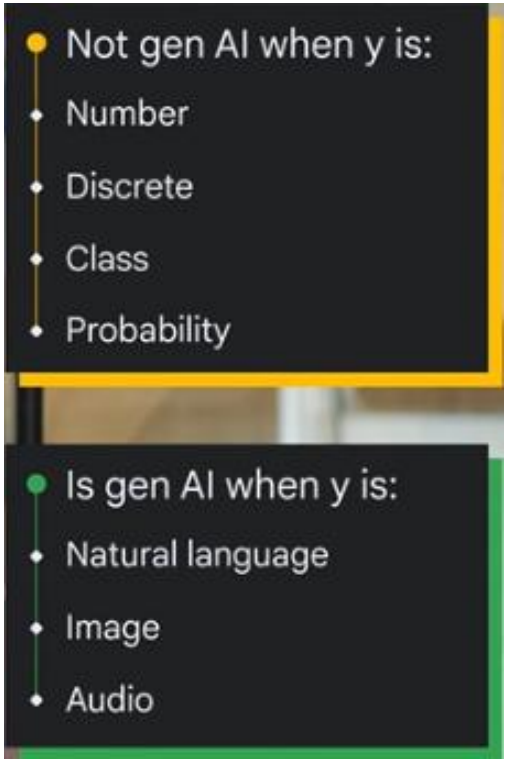


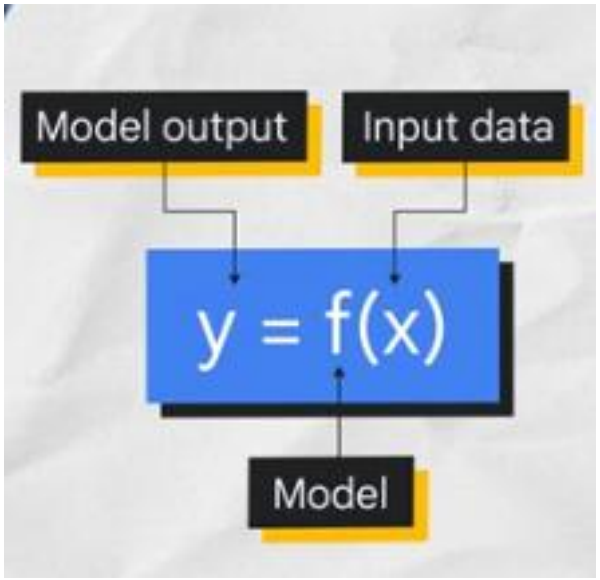
Deep Learning Vs Machine Learning

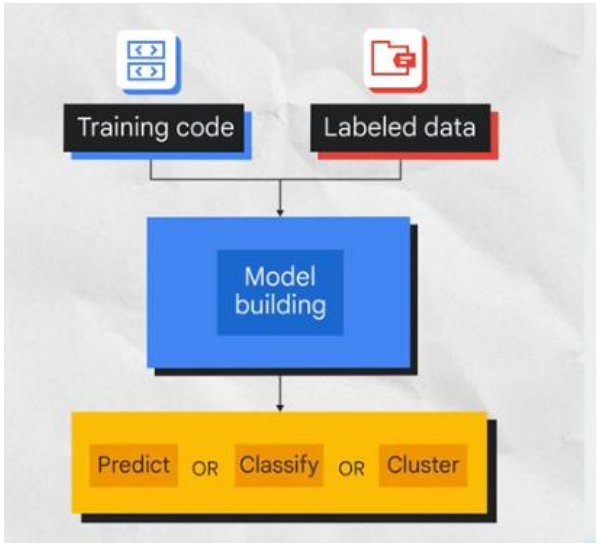


https://builtin.com/machine-learning/deep-learning

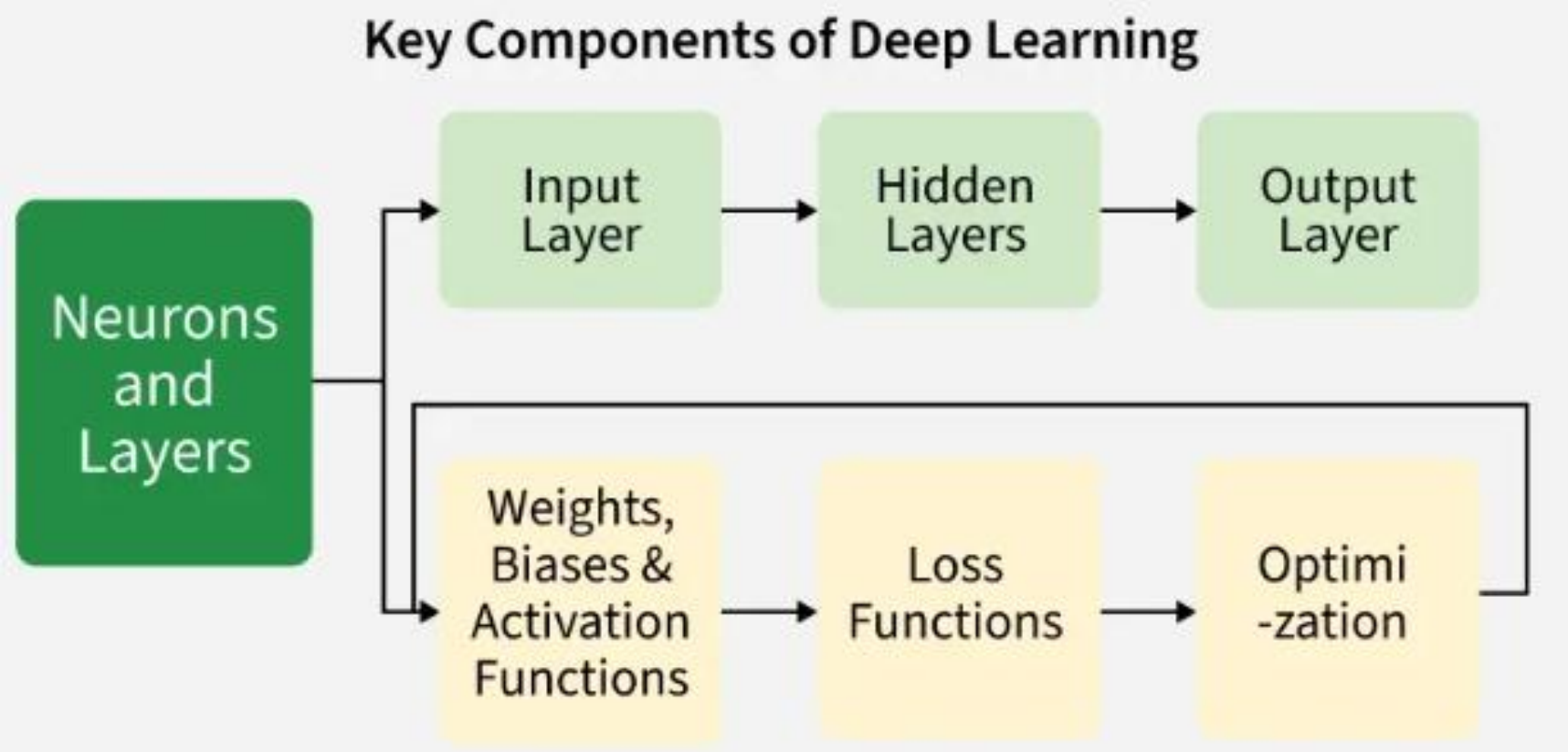
Deep Learning



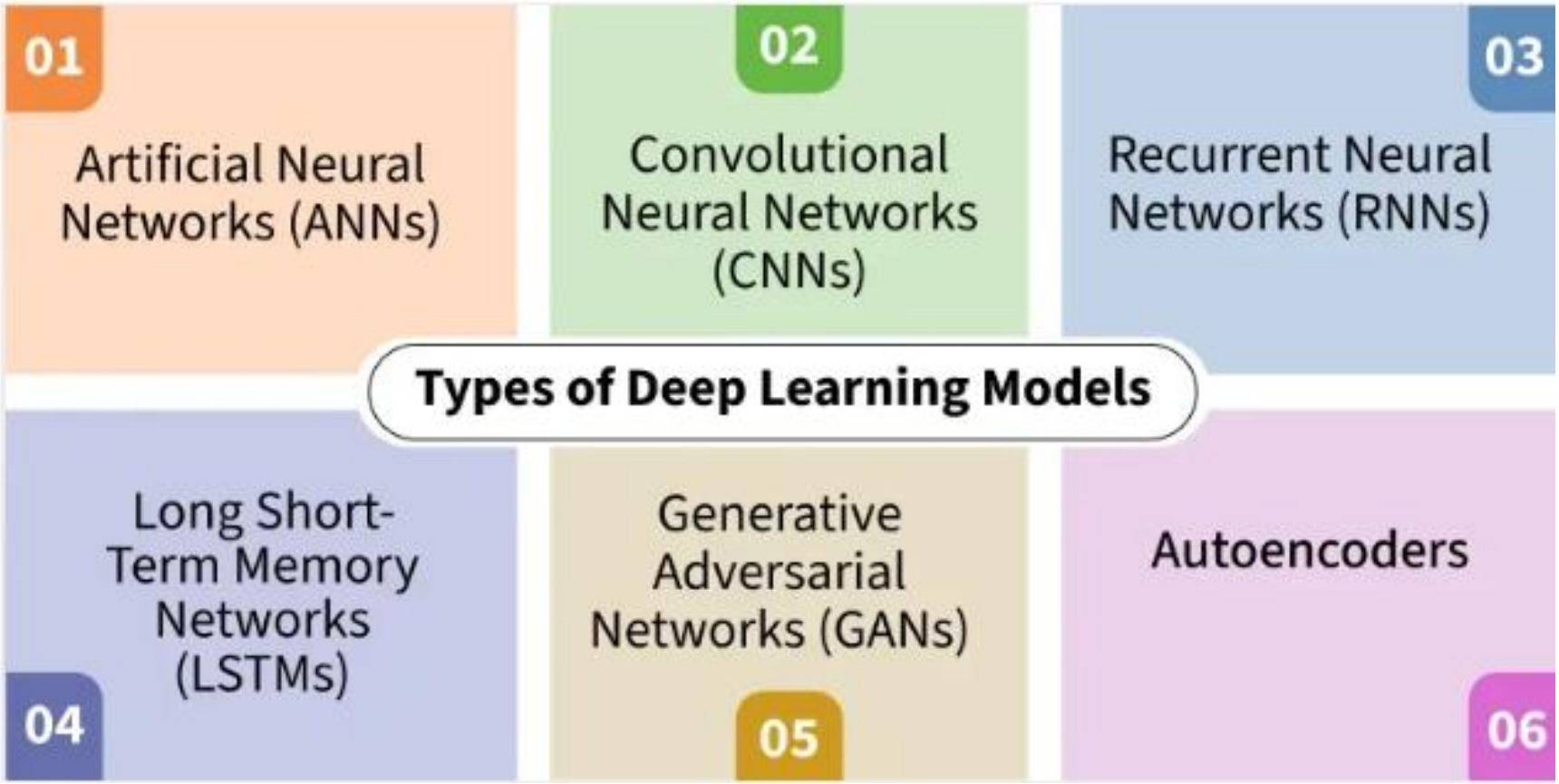




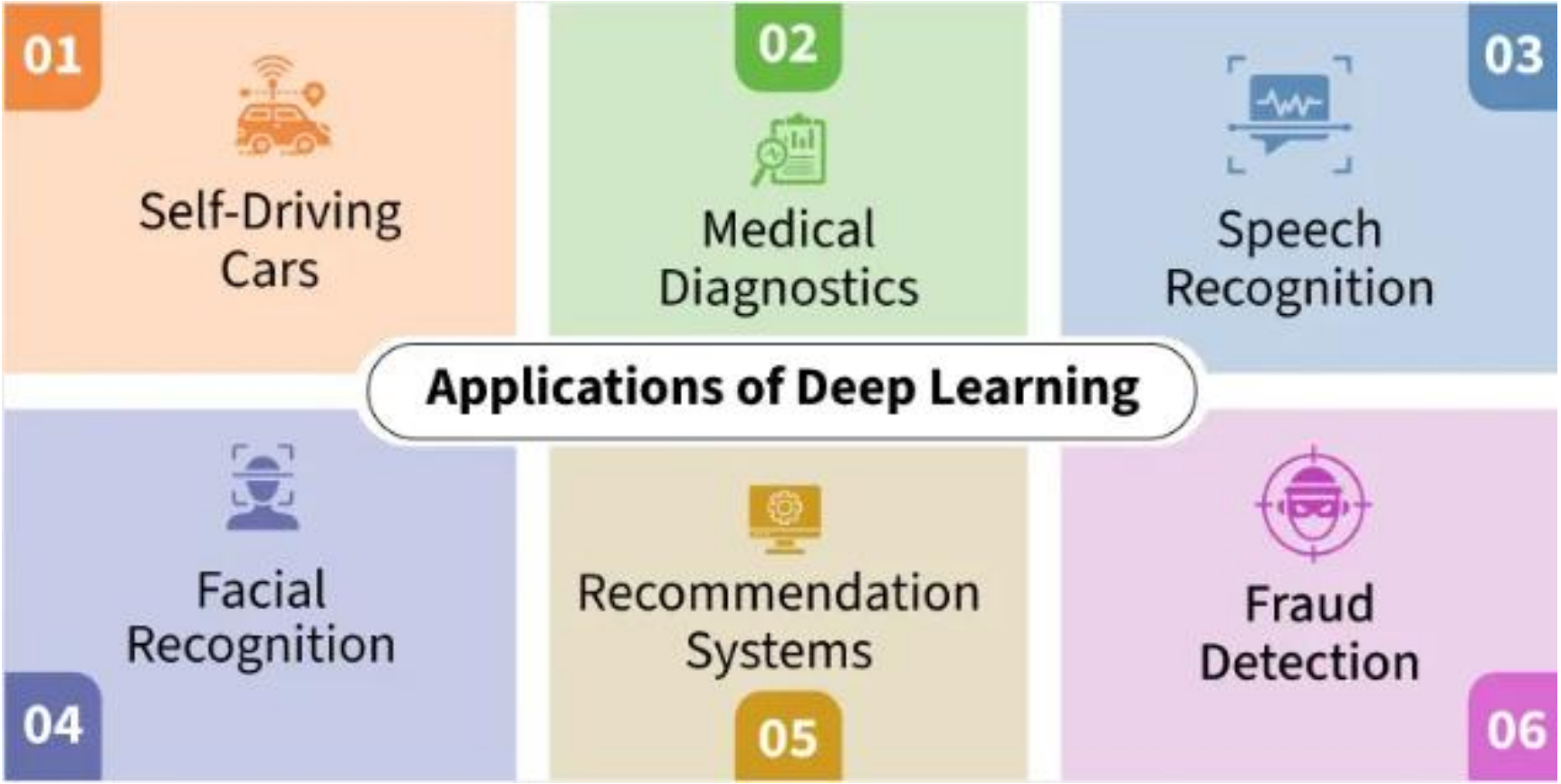
Key Components of Deep Learning



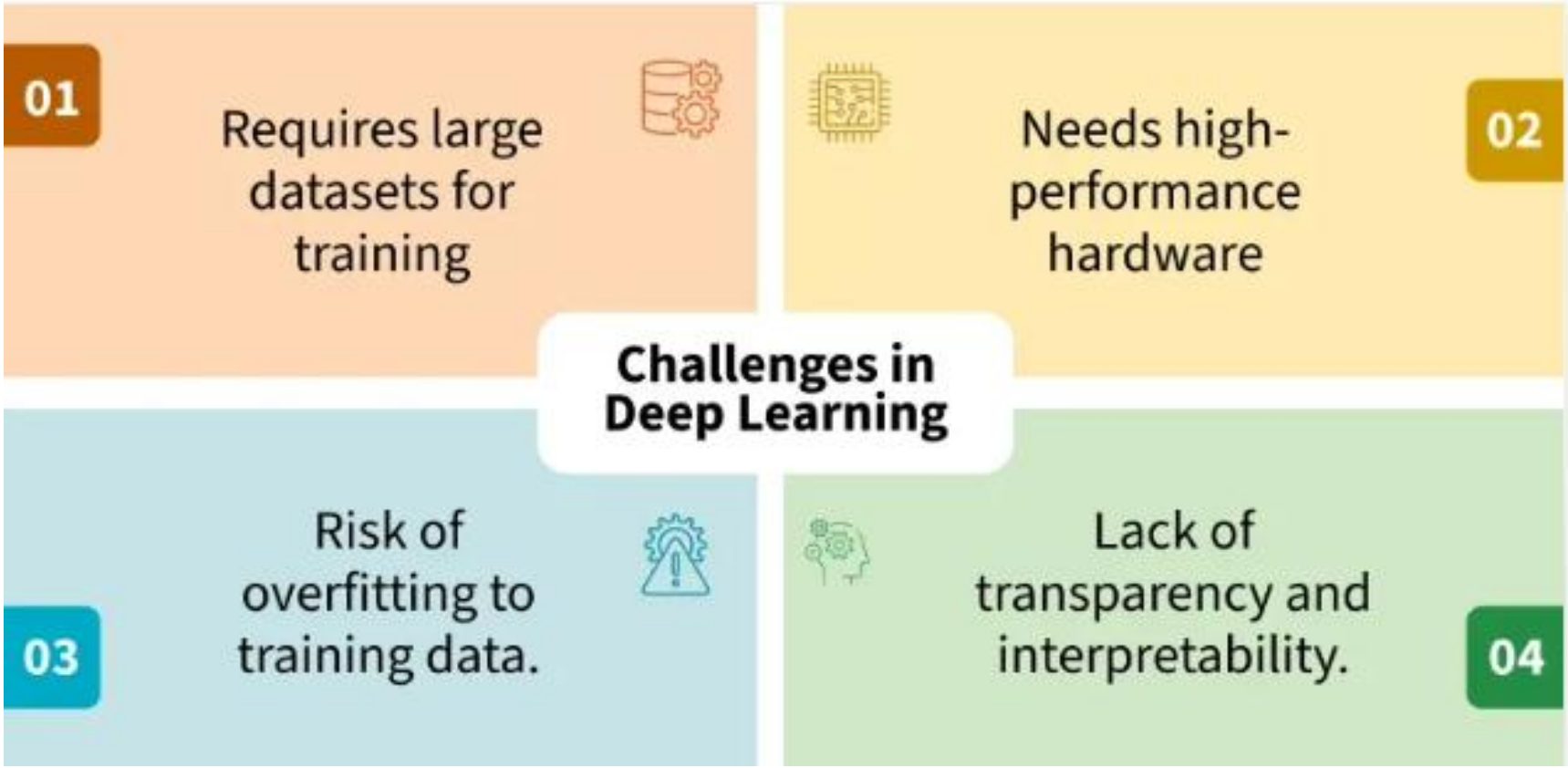
Types of Deep Learning



Deep Learning Applications



Deep Learning Challenges



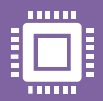
Generative AI



Purpose: To create new content—text, images, audio, code, and more—based on learned patterns.



Examples: ChatGPT, DALL·E, GitHub Copilot, Midjourney.

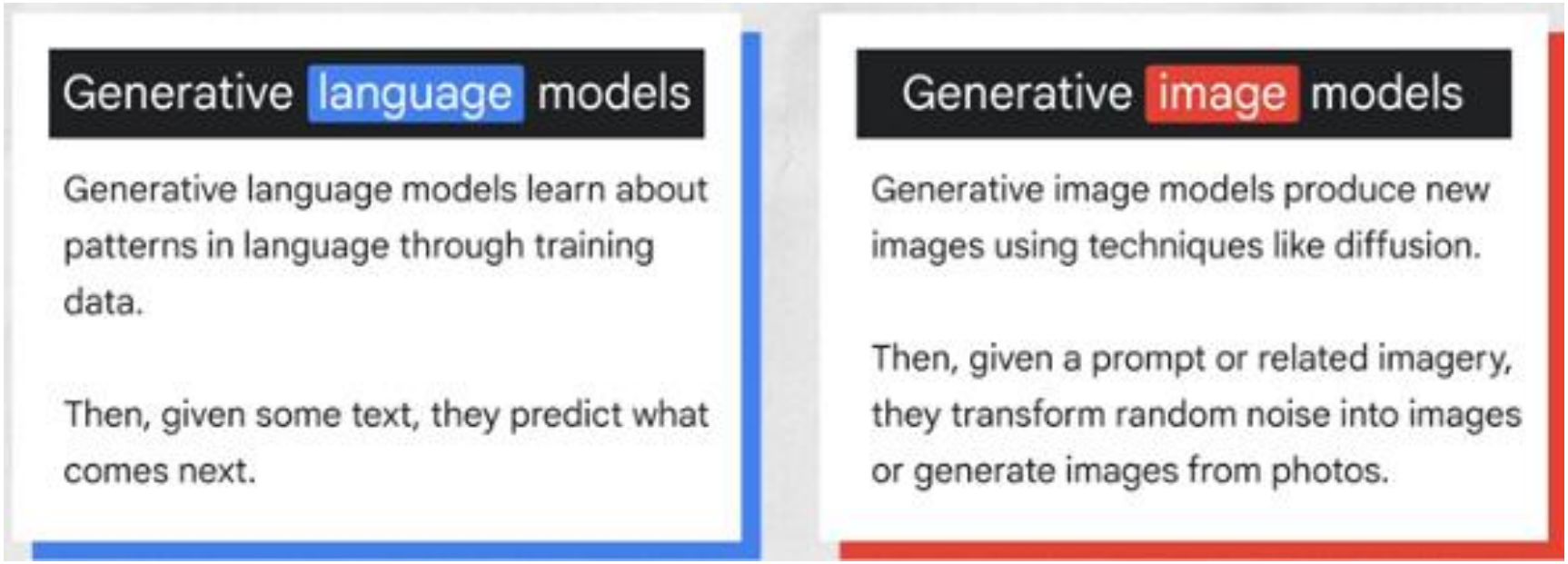


Use Cases: Content creation, code generation, design, simulations, personalization.

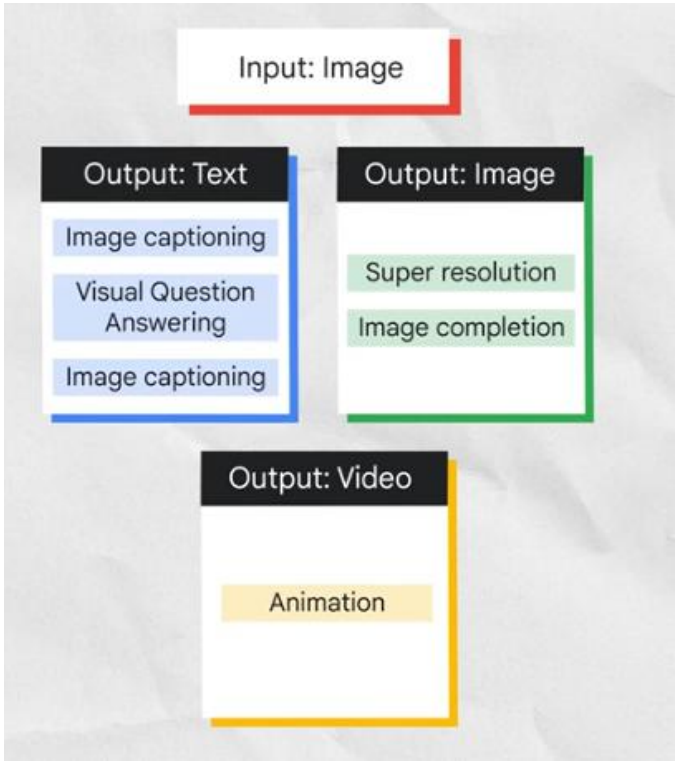


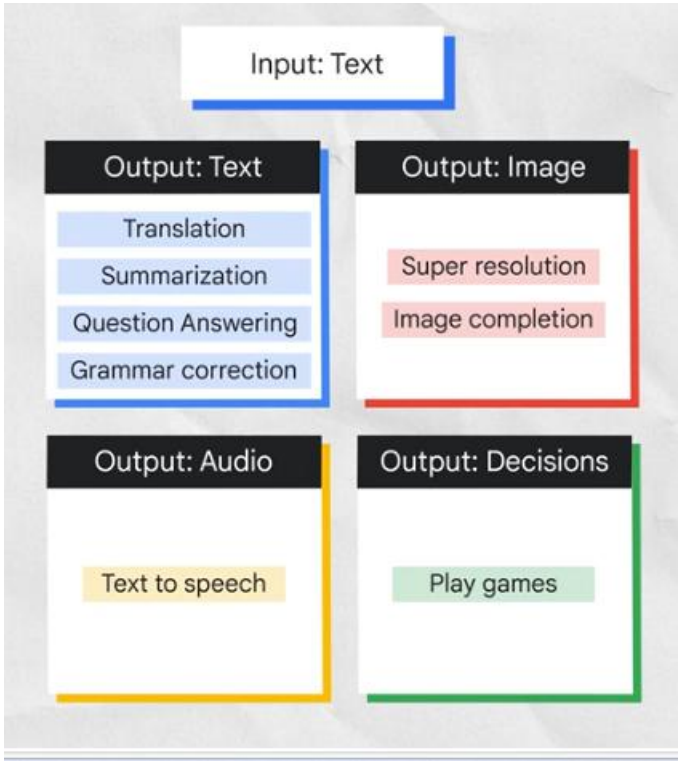
Key Technologies: Large Language Models (LLMs), Diffusion Models, Transformers.

Generative AI

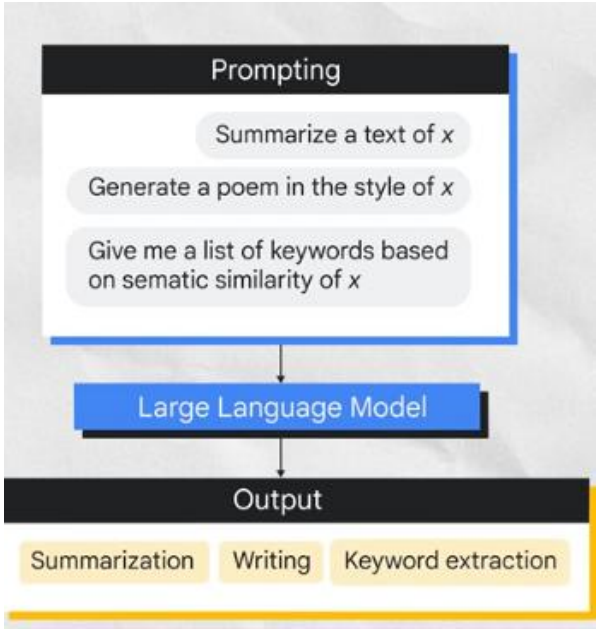


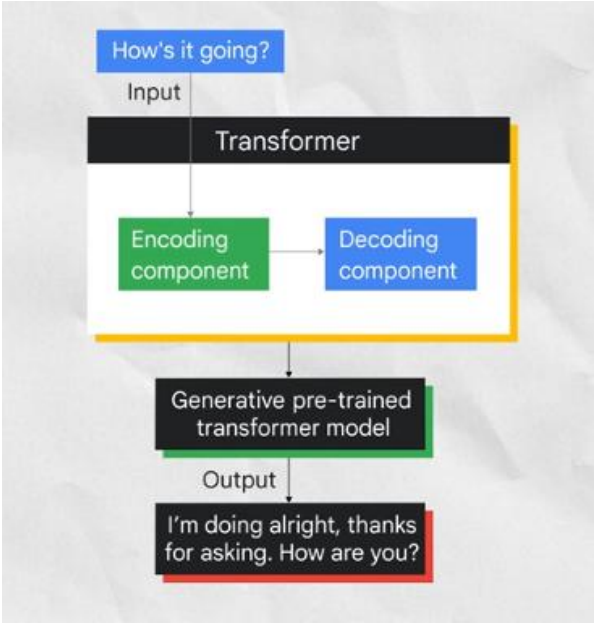
Generative AI

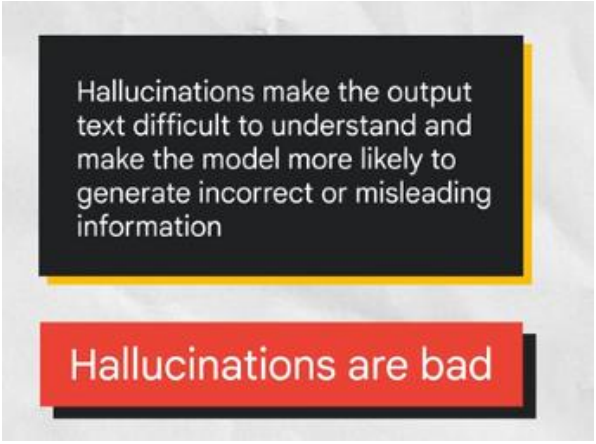




Transformers, Hallucinations & Prompts

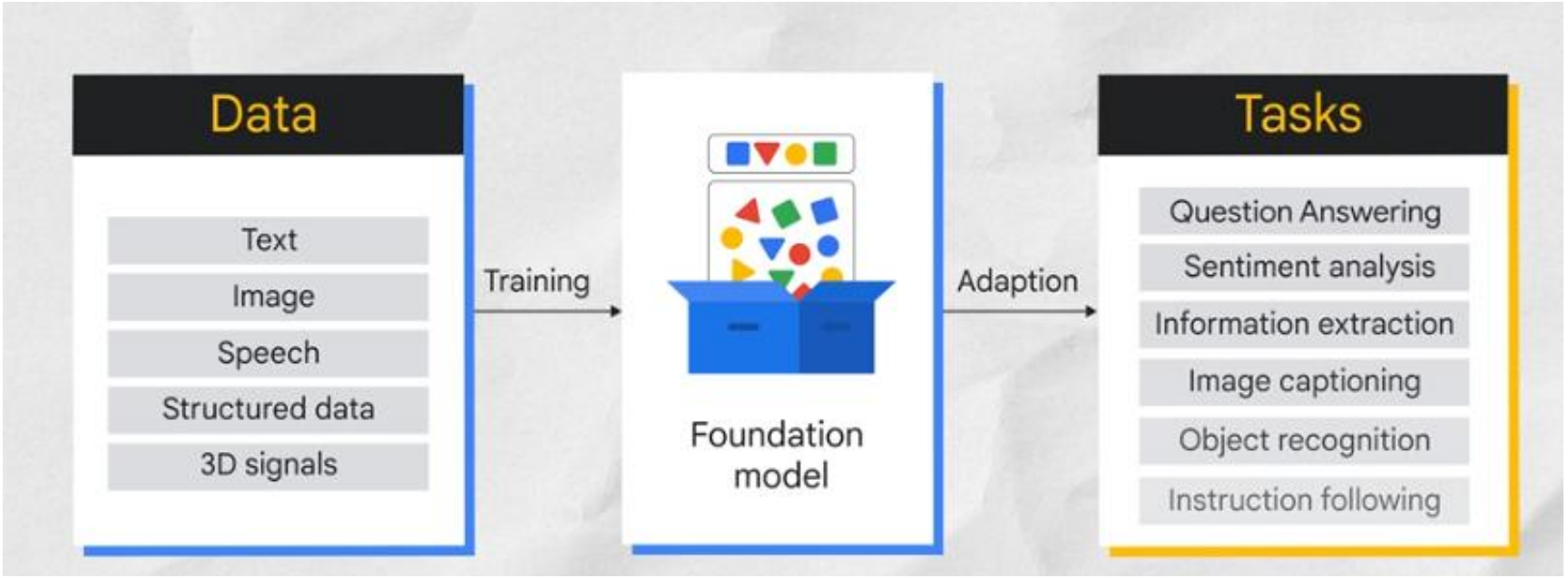




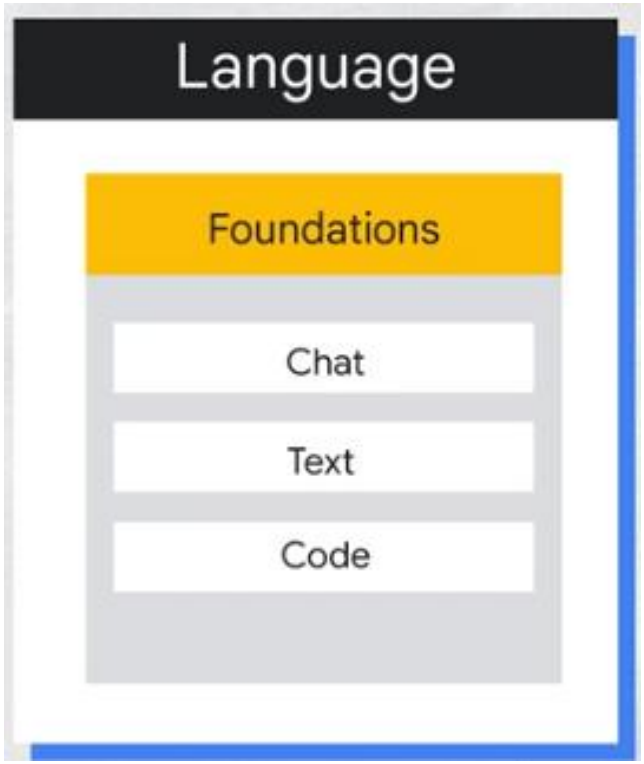


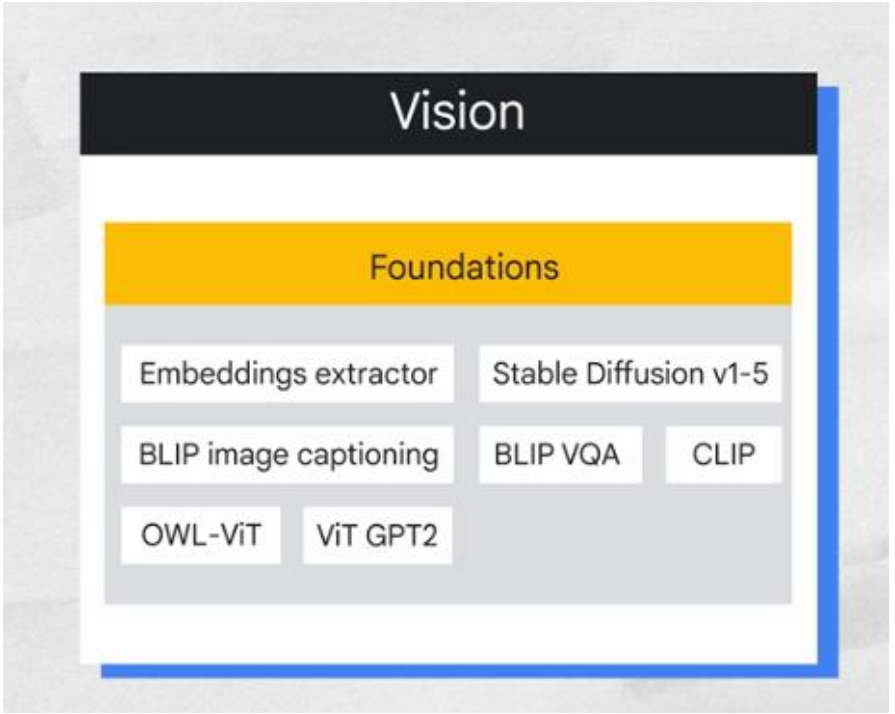
Foundation Model Generative AI

- **Foundation models** are the backbone of modern **Generative AI**. They are trained on massive datasets and can be adapted to perform various tasks across domains — from generating text and images to understanding speech and code. Their versatility and scalability make them foundational to many AI applications.

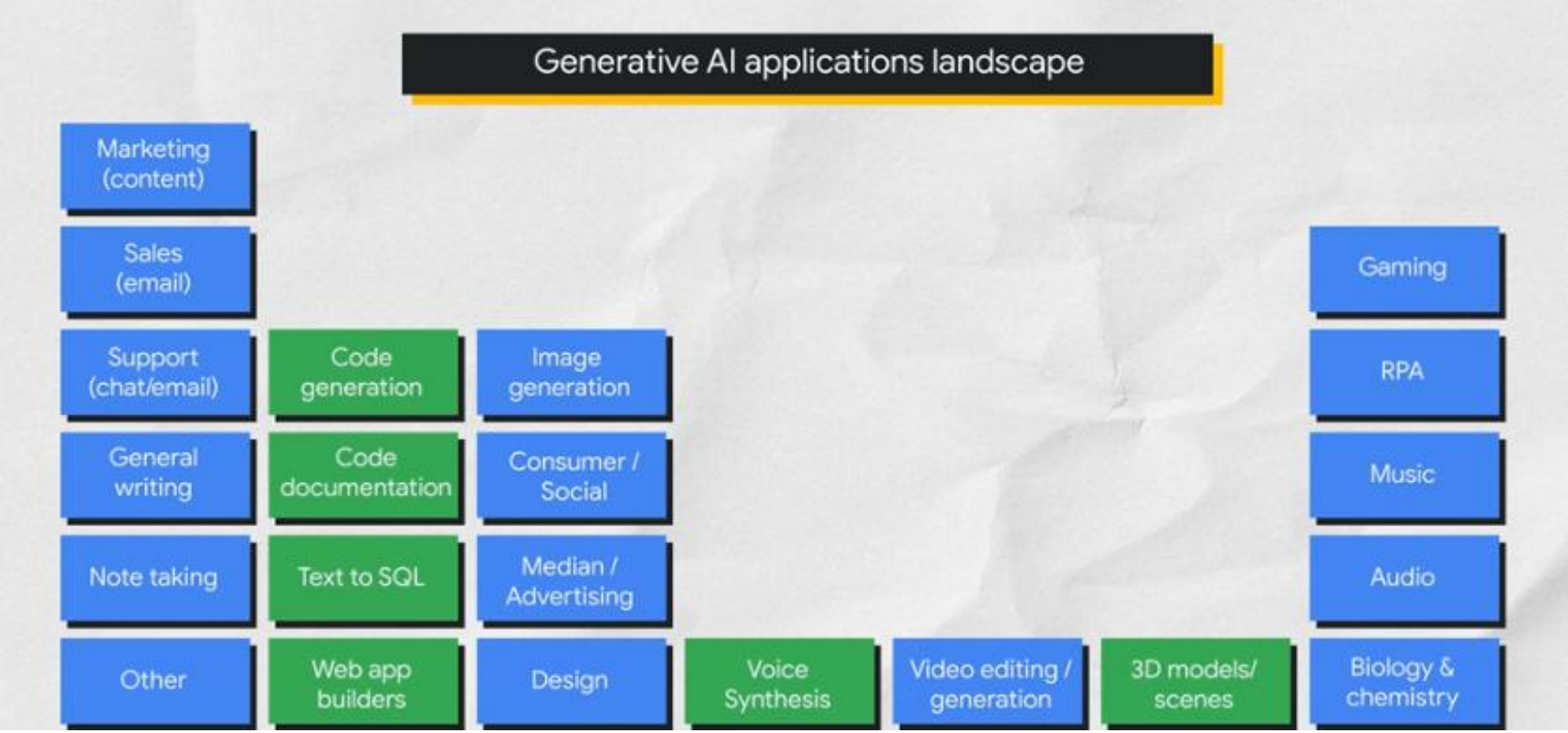


Foundation Model Generative AI





Generative AI Applications Landscape

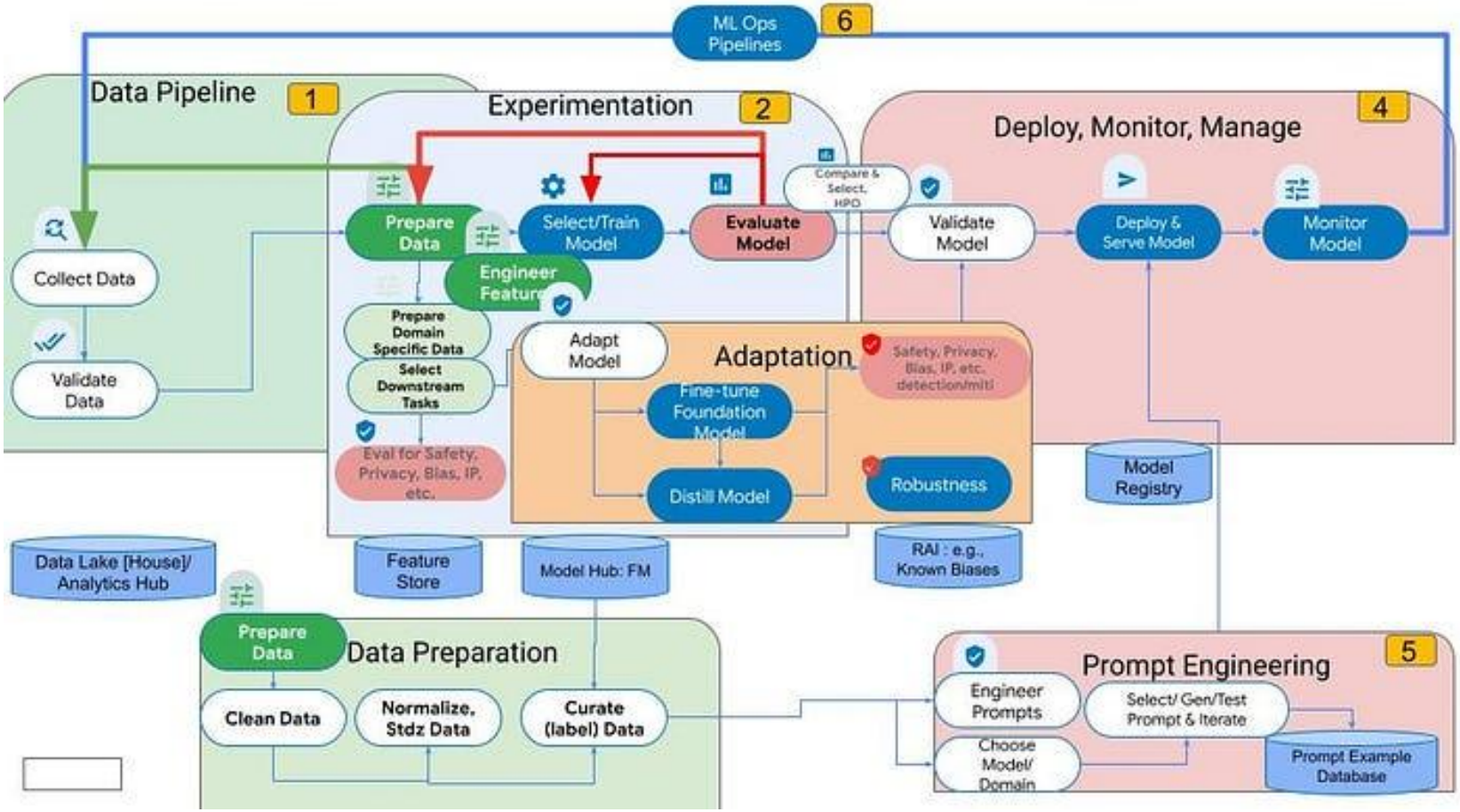


Predictive AI

- **Purpose**: To forecast future outcomes based on historical data.
- **Examples**: Demand forecasting, fraud detection, predictive maintenance.
- **Use Cases**: Business intelligence, healthcare diagnostics, financial risk modeling.
- **Key Technologies**: Machine Learning (regression, classification), Time Series Analysis, Ensemble Models.



Generative AI Lifecycle



https://dr-arsanjani.medium.com/the-generative-ai-life-cycle-fb2271a70349

End-to-End ML Lifecycle Overview

1. Data Pipeline

Collect Data: Gather raw data from various sources (e.g., databases, APIs, sensors).

Validate Data: Ensure completeness, consistency, and quality before use.

2. Experimentation

Prepare Data:

Tailor data for specific domains.

Select downstream tasks (e.g., classification, forecasting).

Evaluate for safety, privacy, bias, and intellectual property (IP) risks.

Engineer Features: Create meaningful inputs for models.

Select/Train Model: Choose appropriate algorithms and train them.

Evaluate Model: Assess performance using metrics and validation techniques.

3. Adaptation

Adapt Model:

Fine-tune Foundation Models for specific use cases.

Distill Models to simplify and optimize.

Enhance **robustness** by detecting and mitigating safety, privacy, bias, and IP issues.

4. Deploy, Monitor, Manage

Validate Model: Compare models and select the highest performing design (HPD).

Deploy & Serve Model: Use model registry for deployment.

Monitor Model: Track performance, drift, and reliability in production.

5. Prompt Engineering

Engineer Prompts: Design prompts tailored to the model and domain.

Select/Generate/Test Prompts: Use a prompt example database to iterate and refine.

6. ML Ops Pipelines

Automate and orchestrate the entire ML lifecycle, ensuring scalability, reproducibility, and governance.

Language Model

What is a language model?

A language model is a machine learning model that aims to predict and generate plausible language. Autocomplete is a language model, for example.

These models work by estimating the probability of a token or sequence of tokens occurring within a longer sequence of tokens. Consider the following sentence:

```
When I hear rain on my roof, I _______ in my kitchen.
```

If you assume that a token is a word, then a language model determines the probabilities of different words or sequences of words to replace that underscore. For example, a language model might determine the following probabilities:

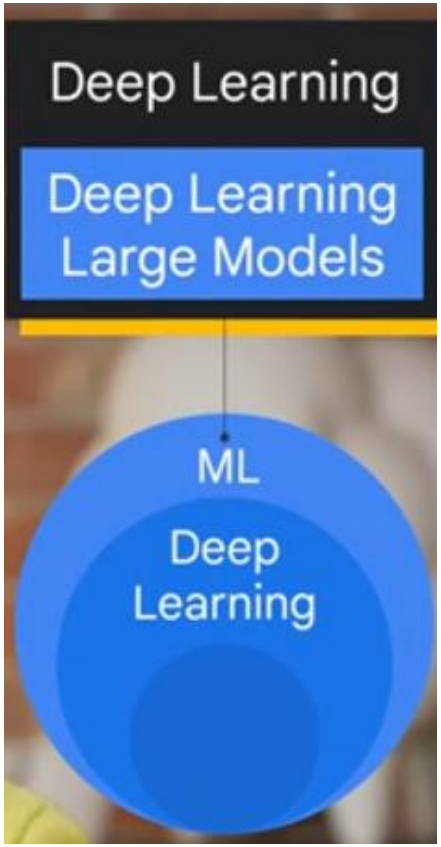
```
cook soup 9.4%
warm up a kettle 5.2%
cower 3.6%
nap 2.5%
relax 2.2%
...
```

A "sequence of tokens" could be an entire sentence or a series of sentences. That is, a language model could calculate the likelihood of different entire sentences or blocks of text.

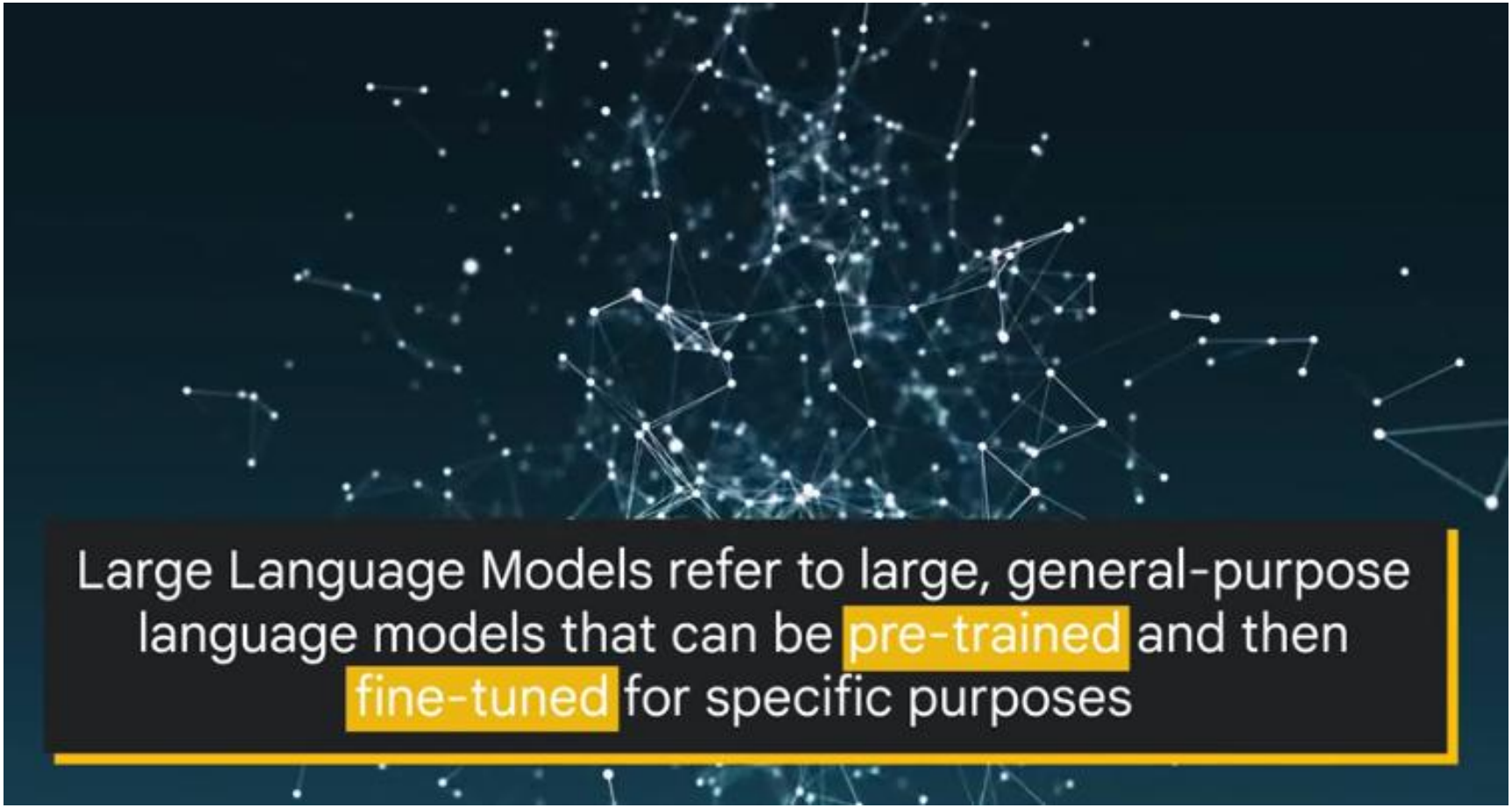
Estimating the probability of what comes next in a sequence is useful for all kinds of things: generating text, translating languages, and answering questions, to name a few.

Large Language Model

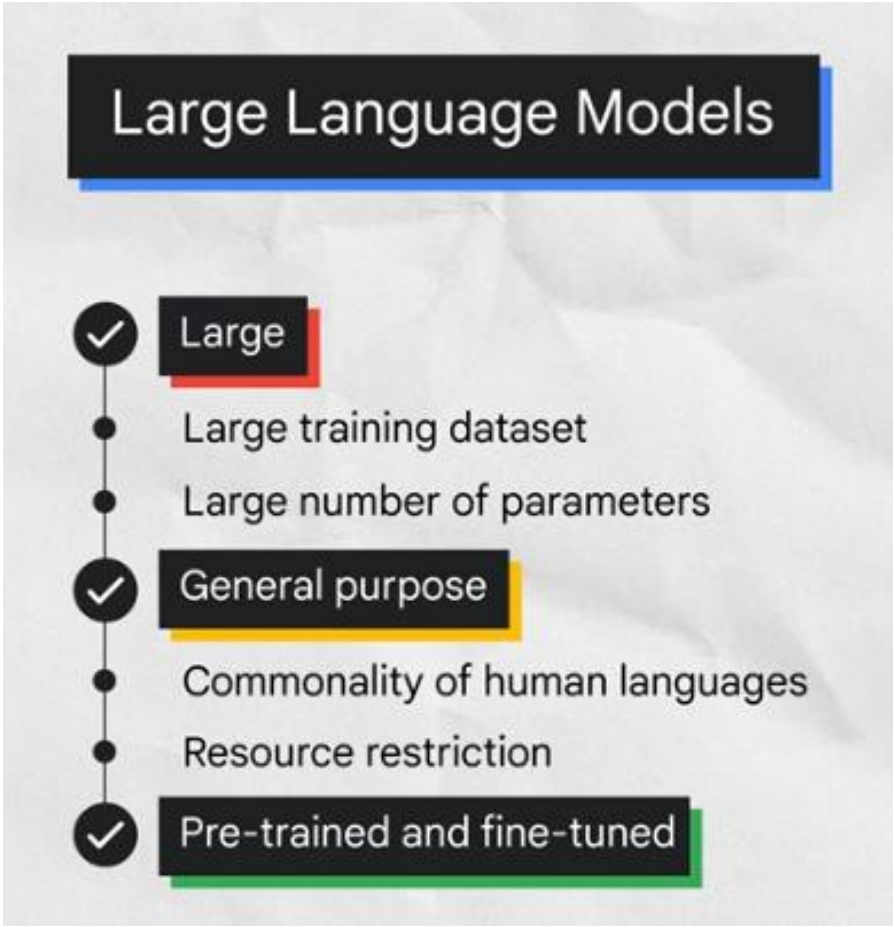
Large Language Models (LLMs) are a subset of Deep Learning



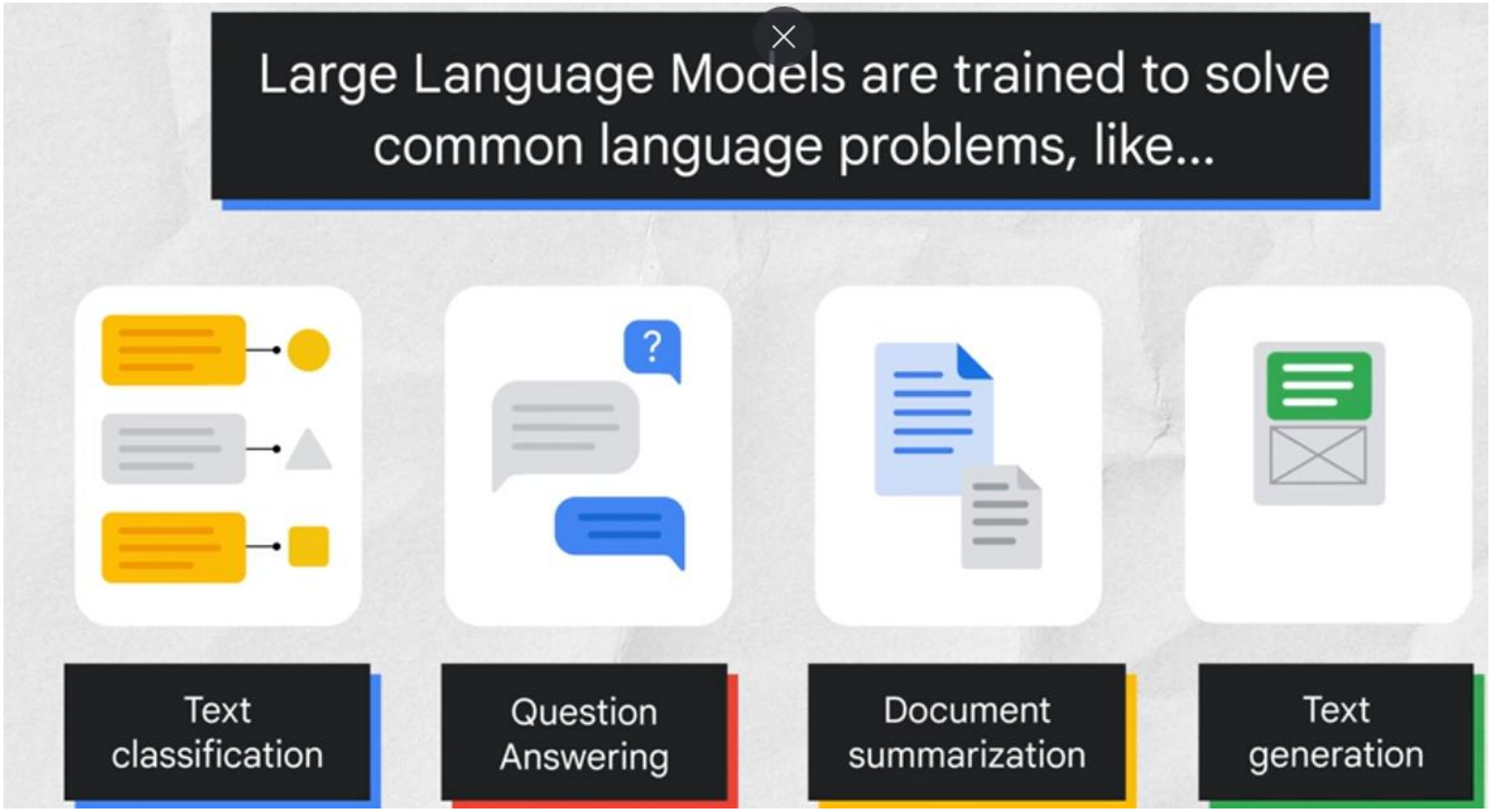
Large Language Model



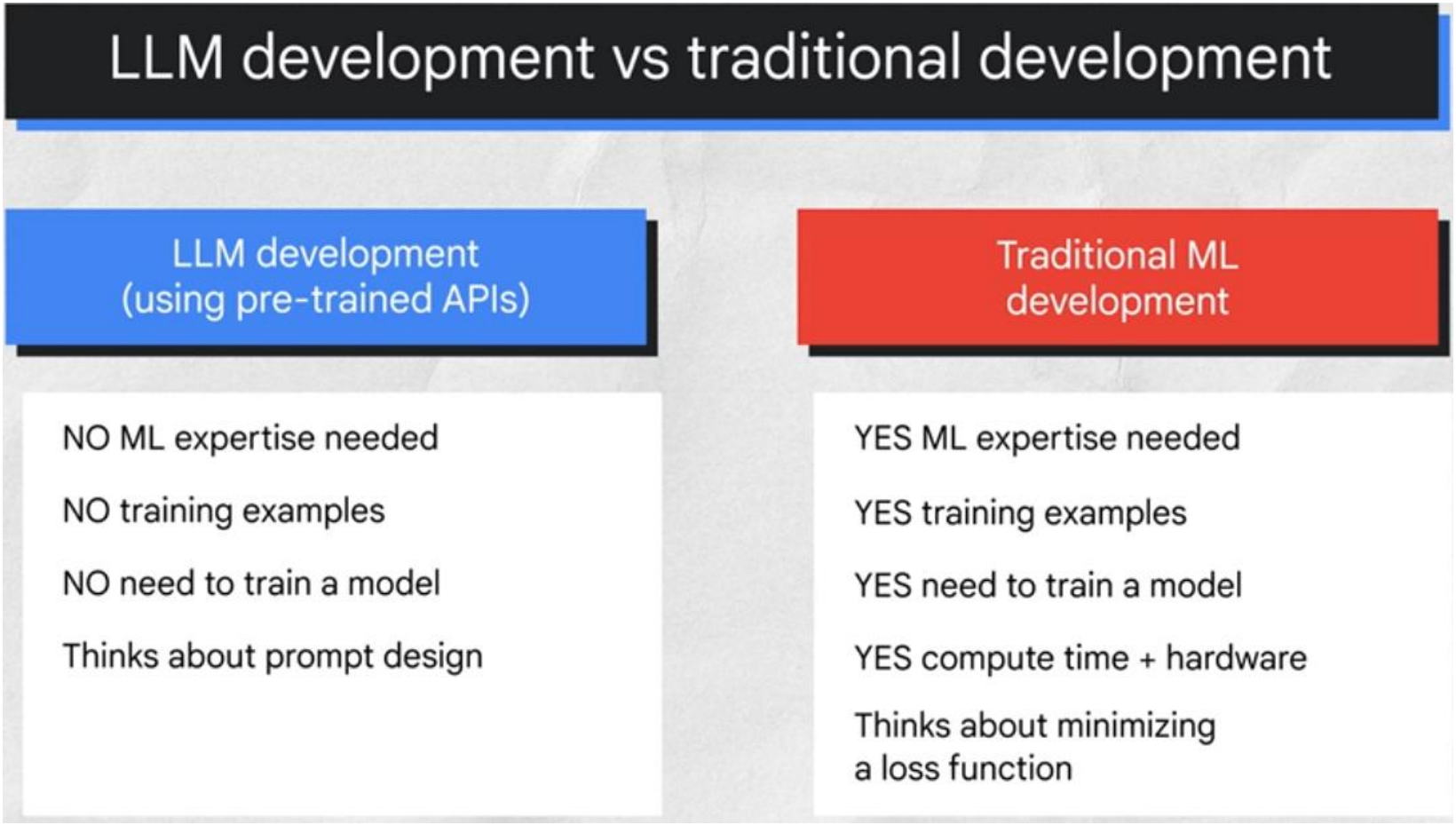
Large Language Model



LLM's


Large Language Models are trained to solve common language problems, like...
Text classification
Question Answering
Document summarization
Text generation

LLM's


LLM development vs traditional development
LLM development (using pre-trained APIs)
NO ML expertise needed
NO training examples
NO need to train a model
Thinks about prompt design
Traditional ML development
YES ML expertise needed
YES training examples
YES need to train a model
YES compute time + hardware
Thinks about minimizing a loss function

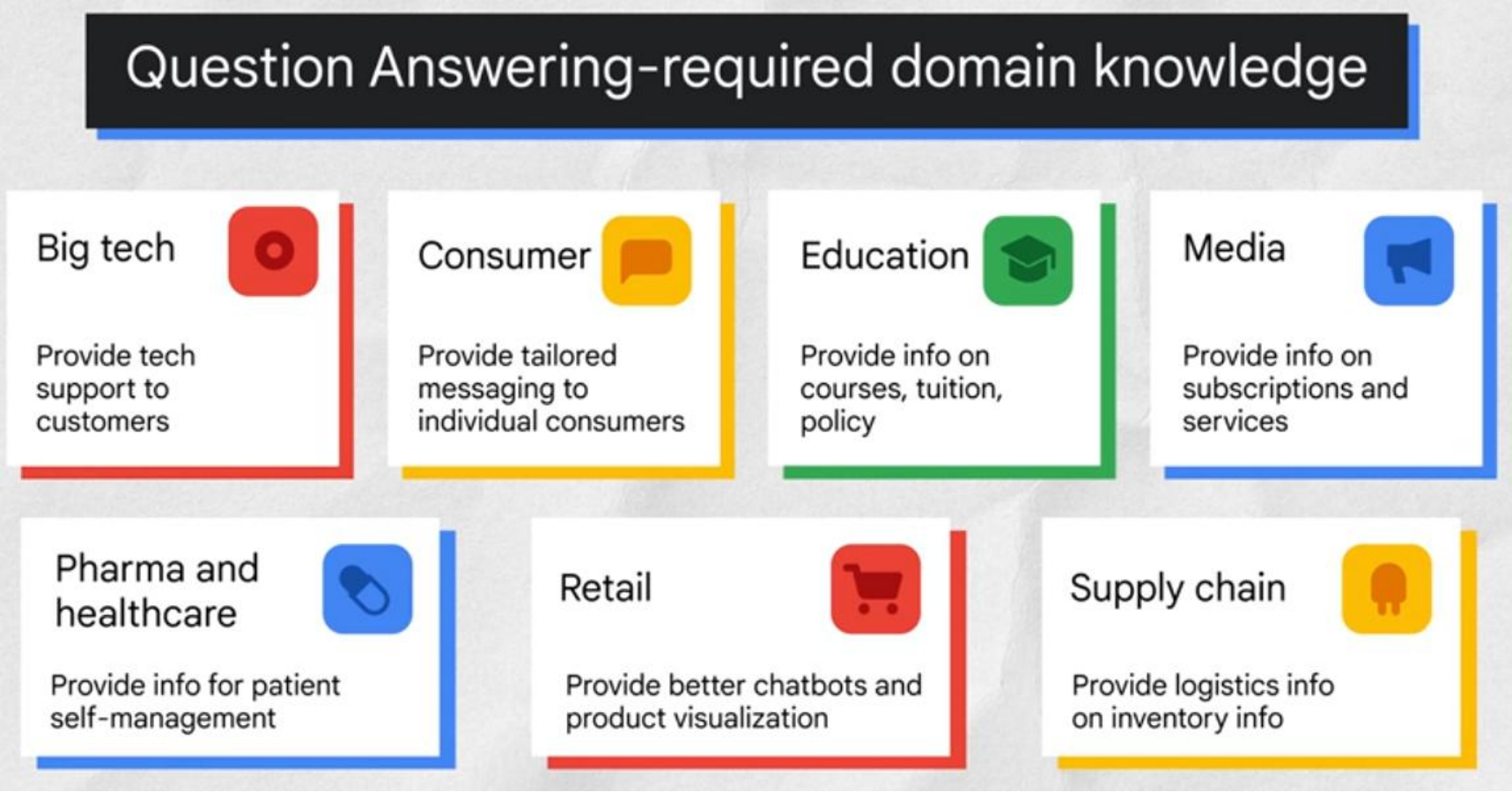
Question Answering

What is Question Answering in Natural Language Processing?

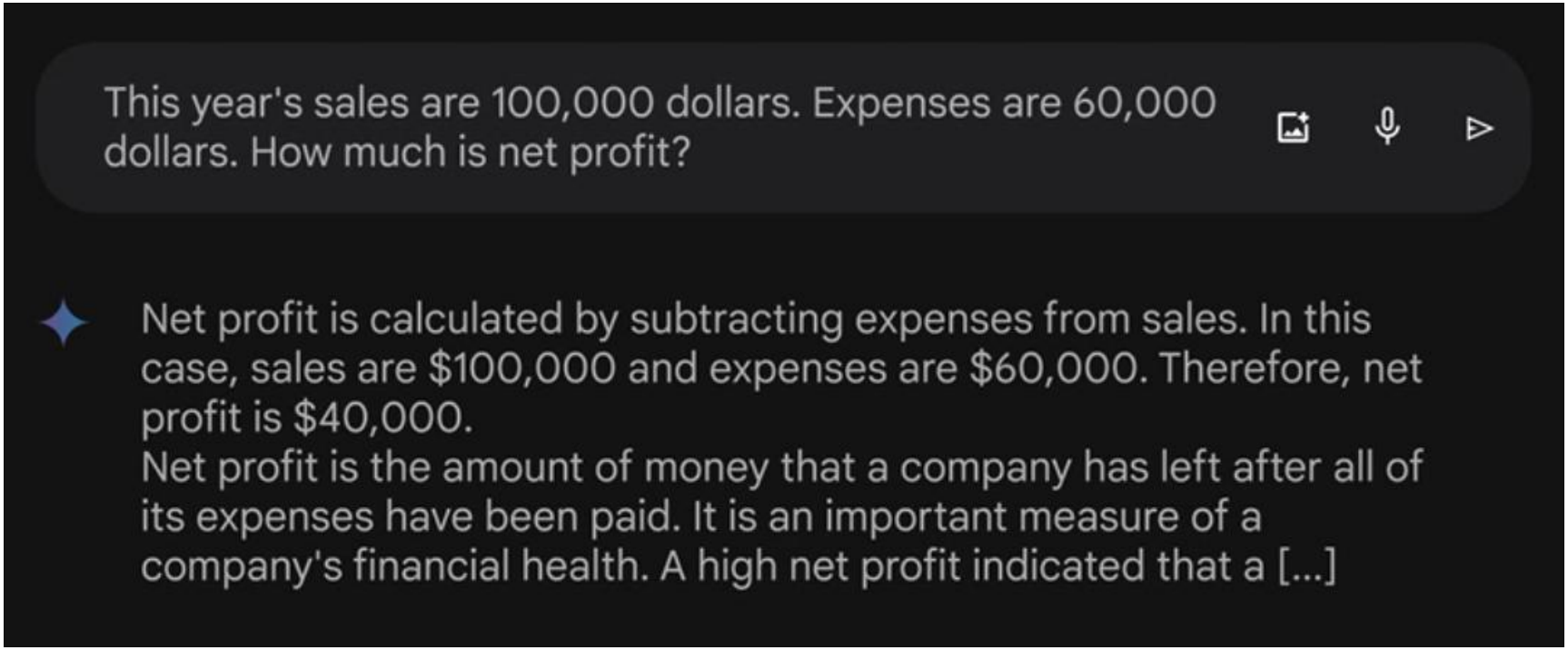
Question Answering (QA) is a subfield of Natural Language Processing that deals with the task of automatically answering questions posed in natural language

QA systems are able to answer a wide range of questions, including factual, definitional, and opinion-based questions

Question Answering

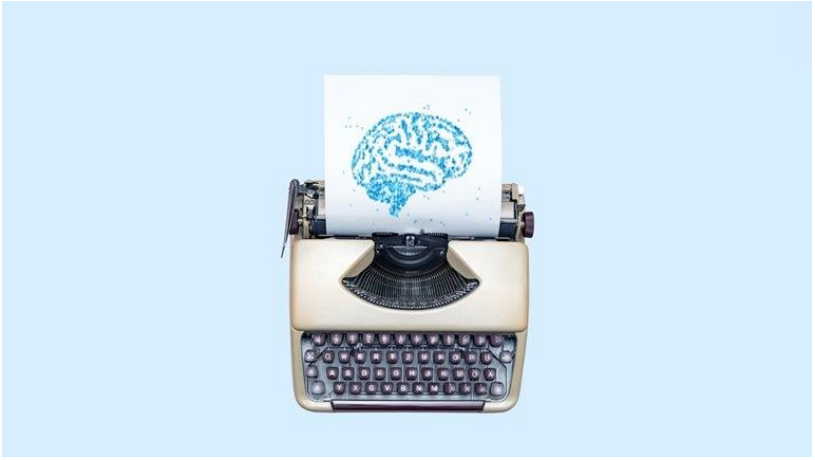


Question Answering - Gemini

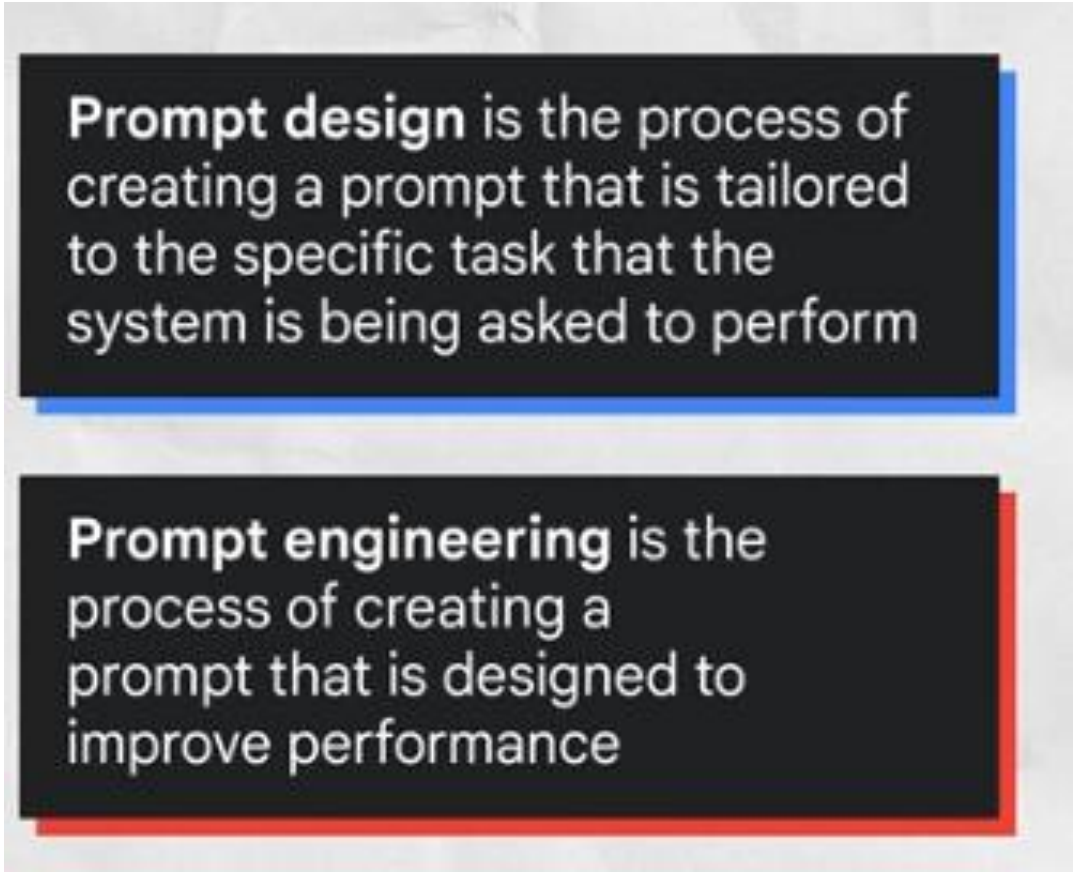


What is Prompt Engineering?

Prompt engineering is the practice of designing inputs for AI tools that will produce optimal outputs.
Just as better ingredients can make for a better dinner, better inputs into a generative AI (gen AI) model can make for better results. These inputs are called prompts, and the practice of writing them is called prompt engineering.



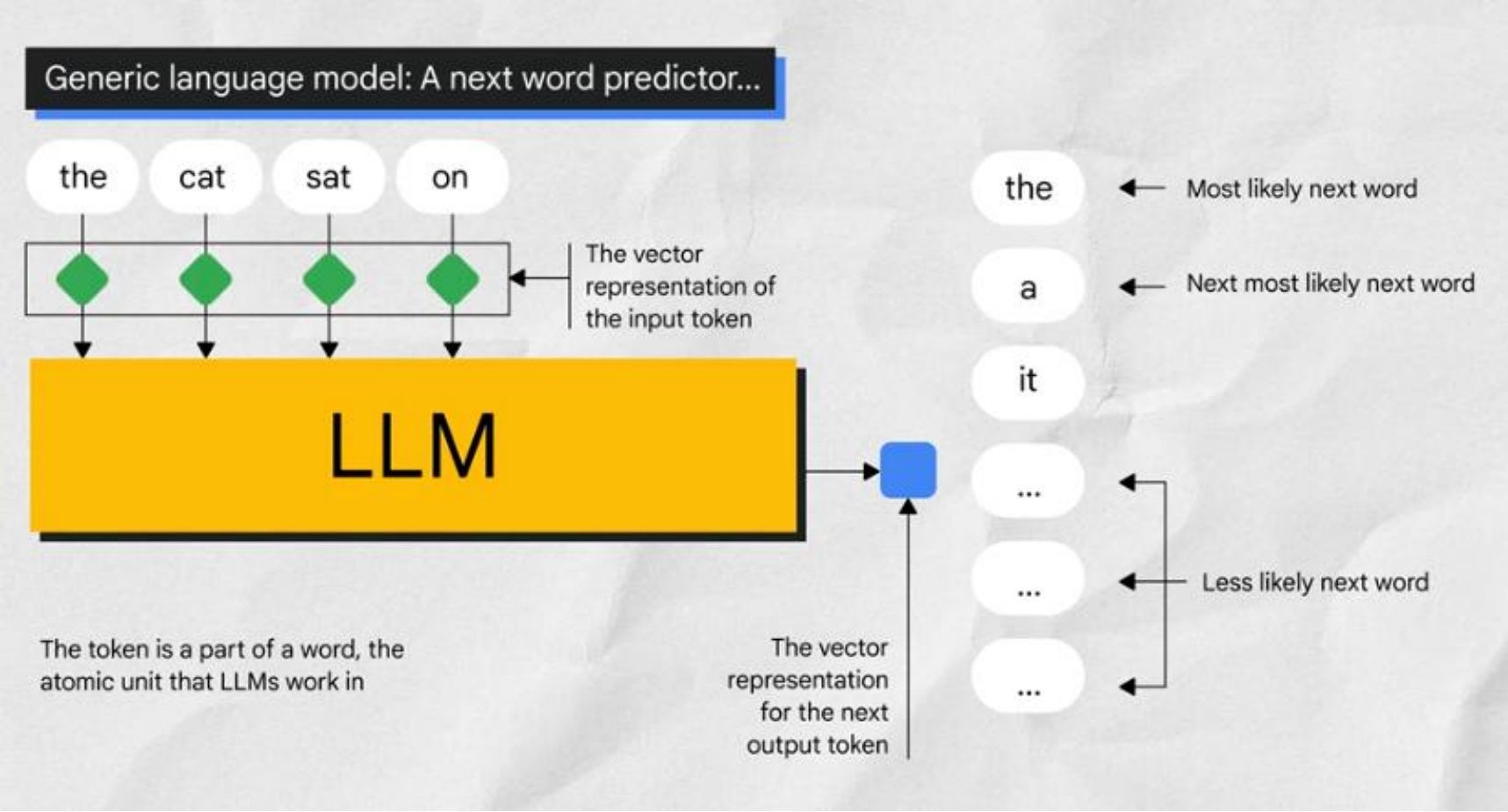
Prompt Design & Engineering



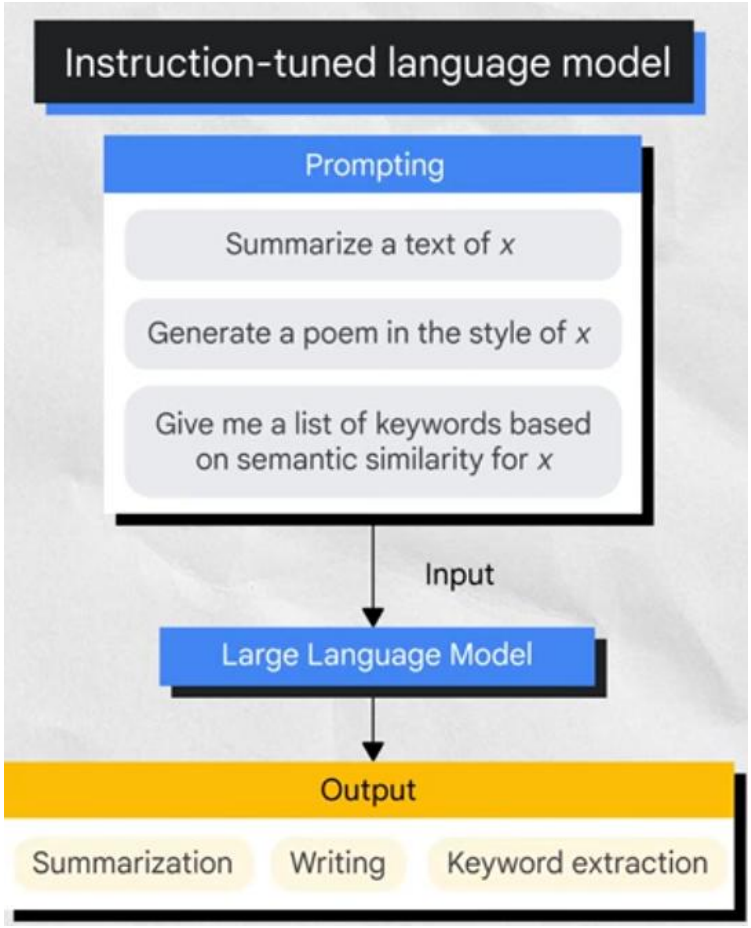
Generic Language Model

Generic (or raw) language models: Predict the next word (technically, token) based on the language in the training data

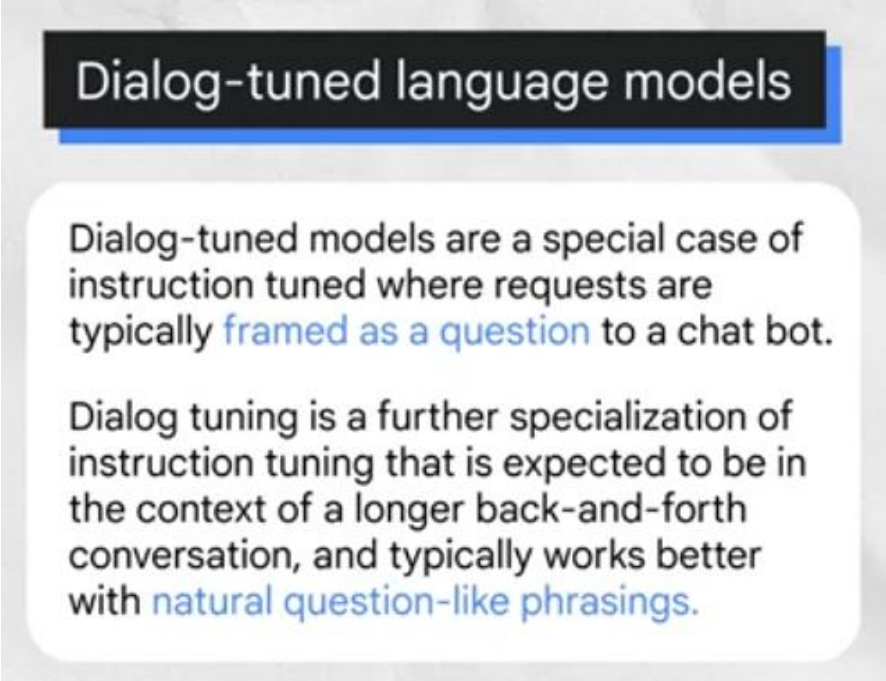
Generic Language Model



Instruction Turned Model

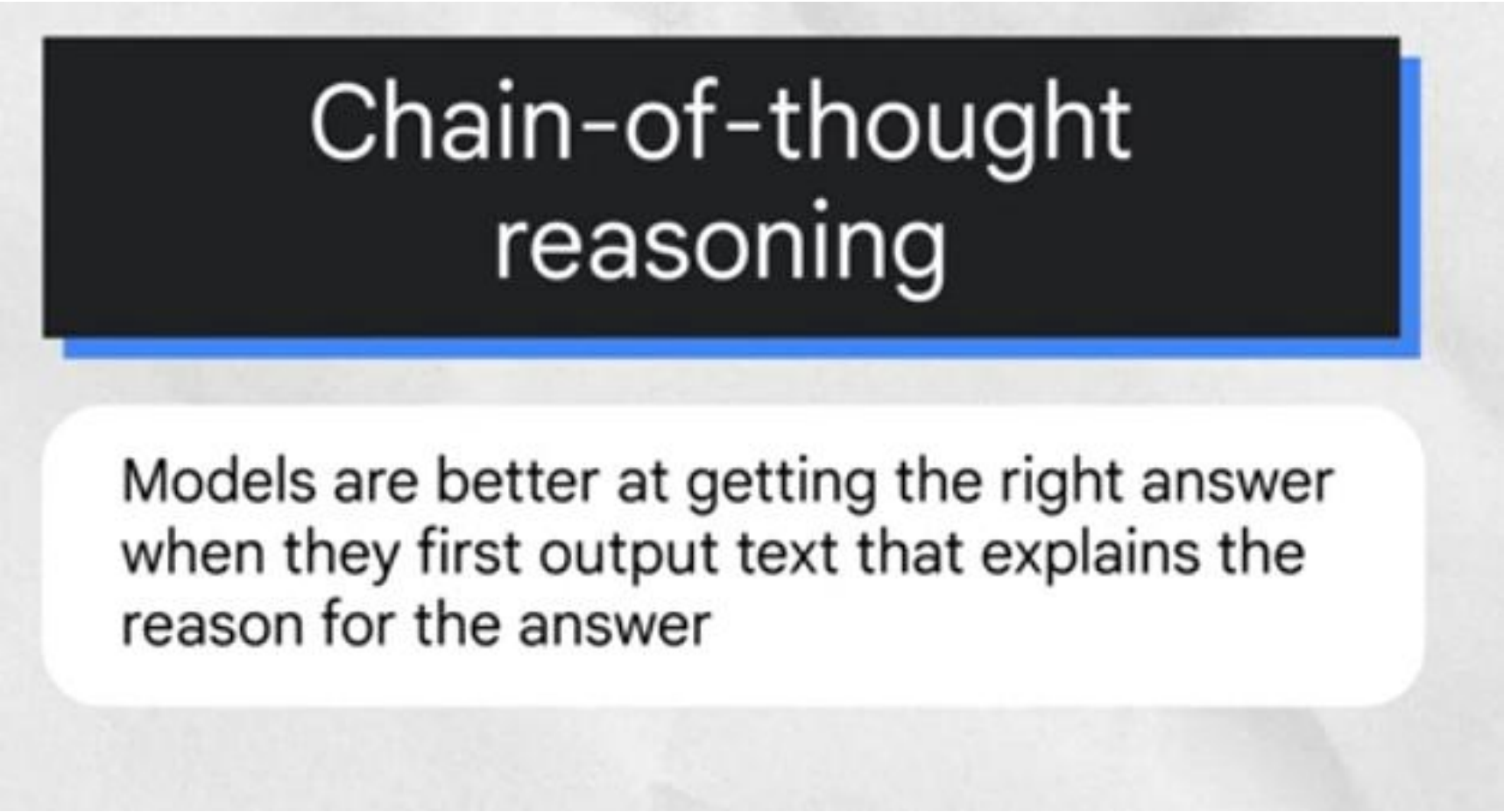


Dialog Turned Model

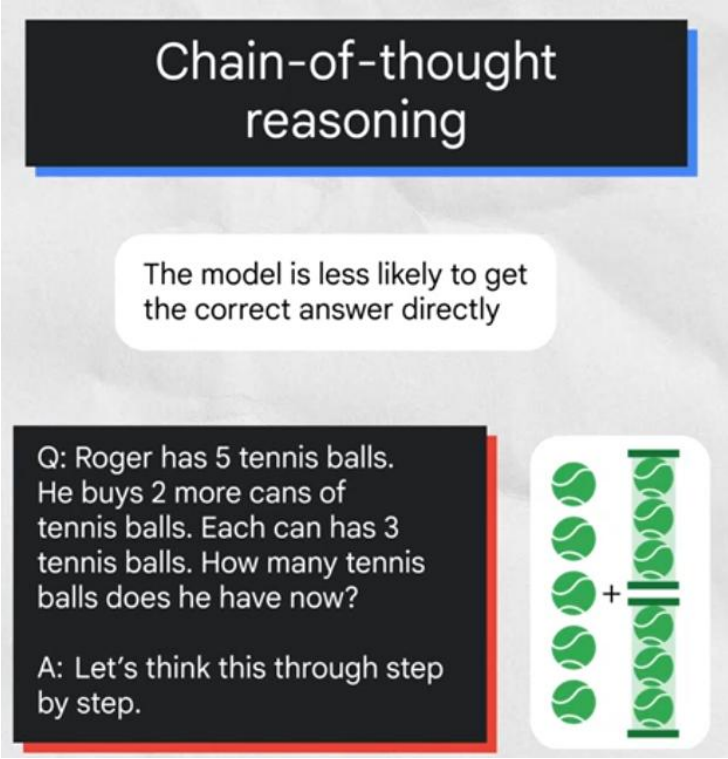


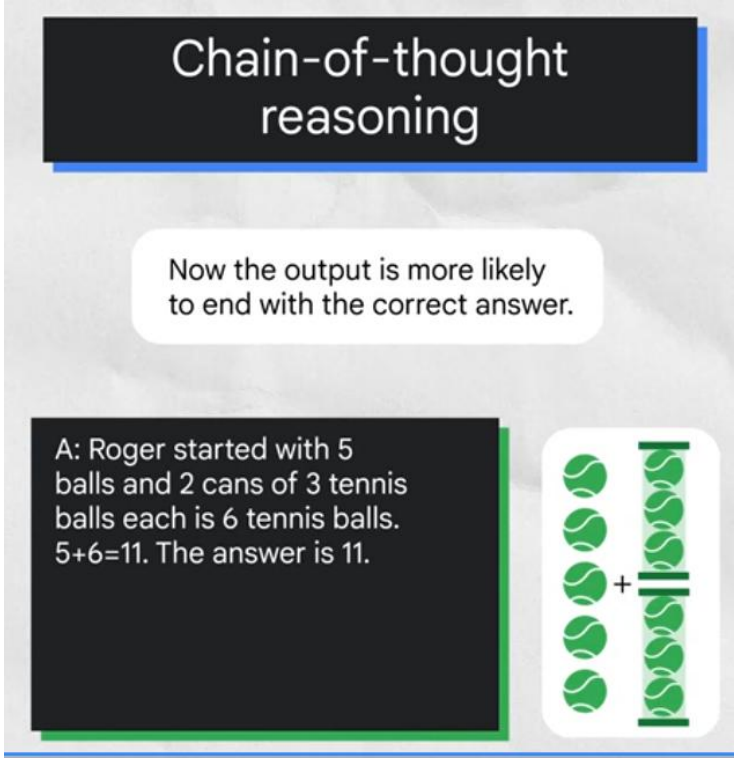
Dialog tuned: Trained to have a dialog by predicting the next response

Chain-of-thought reasoning

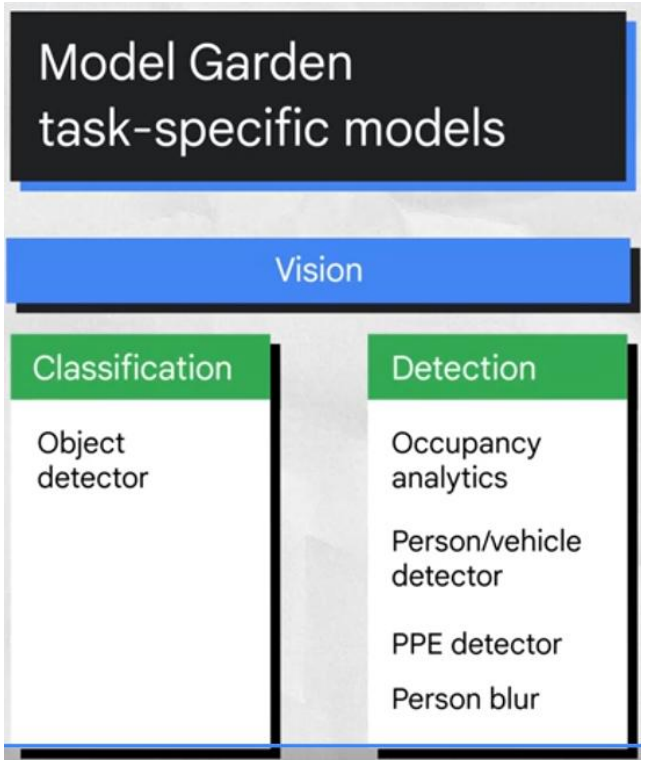


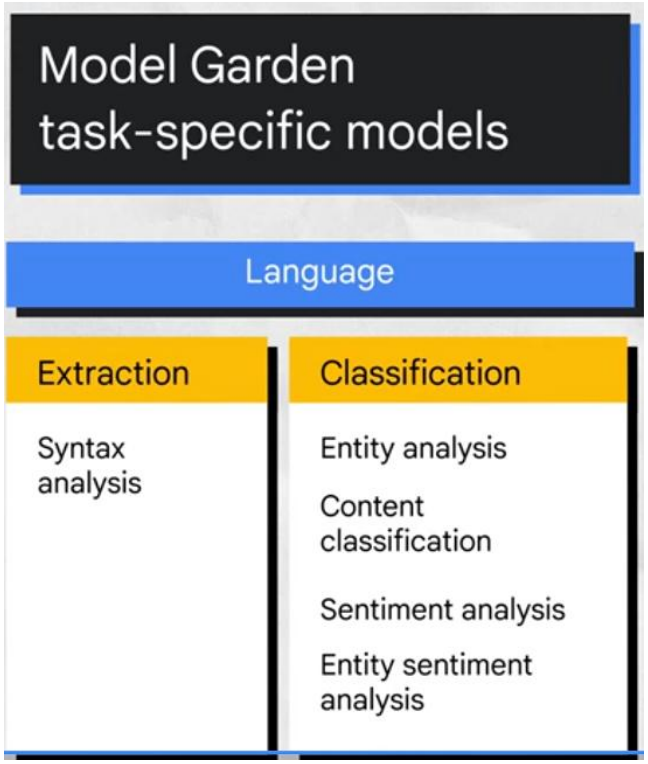
Chain-of-thought reasoning



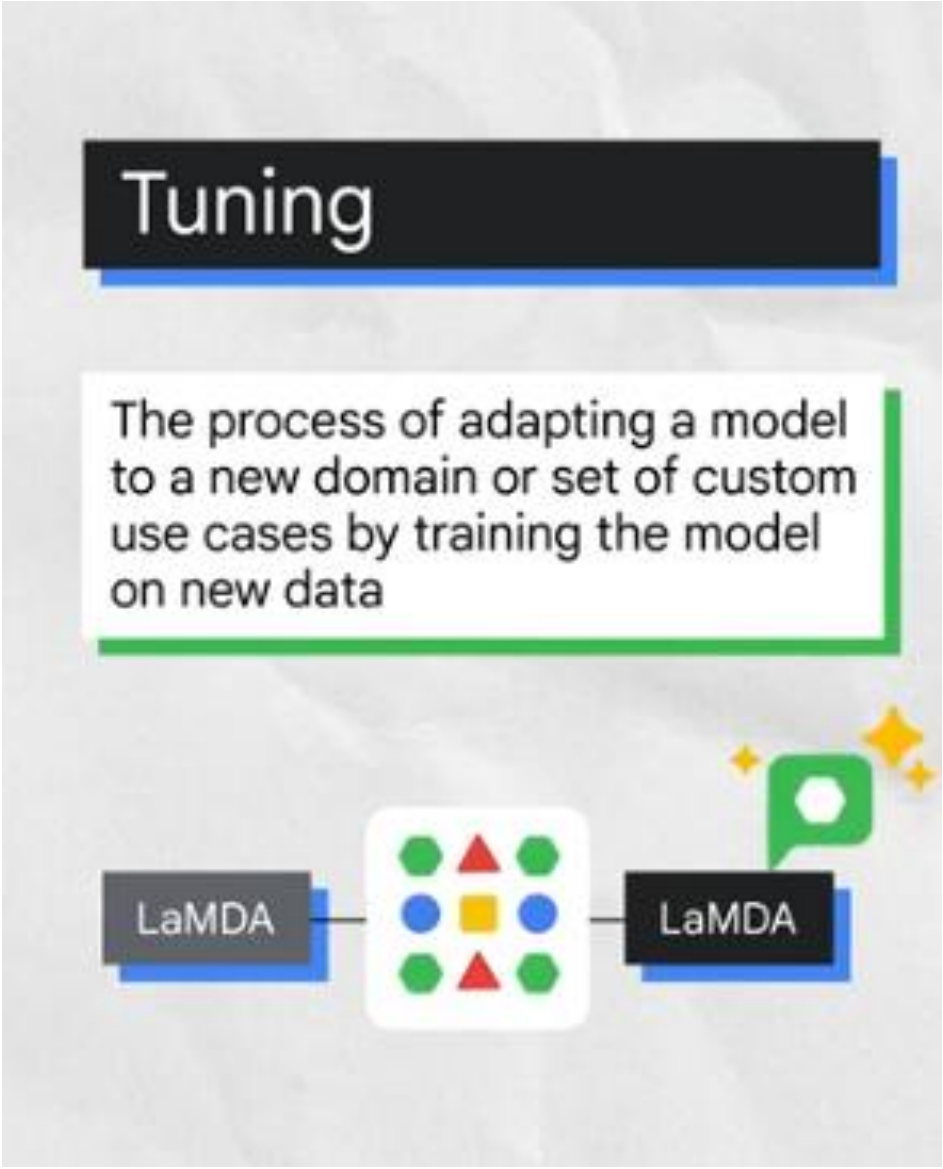


Model Garden Task Specific Models

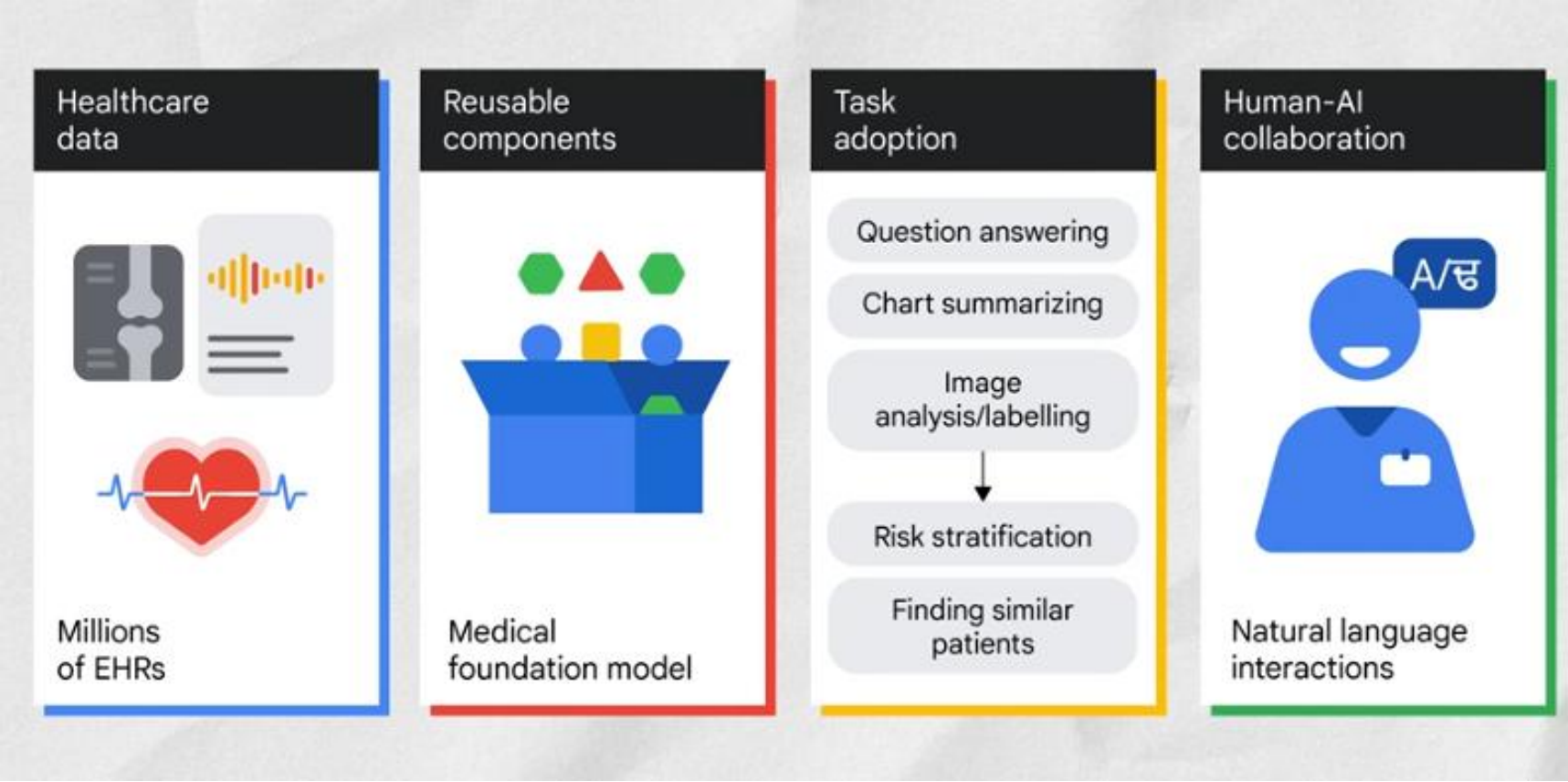




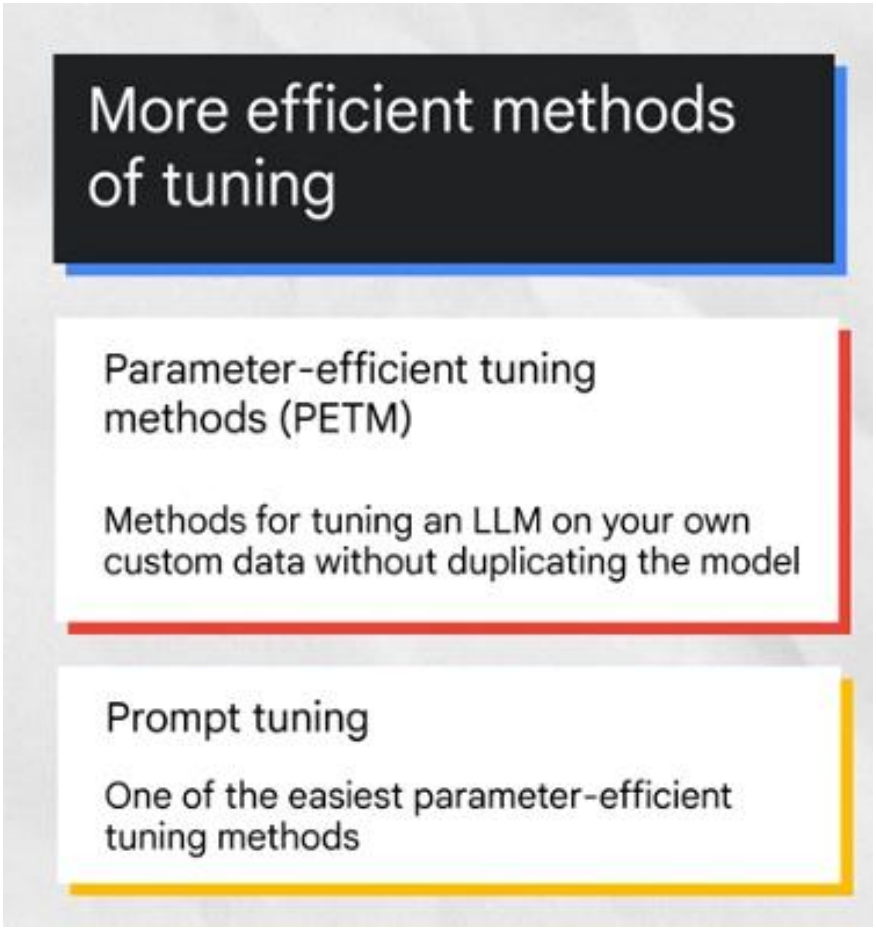
Tuning


Tuning
The process of adapting a model to a new domain or set of custom use cases by training the model on new data
LaMDA
LaMDA

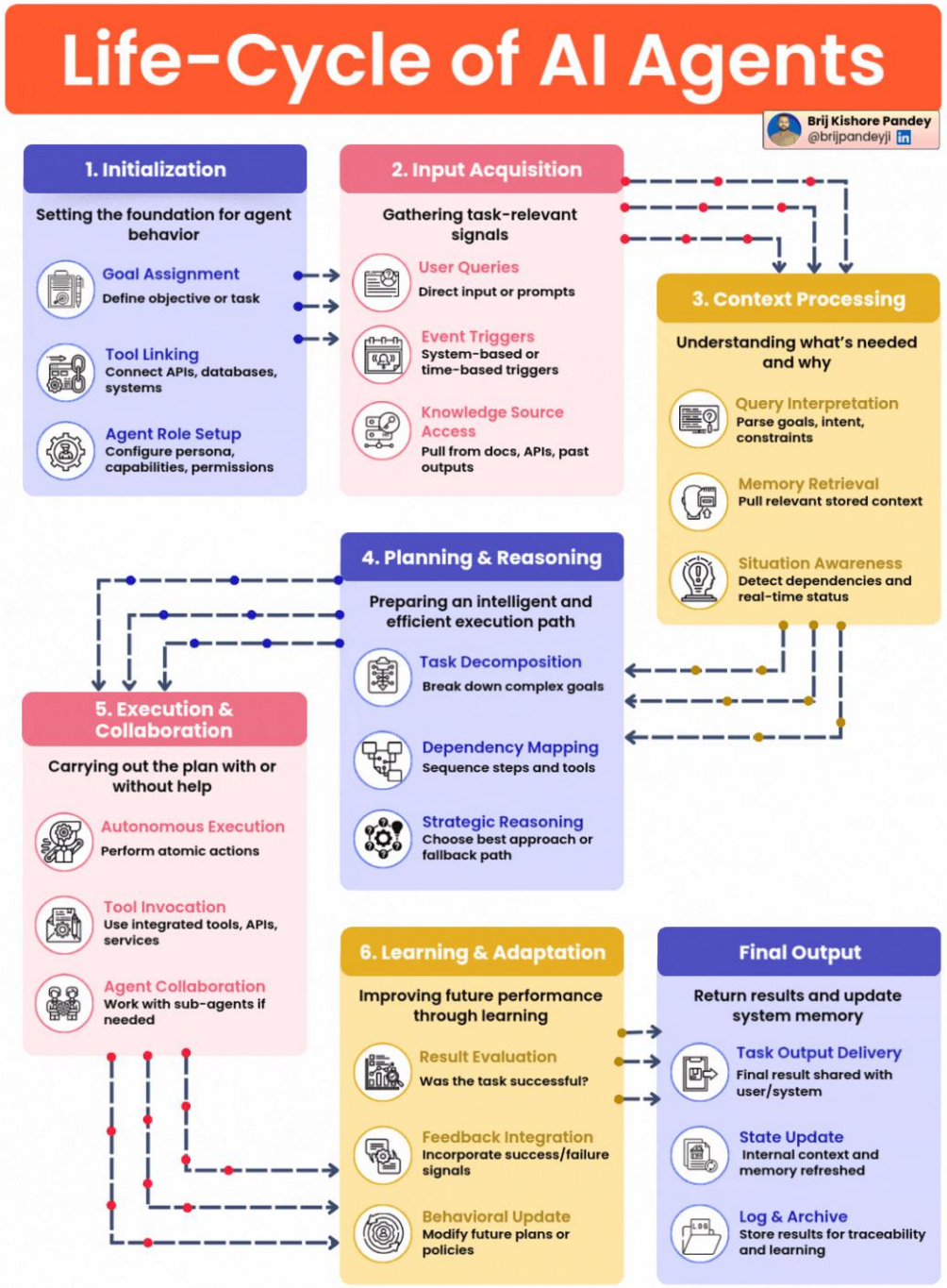
Tuning


Healthcare data
Millions of EHRs
Reusable components
Medical foundation model
Task adoption
Question answering
Chart summarizing
Image analysis/labelling
Risk stratification
Finding similar patients
Human-AI collaboration
A/ह
Natural language interactions

Tuning


More efficient methods of tuning
Parameter-efficient tuning methods (PETM)
Methods for tuning an LLM on your own custom data without duplicating the model
Prompt tuning
One of the easiest parameter-efficient tuning methods

Life-Cycle of AI Agents

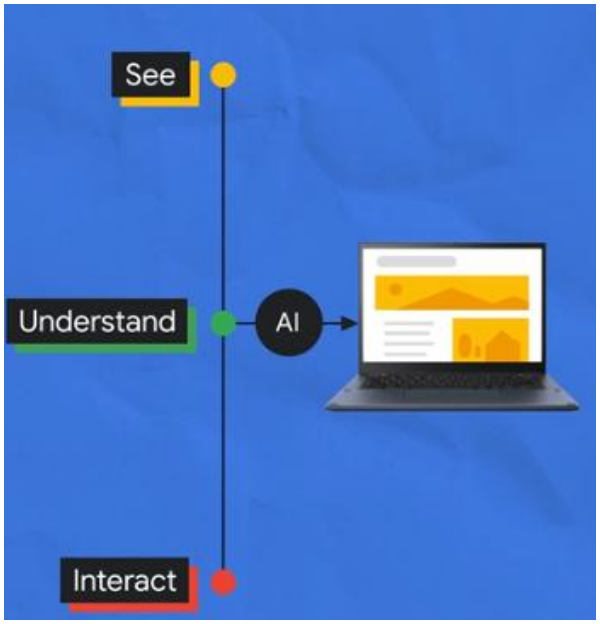


Responsible AI

- **Purpose**: To ensure AI systems are ethical, transparent, fair, and accountable.
- **Focus Areas**:
  - **Fairness**: Avoiding bias in data and models.
  - **Explainability**: Making AI decisions understandable.
  - **Privacy**: Protecting user data and complying with regulations.
  - **Accountability**: Ensuring human oversight and governance.
- **Frameworks**: AI ethics guidelines, model interpretability tools (e.g., SHAP, LIME), regulatory compliance (e.g., GDPR, AI Act).



Responsible AI

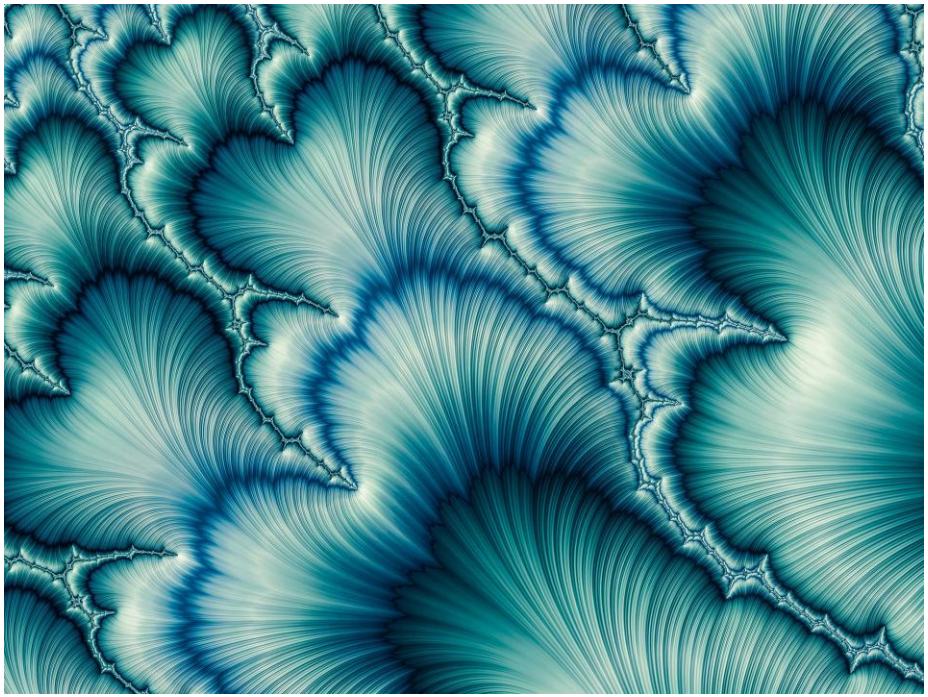


- Built for everyone
- Accountable and safe
- Respects privacy
- Driven by scientific excellence

Developing responsible AI requires an understanding of the possible issues, limitations, or unintended consequences

Real-World Applications of Generative AI

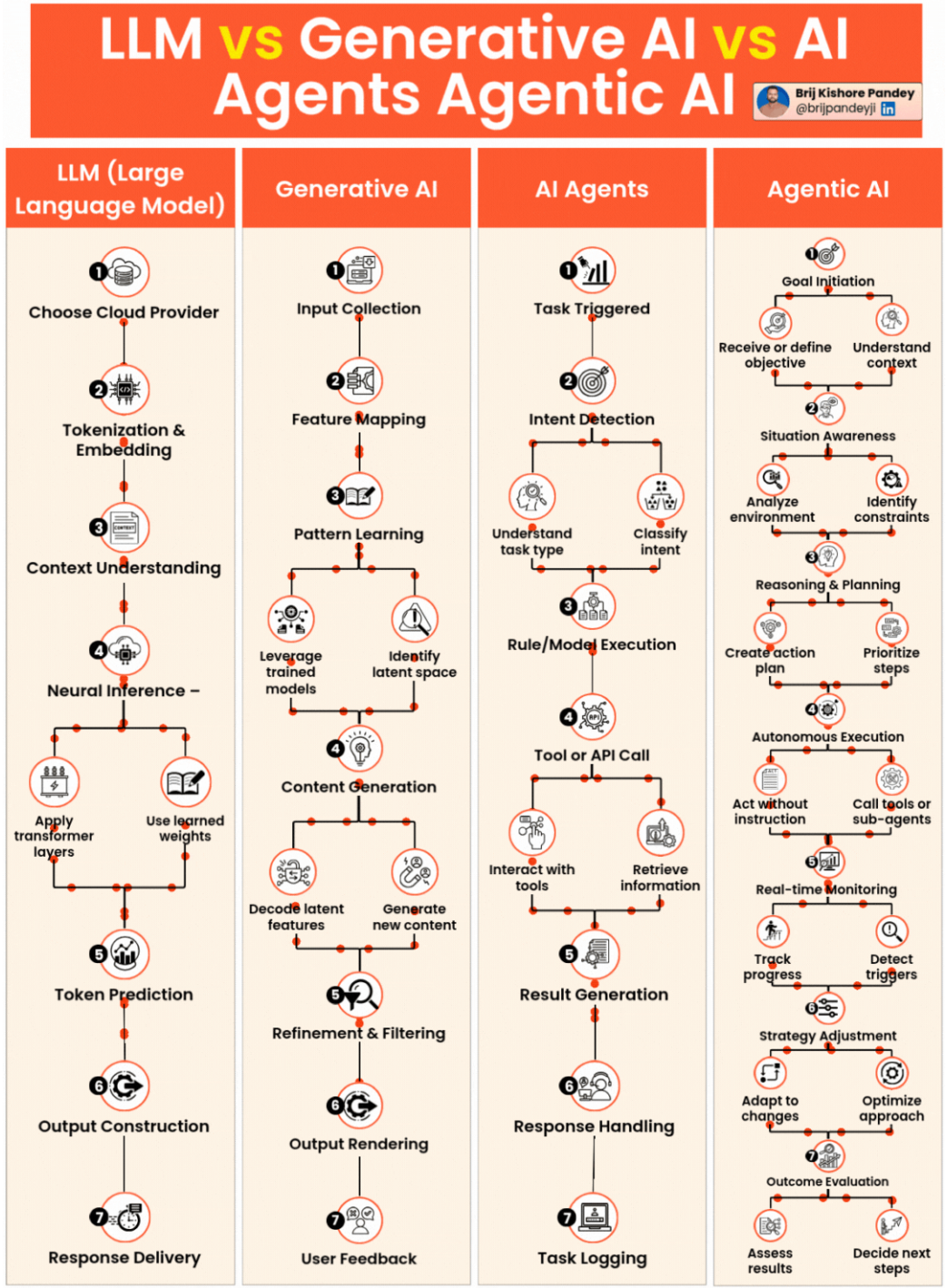
- Generative AI is transforming content creation across industries.
- Applications include automated writing, design, and code generation.
- Enhances creativity by providing novel ideas and solutions.
- Enables personalized and adaptive user experiences.
- Continuously improving through feedback and real-world use.



LLM ≠ Generative AI ≠ AI Agents ≠ Agentic AI

We need to stop grouping them together.

Each serves a different purpose, operates at a different level of complexity, and solves a different class of problems.



Here’s the breakdown

- **LLM**

Predicts tokens by recognizing patterns in data.
It has no memory, no intention, and cannot perform tasks—simply transforming input into output.

- **Generative AI**

Extends LLMs to produce text, code, images, and more.
It grasps latent spaces and can generate original content yet still relies on instructions.

- **AI Agents**

Carry out specific tasks.
They interpret intent, invoke tools or APIs, and manage responses. While modular and functional, they lack autonomy.

- **Agentic AI**

Functions with goals, planning, context, and memory.
It reasons, adapts, delegates to sub-agents, tracks progress, and independently decides next steps without human input.

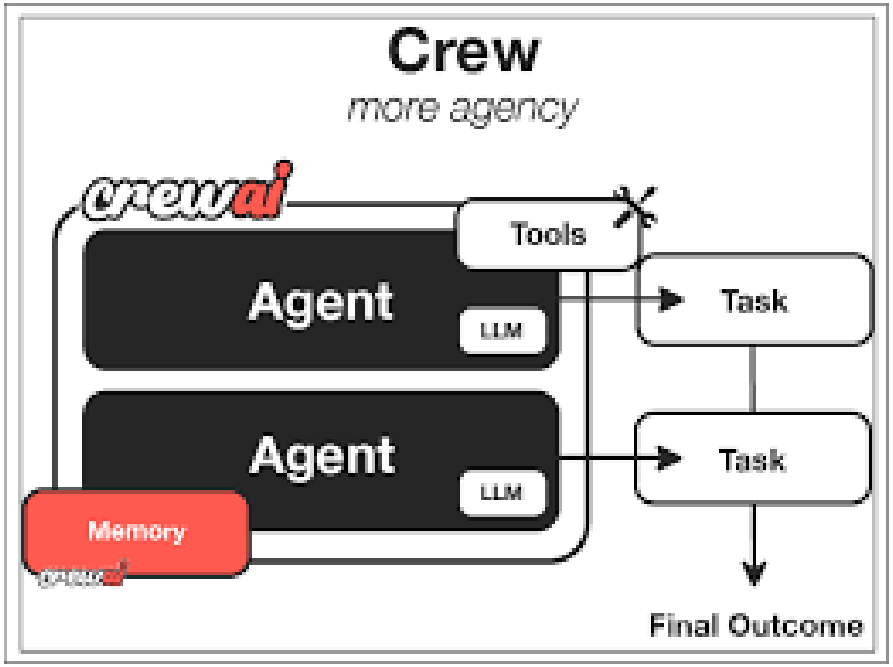
This represents more than just added features—

it marks a transformation in system design: shifting from prediction to orchestration, and from command-based to autonomous operation.

When developing with AI, understanding your system’s place in this hierarchy influences architecture, tools, risks, and overall value.

Crew AI

https://www.crewai.com/



What is CrewAI?

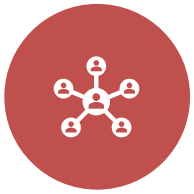
CrewAI is a lean, lightning-fast Python framework built entirely from scratch—completely independent of LangChain or other agent frameworks.

CrewAI empowers developers with both high-level simplicity and precise low-level control, ideal for creating autonomous AI agents tailored to any scenario:

- **CrewAI Crews**: Optimize for autonomy and collaborative intelligence, enabling you to create AI teams where each agent has specific roles, tools, and goals.
- **CrewAI Flows**: Enable granular, event-driven control, single LLM calls for precise task orchestration and supports Crews natively.

With over 100,000 developers certified through our community courses, CrewAI is rapidly becoming the standard for enterprise-ready AI automation.

CrewAI’s main components



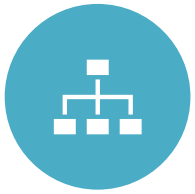
AGENTS: THESE ARE LIKE YOUR DEDICATED TEAM MEMBERS, EACH WITH THEIR ROLE, BACKGROUND STORY, GOAL AND MEMORY



TOOLS: THE EQUIPMENT OUR AGENTS USE TO PERFORM THEIR TASKS EFFICIENTLY; YOU CAN USE ANY EXISTING ONES FROM LANGCHAIN OR QUICKLY WRITE YOUR OWN



TASKS: SMALL AND FOCUSED MISSION A GIVEN AGENT SHOULD ACCOMPLISH

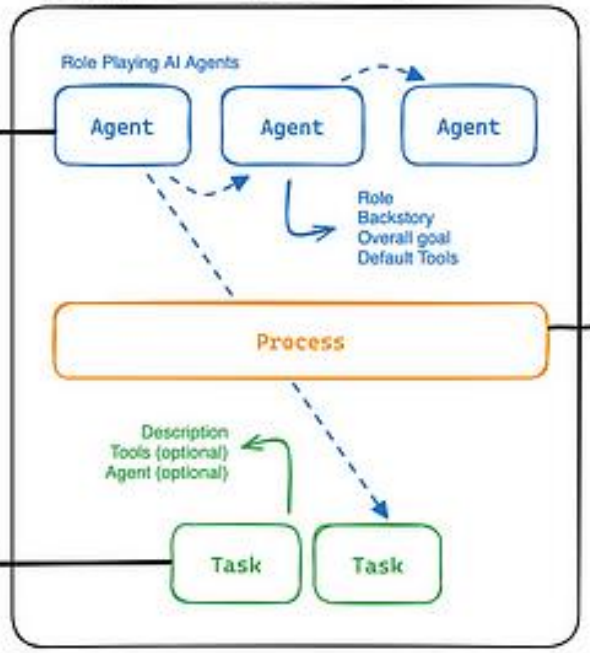


PROCESS: THIS IS THE WORKFLOW OR STRATEGY THE CREW FOLLOWS TO COMPLETE TASKS.



CREW: WHERE AGENTS, TASKS AND A PROCESS MEET, THIS IS THE CONTAINER LAYER WHERE THE WORK HAPPENS

CrewAI - A role playing multi-agent framework



Agents have the inert ability of reach out to another to delegate work or ask questions.

Tasks can override agent tool with specific ones that should be used and also have a specific agent tackle them.

Processes define how agents will work together:

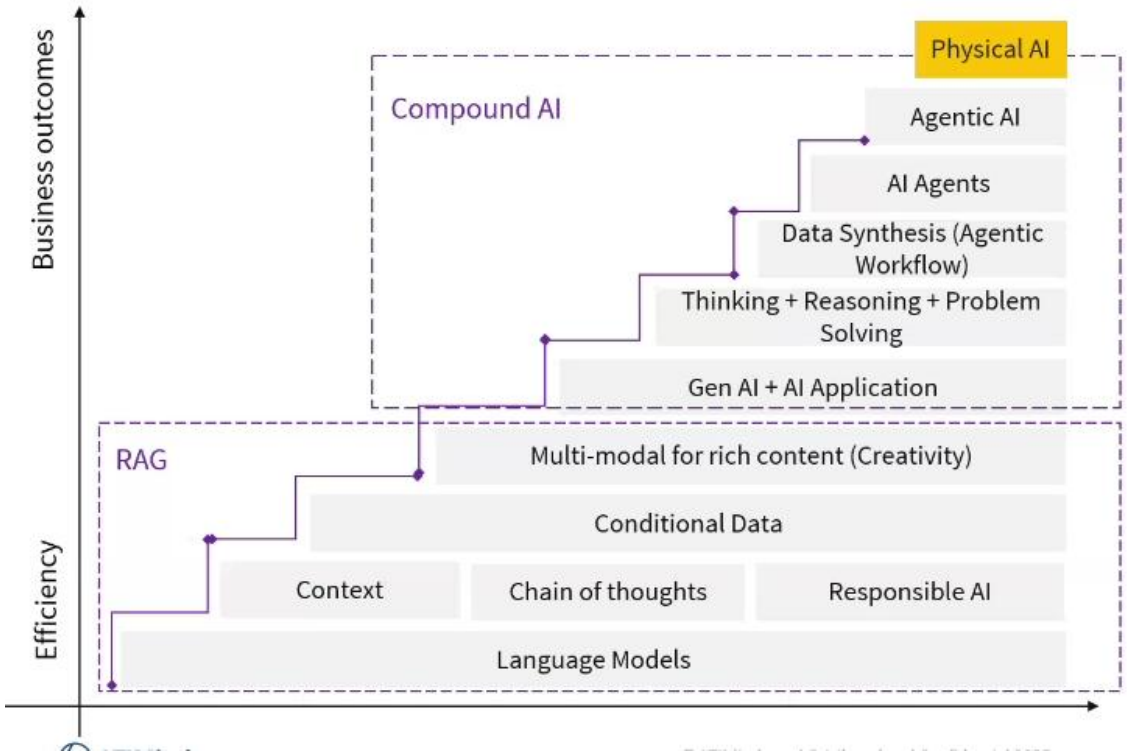
- How tasks will be assigned to Agents.
- How agents interact with each other.
- How agents perform their tasks.

PseudoCode

```
researcher = Agent( ... )
writer = Agent( ... )
tasks = [Task( ... ), Task( ... )]

crew = Crew(
    agents=[researcher, writer],
    tasks=tasks,
    process=Process.sequential
)
```

Fast evolving world of AI: Focus re-shifting from AI Agents to Agentic Workflow



Compute has new challengers

Training vs Inferencing (NVIDIA vs Groq LPU). Microsoft Quantum Chip Majorana 1

Physical AI

Multimodal sensing can understand complex environment and take action at edge

Human role is evolving

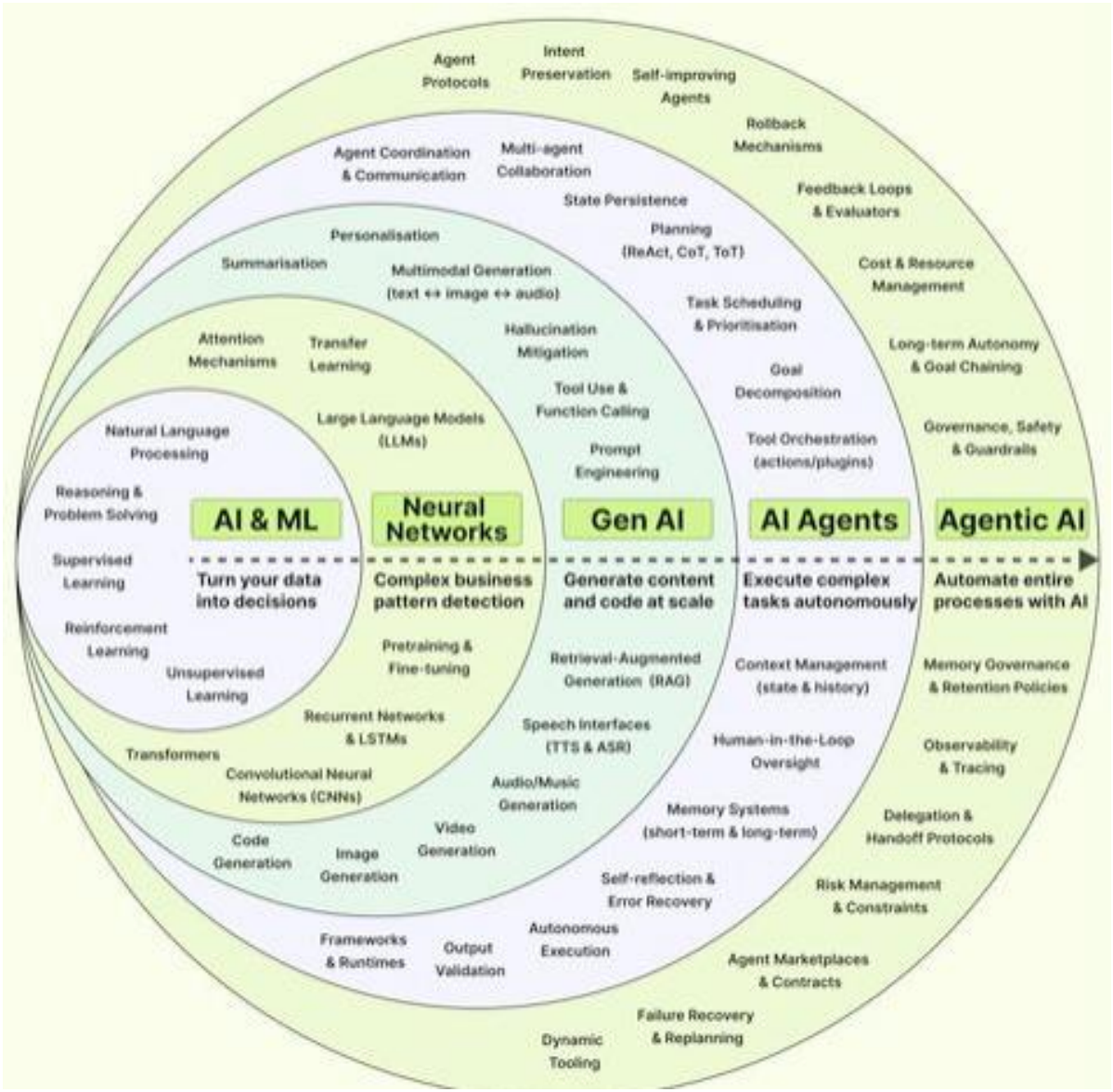
HITL (Human in the loop) to HOTL (Human over the loop) to HOOF (Human out of loop)

Agents are taking over

Agents with deep domain thinking & reasoning (marketing, customer service, supplier management, underwriting, trials etc.). Will require solve for Domain expertise, Marketplace, Observability, FinOps, Orchestration, Interoperability….

Rise of Reasoning & thinking Models

DeepSeek built fraction of cost. Microsoft o series, Google Gemini Flash 2.0. Model Engineering, Distilling, Finetuning for XLM key skills


AI & ML
Neural Networks
Gen AI
AI Agents
Agentic AI
Turn your data into decisions
Complex business pattern detection
Generate content and code at scale
Execute complex tasks autonomously
Automate entire processes with AI
Natural Language Processing
Reasoning & Problem Solving
Supervised Learning
Reinforcement Learning
Unsupervised Learning
Large Language Models (LLMs)
Attention Mechanisms
Transfer Learning
Transformers
Recurrent Networks & LSTMs
Convolutional Neural Networks (CNNs)
Pretraining & Fine-tuning
Prompt Engineering
Tool Use & Function Calling
Hallucination Mitigation
Retrieval-Augmented Generation (RAG)
Speech Interfaces (TTS & ASR)
Audio/Music Generation
Video Generation
Image Generation
Code Generation
Personalisation
Summarisation
Multimodal Generation (text ↔ image ↔ audio)
Tool Orchestration (actions/plugins)
Goal Decomposition
Task Scheduling & Prioritisation
Planning (ReAct, CoT, ToT)
Context Management (state & history)
Human-in-the-Loop Oversight
Memory Systems (short-term & long-term)
Self-reflection & Error Recovery
Autonomous Execution
Output Validation
Frameworks & Runtimes
Agent Coordination & Communication
Multi-agent Collaboration
State Persistence
Feedback Loops & Evaluators
Cost & Resource Management
Long-term Autonomy & Goal Chaining
Governance, Safety & Guardrails
Memory Governance & Retention Policies
Observability & Tracing
Delegation & Handoff Protocols
Risk Management & Constraints
Agent Marketplaces & Contracts
Failure Recovery & Replanning
Dynamic Tooling
Agent Protocols
Intent Preservation
Self-improving Agents
Rollback Mechanisms

Agentic AI isn’t one single tool. Think of AI as layers of capability. Each unlocks different business cases. Each requires different infrastructure, skills, and governance. Here's the breakdown:

1. **Machine Learning** : Turn your data into decisions. Systems that learn from historical data, make predictions, and optimize without explicit programming. → Forecast sales and demand with precision → Detect fraud before it hits your bottom line → Predict customer churn and intervene proactively → Optimize pricing based on market conditions Tools: AWS SageMaker, Google Vertex AI, Azure ML

2. **Neural Networks** & **Deep Learning** : Complex pattern detection at scale. When data gets complex, images, audio, and documents, traditional algorithms fall short. Neural networks recognize what humans can't program. → Inspect product quality with computer vision → Enable voice commands and real-time transcription → Process invoices, contracts, and documents → Power facial recognition for security Tools: TensorFlow, PyTorch, AWS Rekognition

3. **Generative AI** : Generate content and code at scale. This is where most companies are today. AI that creates text, code, images, audio, video from prompts. → Draft marketing content, emails, reports in minutes → Automate meeting notes, summaries, scheduling → Build knowledge bases for customer support → Generate code, documentation, test cases → Create product images and ad creatives Tools: ChatGPT, Claude, Gemini, Midjourney

4. **AI Agents** : Execute complex tasks autonomously. The leap from creation to action. AI Agents plan, use tools, maintain context, and execute multi-step workflows. → Execute IT tasks using APIs and integrations → Generate leads and personalize outreach sequences → Handle customer requests from start to resolution → Research topics and compile comprehensive reports Tools: LangChain, CrewAI, Microsoft Copilot

5. **Agentic AI** : Automate entire processes with AI. The frontier. Networks of agents that collaborate autonomously. They plan long-term, coordinate, self-evaluate, and improve over time. → Modernize legacy software with Agentic coding → Build AI directly into your products → Orchestrate end-to-end business processes Tools: Claude Code, OpenAI Codex, Devin

The Key Differences

- Machine Learning → Analyzes and predicts
- Neural Networks → Recognizes patterns
- Generative AI → Creates content and code AI
- Agents → Execute multi-step tasks with tools
- Agentic AI → Orchestrates entire processes

Github

- https://github.com/SomGitHub2018/Reva_Classes